A Living Review of the Intersection of Nuclear and Particle Physics and Quantum Information Science

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Abstract: Inspired by "A Living Review of Machine Learning for Particle Physics"[1], the goal of this document is to provide a nearly comprehensive list of citations for those developing and applying quantum information approaches to experimental, phenomenological, or theoretical analyses. Applications of quantum information science to high energy physics is a relatively new field of research. As a living document, it will be updated as often as possible with the relevant literature with the latest developments. Suggestions are most welcome.

[1] See https://github.com/iml-wg/HEPML-LivingReview.

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1 Introduction

The purpose of this note is to collect references for quantum information science as applied to particle and nuclear physics. The papers are listed in reverse chronological order. In order to be as useful as possible, this document will continually change. Please check back [2] regularly. You can simply download the .bib file to get all of the latest references. Suggestions are most welcome.

2 Nuclear and Particle Physics in Quantum Information Science

2.1 Reviews and Whitepapers and Proceedings

2.1.1 Reviews

2.1.1.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

[2] See https://github.com/PamelaPajarillo/NUPAQIS-LivingReview.

2.1.1.2 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

2.1.1.3 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.1.1.4 Quantum Sensors with Matter Waves for GW Observation [4]

- **Authors:** Andrea Bertoldi, Philippe Bouyer, Benjamin Canuel
- **Posted on arXiv:** 27 October 2020
- **Published in nan:** 2022

2.1.1.5 Quantum Computing Applications in Future Colliders [5]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.1.1.6 Many-Body Collective Neutrino Oscillations: Recent Developments [6]

- **Authors:** Amol V. Patwardhan, Michael J. Cervia, Ermal Rrapaj, Pooja Siwach, A.B. Balantekin
- **Posted on arXiv:** 31 December 2022
- **Published in nan:** 2023

2.1.1.7 Review on Quantum Computing for Lattice Field Theory [7]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in nan:** 01 February 2023

2.1.1.8 Quantum information and quantum simulation of neutrino physics [8]

- **Authors:** A.B. Balantekin, Michael J. Cervia, Amol V. Patwardhan, Ermal Rrapaj, Pooja Siwach
- **Posted on arXiv:** 01 May 2023
- **Published in Eur.Phys.J.A:** 17 August 2023

2.1.1.9 Machine learning for anomaly detection in particle physics [9]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klaeboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

2.1.1.10 Quantum entanglement and Bell inequality violation at colliders [10]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

2.1.1.11 Quantum Complexity and New Directions in Nuclear Physics and High-Energy Physics Phenomenology [11]

- **Authors:** Caroline E.P. Robin, Martin J. Savage
- **Posted on arXiv:** 29 April 2026

2.1.2 Whitepapers and Proceedings

2.1.2.1 Quantum Sensing for High Energy Physics [12]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs,

Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

2.1.2.2 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar

- **Posted on arXiv:** 13 March 2019

2.1.2.3 AEDGE: Atomic Experiment for Dark Matter and Gravity Exploration in Space [14]

- **Authors:** Yousef Abou El-Neaj, Cristiano Alpigiani, Sana Amairi-Pyka, Henrique Araújo, Antun Balaž, Angelo Bassi, Lars Bathe-Peters, Baptiste Battelier, Aleksandar Belić, Elliot Bentine, José Bernabeu, Andrea Bertoldi, Robert Bingham, Diego Blas, Vasiliki Bolpasi, Kai Bongs, Sougato Bose, Philippe Bouyer, Themis Bowcock, William Bowden, Oliver Buchmueller, Clare Burrage, Xavier Calmet, Benjamin Canuel, Laurentiu-Ioan Caramete, Andrew Carroll, Giancarlo Cella, Vassilis Charmandaris, Swapan Chattopadhyay, Xuzong Chen, Maria Luisa Chiofalo, Jonathon Coleman, Joseph Cotter, Yanou Cui, Andrei Derevianko, Albert De Roeck, Goran S. Djordjevic, Peter Dornan, Michael Doser, Ioannis Drougkakis, Jacob Dunningham, Ioana Dutan, Sajan Easo, Gedminas Elertas, John Ellis, Mai El Sawy, Farida Fassi, Daniel Felea, Chen-Hao Feng, Robert Flack, Chris Foot, Ivette Fuentes, Naceur Gaaloul, Alexandre Gauguet, Remi Geiger, Valerie Gibson, Gian Giudice, Jon Goldwin, Oleg Grachov, Peter W. Graham, Dario Grasso, Maurits van der Grinten, Mustafa Gündogan, Martin G. Haehnelt, Tiffany Harte, Aurelien Hees, Richard Hobson, Jason Hogan, Bodil Holst, Michael

Holynski, Mark Kasevich, Bradley J. Kavanagh, Wolf von Klitzing, Tim Kovachy, Benjamin Krikler, Markus Krutzik, Marek Lewicki, Yu-Hung Lien, Miaoyuan Liu, Giuseppe Gaetano Luciano, Alain Magnon, Mohammed Attia Mahmoud, Sarah Malik, Christopher McCabe, Jeremiah Mitchell, Julia Pahl, Debapriya Pal, Saurabh Pandey, Dimitris Papazoglou, Mauro Paternostro, Bjoern Penning, Achim Peters, Marco Prevedelli, Vishnupriya Puthiya-Veettil, John Quenby, Ernst Rasel, Sean Ravenhall, Jack Ringwood, Albert Roura, Dylan Sabulsky, Muhammed Sameed, Ben Sauer, Stefan Alaric Schäffer, Stephan Schiller, Vladimir Schkolnik, Dennis Schlippert, Christian Schubert, Haifa Rejeb Sfar, Armin Shayeghi, Ian Shipsey, Carla Signorini, Yeshpal Singh, Marcelle Soares-Santos, Fiodor Sorrentino, Timothy Sumner, Konstantinos Tassis, Silvia Tentindo, Guglielmo Maria Tino, Jonathan N. Tinsley, James Unwin, Tristan Valenzuela, Georgios Vasilakis, Ville Vaskonen, Christian Vogt, Alex Webber-Date, Andŕ e} Wenzlawski, Patrick Windpassinger, Marian Woltmann, Efe Yazgan, Ming-Sheng Zhan, Xinhao Zou, Jure Zupan

- **Posted on arXiv:** 02 August 2019
- **Published in EPJ Quant.Technol.:** 04 March 2020

2.1.2.4 Mechanical quantum sensing in the search for dark matter [15]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.1.2.5 Standard model physics and the digital quantum revolution: thoughts about the interface [16]

- **Authors:** Natalie Klco, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 10 July 2021
- **Published in Rept.Prog.Phys.:** 19 May 2022

2.1.2.6 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [17]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

2.1.2.7 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [18]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.1.2.8 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [19]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.1.2.9 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [20]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

2.1.2.10 Snowmass2021 Cosmic Frontier White Paper: Future Gravitational-Wave Detector Facilities [21]

- **Authors:** Stefan W. Ballmer, Rana Adhikari, Leonardo Badurina, Duncan A. Brown, Swapan Chattopadhyay, Matthew Evans, Peter Fritschel, Evan Hall, Jason M. Hogan, Karan Jani, Tim Kovachy, Kevin Kuns, Ariel Schwartzman, Daniel Sigg, Bram Slagmolen, Salvatore Vitale, Christopher Wipf
- **Posted on arXiv:** 15 March 2022

2.1.2.11 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo

- **Posted on arXiv:** 15 March 2022

2.1.2.12 New Horizons: Scalar and Vector Ultralight Dark Matter [23]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

2.1.2.13 Quantum Networks for High Energy Physics [24]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

2.1.2.14 Quantum Simulation for High-Energy Physics [25]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti

- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.1.2.15 Quantum computing hardware for HEP algorithms and sensing [26]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti
- **Posted on arXiv:** 18 April 2022

2.1.2.16 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [27]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

2.1.2.17 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne,

Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao

- **Posted on arXiv:** 28 February 2023

2.1.2.18 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023

- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.1.2.19 Quantum Sensors for High Energy Physics [30]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

2.2 Anomaly Detection

2.2.1 Quantum Computing Paradigms - Continuous Variable Quantum Computing

2.2.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [31]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.2.2 Quantum Computing Paradigms - Quantum Annealing

2.2.2.1 A Quantum Algorithm for Model-Independent Searches for New Physics [32]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

2.2.3 Quantum Machine Learning - Supervised Methods

2.2.3.1 Unravelling physics beyond the standard model with classical and quantum anomaly detection [33]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

2.2.4 Quantum Machine Learning - Unsupervised Methods

2.2.4.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [34]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

2.2.5 Quantum Machine Learning - Variational Quantum Algorithms

2.2.5.1 Anomaly detection in high-energy physics using a quantum autoencoder [35]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

2.2.5.2 Quantum anomaly detection for collider physics [36]

- **Authors:** Sulaiman Alvi, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 16 June 2022
- **Published in JHEP:** 22 February 2023

2.2.6 Crosslists

2.2.6.1 Machine learning for anomaly detection in particle physics [9]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klaeboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

2.3 Beyond the Standard Model

2.3.1 Quantum Algorithms - Grover's Search Algorithm

2.3.1.1 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [37]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

2.3.2 Quantum Algorithms - Quantum Simulations

2.3.2.1 A Quantum Computational Determination of the Weak Mixing Angle in the Standard Model [38]

- **Authors:** Qiaofeng Liu, Ian Low, Zhewei Yin
- **Posted on arXiv:** 22 September 2025

2.3.3 Quantum Entanglement and Bell Inequalities

2.3.3.1 Entanglement suppression, enhanced symmetry, and a standard-model-like Higgs boson [39]

- **Authors:** Marcela Carena, Ian Low, Carlos E.M. Wagner, Ming-Lei Xiao
- **Posted on arXiv:** 16 July 2023
- **Published in Phys.Rev.D:** 01 March 2024

2.3.3.2 Flavor patterns of fundamental particles from quantum entanglement? [40]

- **Authors:** Jesse Thaler, Sokratis Trifinopoulos
- **Posted on arXiv:** 30 October 2024
- **Published in Phys.Rev.D:** 01 March 2025

2.3.4 Quantum Machine Learning - Variational Quantum Algorithms

2.3.4.1 Hybrid Quantum-Classical Graph Convolutional Network [41]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

2.3.5 Quantum Measurement and Tomography

2.3.5.1 Emergent symmetry in a two-Higgs-doublet model from quantum information and nonstabilizerness [42]

- **Authors:** Giorgio Busoni, John Gargalionis, Ewan N.V. Wallace, Martin J. White
- **Posted on arXiv:** 02 June 2025
- **Published in Phys. Rev. D. 112 (2025) 035022:** 01 August 2025

2.3.6 Crosslists

2.3.6.1 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

2.3.6.2 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

2.3.6.3 Event Classification with Quantum Machine Learning in High-Energy Physics [43]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.3.6.4 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [44]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.3.6.5 Quantum convolutional neural networks for high energy physics data analysis [45]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.3.6.6 Quantum algorithm for the classification of supersymmetric top quark events [46]

- **Authors:** Pedrame Bargassa, Timothée Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothée Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.3.6.7 Unravelling physics beyond the standard model with classical and quantum anomaly detection [33]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

2.3.6.8 Quantum anomaly detection in the latent space of proton collision events at the LHC [34]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

2.3.6.9 Quantum Sensors for High Energy Physics [30]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.3.6.10 Quantum Metric Learning for New Physics Searches at the LHC [47]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

2.4 Dark Matter - Particle-like Dark Matter

2.4.1 Quantum Algorithms - Quantum Simulations

2.4.1.1 Quantum simulations of dark sector showers [48]

- **Authors:** So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 26 April 2022
- **Published in Phys.Lett.B:** 26 September 2022

2.4.2 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.4.2.1 Trapped Electrons and Ions as Particle Detectors [49]

- **Authors:** Daniel Carney, Hartmut Häffner, David C. Moore, Jacob M. Taylor
- **Posted on arXiv:** 12 April 2021
- **Published in Phys Rev Lett, Vol. 127, No. 6, 2021. "Editor's suggestion":** 05 August 2021

2.4.2.2 Millicharged Dark Matter Detection with Ion Traps [50]

- **Authors:** Dmitry Budker, Peter W. Graham, Harikrishnan Ramani, Ferdinand Schmidt-Kaler, Christian Smorra, Stefan Ulmer
- **Posted on arXiv:** 11 August 2021
- **Published in PRX Quantum:** 01 February 2022

2.4.3 Quantum Sensors - Optomechanical Sensors

2.4.3.1 Search for Kilogram-scale Dark Matter with Precision Displacement Sensors [51]

- **Authors:** Akio Kawasaki
- **Posted on arXiv:** 26 August 2018
- **Published in Phys. Rev. D 99, 023005 (2019):** 07 January 2019

2.4.3.2 Proposal for gravitational direct detection of dark matter [52]

- **Authors:** Daniel Carney, Sohitri Ghosh, Gordan Krnjaic, Jacob M. Taylor
- **Posted on arXiv:** 01 March 2019
- **Published in Phys.Rev.D:** 14 October 2020

2.4.3.3 Search for composite dark matter with optically levitated sensors [53]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

2.4.3.4 Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors [54]

- **Authors:** Gadi Afek, Daniel Carney, David C. Moore
- **Posted on arXiv:** 05 November 2021
- **Published in Phys.Rev.Lett.:** 09 March 2022

2.4.3.5 Models of ultraheavy dark matter visible to macroscopic mechanical sensing arrays [55]

- **Authors:** Carlos Blanco, Bahaa Elshimy, Rafael F. Lang, Robert Orlando
- **Posted on arXiv:** 29 December 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.4.3.6 Optomechanical dark matter instrument for direct detection [56]

- **Authors:** Christopher G. Baker, Warwick P. Bowen, Peter Cox, Matthew J. Dolan, Maxim Goryachev, Glen Harris
- **Posted on arXiv:** 16 June 2023
- **Published in Phys. Rev. D 110, 043005 (2024):** 02 August 2024

2.4.3.7 Probing levitodynamics with multi-stochastic forces and the simple applications on the dark matter detection in optical levitation experiment [57]

- **Authors:** Xi Cheng, Ji-Heng Guo, Wenyu Wang, Bin Zhu
- **Posted on arXiv:** 07 December 2023
- **Published in Nucl. Phys. B 1010 (2025) 116780:** 19 December 2024

2.4.3.8 Dark Matter Searches with Levitated Sensors [58]

- **Authors:** Eva Kilian, Markus Rademacher, Jonathan M.H. Gosling, Julian H. Iacoponi, Fiona Alder, Marko Toroš, Antonio Pontin, Chamkaur Ghag, Sougato Bose, Tania S. Monteiro, P.F. Barker
- **Posted on arXiv:** 31 January 2024
- **Published in AVS Quantum Sci. 6, 030503 (2024):** 2024

2.4.3.9 Probing light particles with optically trapped sensors through nucleon scattering [59]

- **Authors:** Bhaskar Dutta, Dilip Kumar Ghosh, Sk Jeesun
- **Posted on arXiv:** 31 January 2025
- **Published in Phys.Rev.D:** 01 January 2026

2.4.3.10 Mechanical sensors for ultraheavy dark matter searches via long-range forces [60]

- **Authors:** Juehang Qin, Dorian W.P. Amaral, Sunil A. Bhave, Erqian Cai, Daniel Carney, Rafael F. Lang, Shengchao Li, Alberto M. Marino, Giacomo Marocco, Claire Marvinney, Jared R. Newton, Jacob M. Taylor, Christopher Tunnell
- **Posted on arXiv:** 14 March 2025
- **Published in Phys.Rev.D 112 (2025) 7, 072003:** 01 October 2025

2.4.4 Quantum Sensors - Solid State Sensors

2.4.4.1 Dark matter direct detection with quantum dots [61]

- **Authors:** Carlos Blanco, Rouven Essig, Marivi Fernandez-Serra, Harikrishnan Ramani, Oren Slone
- **Posted on arXiv:** 10 August 2022
- **Published in Phys.Rev.D:** 01 May 2023

2.4.4.2 Dark Matter Induced Power in Quantum Devices [62]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 17 October 2022
- **Published in Phys. Rev. Lett. 132, 121801 (2024):** 22 March 2024

2.4.4.3 Transmon Qubit constraints on dark matter-nucleon scattering [63]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 30 April 2024
- **Published in JHEP:** 25 July 2024

2.4.4.4 Quantum parity detectors: A qubit-based particle-detection scheme with meV thresholds for rare-event searches [64]

- **Authors:** Karthik Ramanathan, Brandon J. Sandoval, John E. Parker, Lalit M. Joshi, Andrew D. Beyer, Pierre M. Echternach, Serge Rosenblum, Sunil R. Golwala
- **Posted on arXiv:** 27 May 2024
- **Published in APS Open Science:** 01 May 2026

2.4.5 Crosslists

2.4.5.1 Quantum Sensing for High Energy Physics [12]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik

Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

2.4.5.2 Mechanical quantum sensing in the search for dark matter [15]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao

- **Posted on arXiv:** 13 August 2020

- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.4.5.3 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [19]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova

- **Posted on arXiv:** 14 March 2022

2.4.5.4 Quantum Networks for High Energy Physics [24]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

2.4.5.5 Quantum Sensors for High Energy Physics [30]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

2.5 Dark Matter - Wave-like Dark Matter

2.5.1 Quantum Error Correction and Mitigation

2.5.1.1 Quantum-Error-Correction-Inspired Noise Mitigation for Wavelike Dark Matter Searches with Quantum Sensors [65]

- **Authors:** Hajime Fukuda, Takeo Moroi, Thanaporn Sichanugrist

- **Posted on arXiv:** 05 November 2025

- **Published in Phys.Rev.Lett.:** 08 July 2026

2.5.2 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.5.2.1 The Global Network of Optical Magnetometers for Exotic physics (GNOME): A novel scheme to search for physics beyond the Standard Model [66]

- **Authors:** Szymon Pustelny, Derek F. Jackson Kimball, Chris Pankow, Micah P. Ledbetter, Przemyslaw Wlodarczyk, Piotr Wcislo, Maxim Pospelov, Joshua R. Smith, Jocelyn Read, Wojciech Gawlik, Dmitry Budker

- **Posted on arXiv:** 22 March 2013

- **Published in Annalen der Physik 525, 659 (2013):** 2013

2.5.2.2 The cosmic axion spin precession experiment (CASPEr): a dark-matter search with nuclear magnetic resonance [67]

- **Authors:** Antoine Garcon, Deniz Aybas, John W. Blanchard, Gary Centers, Nataniel L. Figueroa, Peter W. Graham, Derek F. Jackson Kimball, Surjeet Rajendran, Marina Gil Sendra, Alexander O. Sushkov, Lutz Trahms, Tao Wang, Arne Wickenbrock, Teng Wu, Dmitry Budker

- **Posted on arXiv:** 16 July 2017
- **Published in Quantum Science and Technology 2017:** 15 December 2017

2.5.2.3 Overview of the Cosmic Axion Spin Precession Experiment (CASPEr) [68]

- **Authors:** Derek F. Jackson Kimball, S. Afach, D. Aybas, J.W. Blanchard, D. Budker, G. Centers, M. Engler, N.L. Figueroa, A. Garcon, P.W. Graham, H. Luo, S. Rajendran, M.G. Sendra, A.O. Sushkov, T. Wang, A. Wickenbrock, A. Wilzewski, T. Wu
- **Posted on arXiv:** 09 November 2017
- **Published in Springer Proc.Phys.:** 2020

2.5.2.4 Precision Metrology Meets Cosmology: Improved Constraints on Ultralight Dark Matter from Atom-Cavity Frequency Comparisons [69]

- **Authors:** Colin J. Kennedy, Eric Oelker, John M. Robinson, Tobias Bothwell, Dhruv Kedar, William R. Milner, G. Edward Marti, Andrei Derevianko, Jun Ye
- **Posted on arXiv:** 20 August 2020
- **Published in Phys. Rev. Lett. 125, 201302 (2020):** 12 November 2020

2.5.2.5 Search for Axionlike Dark Matter Using Solid-State Nuclear Magnetic Resonance [70]

- **Authors:** Deniz Aybas, Janos Adam, Emmy Blumenthal, Alexander V. Gramolin, Dorian Johnson, Annalies Kleyheeg, Samer Afach, John W. Blanchard, Gary P. Centers, Antoine Garcon, Martin Engler, Nataniel L. Figueroa, Marina Gil Sendra, Arne Wickenbrock, Matthew Lawson, Tao Wang, Teng Wu, Haosu Luo, Hamdi Mani, Philip Mauskopf, Peter W. Graham, Surjeet Rajendran, Derek F. Jackson Kimball, Dmitry Budker, Alexander O. Sushkov
- **Posted on arXiv:** 04 January 2021
- **Published in Phys. Rev. Lett. 126, 141802 (2021):** 10 April 2021

2.5.2.6 Search for Dark-Matter-Induced Oscillations of Fundamental Constants Using Molecular Spectroscopy [71]

- **Authors:** R. Oswald, A. Nevsky, V. Vogt, S. Schiller, N.L. Figueroa, K. Zhang, O. Tretiak, D. Antypas, D. Budker, A. Banerjee, G. Perez
- **Posted on arXiv:** 12 November 2021
- **Published in Phys.Rev.Lett.:** 15 July 2022

2.5.2.7 Measuring the stability of fundamental constants with a network of clocks [72]

- **Authors:** G. Barontini, L. Blackburn, V. Boyer, F. Butuc-Mayer, X. Calmet, J.R. Crespo Lopez-Urrutia, E.A. Curtis, B. Darquie, J. Dunningham, N.J. Fitch, E.M. Forgan, K. Georgiou, P. Gill, R.M. Godun, J. Goldwin, V. Guarrera, A.C. Harwood, I.R. Hill, R.J. Hendricks, M. Jeong, M.Y.H. Johnson, M. Keller, L.P. Kozhiparambil Sajith, F. Kuipers, H.S. Margolis, C. Mayo, P. Newman, A.O. Parsons, L. Prokhorov, B.I. Robertson, J. Rodewald, M.S. Safronova, B.E. Sauer, M. Schioppo, N. Sherrill, Y.V. Stadnik, K. Szymaniec, M.R. Tarbutt, R.C. Thompson, A. Tofful, J. Tunesi, A. Vecchio, Y. Wang, S. Worm
- **Posted on arXiv:** 20 December 2021
- **Published in EPJ Quantum Technology volume 9, Article number: 12 (2022):** 11 May 2022

2.5.2.8 Intensity interferometry for ultralight bosonic dark matter detection [73]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball
- **Posted on arXiv:** 05 February 2022
- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

2.5.2.9 One-Electron Quantum Cyclotron as a Milli-eV Dark-Photon Detector [74]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Roni Harnik, Thomas G. Myers, Harikrishnan Ramani, Benedict A.D. Sukra, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 12 August 2022
- **Published in Phys. Rev. Lett. 129, 261801 (2022):** 23 December 2022

2.5.2.10 Search for Ultralight Dark Matter from Long-Term Frequency Comparisons of Optical and Microwave Atomic Clocks [75]

- **Authors:** Takumi Kobayashi, Akifumi Takamizawa, Daisuke Akamatsu, Akio Kawasaki, Akiko Nishiyama, Kazumoto Hosaka, Yusuke Hisai, Masato Wada, Hajime Inaba, Takehiko Tanabe, Masami Yasuda
- **Posted on arXiv:** 12 December 2022
- **Published in Phys. Rev. Lett. 129, 241301 (2022):** 07 December 2022

2.5.2.11 Improved Limits on the Coupling of Ultralight Bosonic Dark Matter to Photons from Optical Atomic Clock Comparisons [76]

- **Authors:** M. Filzinger, S. Dörscher, R. Lange, J. Klose, M. Steinel, E. Benkler, E. Peik, C. Lisdat, N. Huntemann
- **Posted on arXiv:** 09 January 2023
- **Published in Phys.Rev.Lett.:** 22 June 2023

2.5.2.12 Oscillating Nuclear Charge Radii as Sensors for Ultralight Dark Matter [77]

- **Authors:** Abhishek Banerjee, Dmitry Budker, Melina Filzinger, Nils Huntemann, Gil Paz, Gilad Perez, Sergey Porsev, Marianna Safronova
- **Posted on arXiv:** 25 January 2023
- **Published in Phys. Rev. Lett. 135 (2025) 22, 223001:** 24 November 2025

2.5.2.13 Analysis of atomic-clock data to constrain variations of fundamental constants [78]

- **Authors:** Nathaniel Sherrill, Adam O. Parsons, Charles F.A. Baynham, William Bowden, E. Anne Curtis, Richard Hendricks, Ian R. Hill, Richard Hobson, Helen S. Margolis, Billy I. Robertson, Marco Schioppo, Krzysztof Szymaniec, Alexandra Tofful, Jacob Tunesi, Rachel M. Godun, Xavier Calmet
- **Posted on arXiv:** 09 February 2023
- **Published in New J.Phys.:** 06 September 2023

2.5.2.14 Frequency ratio measurements at 18-digit accuracy using an optical clock network [79]

- **Authors:** Kyle Beloy, Martha I. Bodine, Tobias Bothwell, Samuel M. Brewer, Sarah L. Bromley, Jwo-Sy Chen, Jean-Daniel Deschênes, Scott A. Diddams, Robert J. Fasano, Tara M. Fortier, Youssef S. Hassan, David B. Hume, Dhruv Kedar, Colin J. Kennedy, Isaac Khader, Amanda Koepke, David R. Leibrandt, Holly Leopardi, Andrew D. Ludlow, William F. McGrew, William R. Milner, Nathan R. Newbury, Daniele Nicolodi, Eric Oelker, Thomas E. Parker, John M. Robinson, Stefania Romisch, Stefan A. Schäffer, Jeffrey A. Sherman, Laura C. Sinclair, Lindsay Sonderhouse, William C. Swann, Jian Yao, Jun Ye, Xiaogang Zhang
- **Posted on arXiv:** 29 May 2020
- **Published in Nature:** 24 March 2021

2.5.2.15 Detection of bosenovae with quantum sensors on Earth and in space [80]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

2.5.2.16 Rydberg-atom-based single-photon detection for haloscope axion searches [81]

- **Authors:** Eleanor Graham, Sumita Ghosh, Yuqi Zhu, Xiran Bai, Sidney B. Cahn, Elsa Durcan, Michael J. Jewell, Danielle H. Speller, Sabrina M. Zacarias, Laura T. Zhou, Reina H. Maruyama
- **Posted on arXiv:** 23 October 2023
- **Published in Phys.Rev.D:** 01 February 2024

2.5.2.17 Quantum entanglement of ions for light dark matter detection [82]

- **Authors:** Asuka Ito, Ryuichiro Kitano, Wakutaka Nakano, Ryoto Takai
- **Posted on arXiv:** 20 November 2023
- **Published in JHEP:** 16 February 2024

2.5.2.18 Nuclear interferometer for ultralight dark matter detection [83]

- **Authors:** Hannah Banks, Elina Fuchs, Matthew McCullough
- **Posted on arXiv:** 15 July 2024
- **Published in Phys. Rev. D, 2026:** 01 July 2026

2.5.2.19 Highly excited electron cyclotron for QCD axion and dark-photon detection [84]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Harikrishnan Ramani, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 07 October 2024
- **Published in Phys.Rev.D 111 (2025) 7, 075022:** 01 April 2025

2.5.2.20 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [85]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

2.5.2.21 Sensitively searching for microwave dark photons with atomic ensembles [86]

- **Authors:** Suirong He, De He, Yufen Li, Li Gao, Xianing Feng, Hao Zheng, L.F. Wei
- **Posted on arXiv:** 01 December 2024
- **Published in Phys.Rev.D:** 15 September 2025

2.5.2.22 Searches for exotic spin-dependent interactions with spin sensors [87]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

2.5.2.23 Spin squeezing of macroscopic nuclear spin ensembles [88]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

2.5.2.24 Detecting Dark Matter Using Optically Trapped Rydberg Atom Tweezer Arrays [89]

- **Authors:** So Chigusa, Taiyo Kasamaki, Toshi Kusano, Takeo Moroi, Kazunori Nakayama, Naoya Ozawa, Yoshiro Takahashi, Atsuhiro Umemoto, Amar Vutha
- **Posted on arXiv:** 17 July 2025
- **Published in Phys. Rev. Lett. 136, 151801 (2026):** 14 April 2026

2.5.2.25 On the Speed-up of Wave-like Dark Matter Searches with Entangled Qubits [90]

- **Authors:** Arushi Bodas, Sohitri Ghosh, Roni Harnik
- **Posted on arXiv:** 13 October 2025

2.5.2.26 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [91]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

2.5.2.27 A thorium-229 optical nuclear clock with feedback loop [92]

- **Authors:** L. Toscani De Col, T. Riebner, I. Morawetz, F. Schneider, N. Sempelmann, J. Schlachet-Lépinay, F. Schaden, M. Bartokos, G.A. Kazakov, K. Beeks, B. Gerstenecker, M. Pimon, S. Lahs, A. Hellerschmied, T. Lercher, J. Premper, A. Niessner, M. Matus, H. Denker, M. Cizek, O. Cip, V. Lal, G. Zitzer, V. Petrov, J. Tiedau, M.V. Okhapkin, E. Peik, T. Schumm
- **Posted on arXiv:** 03 June 2026

2.5.3 Quantum Sensors - Optical/Photonic Sensors

2.5.3.1 Analysis of single-photon and linear amplifier detectors for microwave cavity dark matter axion searches [93]

- **Authors:** S.K. Lamoreaux, K.A. van Bibber, K.W. Lehnert, G. Carosi
- **Posted on arXiv:** 15 June 2013
- **Published in Phys.Rev.D:** 23 August 2013

2.5.3.2 Laser Interferometers as Dark Matter Detectors [94]

- **Authors:** Evan D. Hall, Rana X. Adhikari, Valery V. Frolov, Holger Müller, Maxim Pospelov, Rana X Adhikari
- **Posted on arXiv:** 03 May 2016
- **Published in Phys.Rev.D:** 23 October 2018

2.5.3.3 Accelerating dark-matter axion searches with quantum measurement technology [95]

- **Authors:** Huaixiu Zheng, Matti Silveri, R.T. Brierley, S.M. Girvin, K.W. Lehnert
- **Posted on arXiv:** 08 July 2016

2.5.3.4 A Search for Invisible Axion Dark Matter with the Axion Dark Matter Experiment [96]

- **Authors:** N. Du, N. Force, R. Khatiwada, E. Lentz, R. Ottens, L.J. Rosenberg, G. Rybka, G. Carosi, N. Woollett, D. Bowring, A.S. Chou, A. Sonnenschein, W. Wester, C. Boutan, N.S. Oblath, R. Bradley, E.J. Daw, A.V. Dixit, J. Clarke, S.R. O'Kelley, N. Crisosto, J.R. Gleason, S. Jois, P. Sikivie, I. Stern, N.S. Sullivan, D.B. Tanner, G.C. Hilton
- **Posted on arXiv:** 16 April 2018
- **Published in Phys.Rev.Lett.:** 09 April 2018

2.5.3.5 Searching for Ultralight Dark Matter with Optical Cavities [97]

- **Authors:** Andrew A. Geraci, Colin Bradley, Dongfeng Gao, Jonathan Weinstein, Andrei Derevianko
- **Posted on arXiv:** 01 August 2018
- **Published in Phys. Rev. Lett. 123, 031304 (2019):** 18 July 2019

2.5.3.6 Squeezed vacuum used to accelerate the search for a weak classical signal [98]

- **Authors:** M. Malnou, D.A. Palken, B.M. Brubaker, Leila R. Vale, Gene C. Hilton, K.W. Lehnert
- **Posted on arXiv:** 17 September 2018
- **Published in Phys.Rev.X:** 04 May 2019

2.5.3.7 Extended Search for the Invisible Axion with the Axion Dark Matter Experiment [99]

- **Authors:** T. Braine, R. Cervantes, N. Crisosto, N. Du, S. Kimes, L.J. Rosenberg, G. Rybka, J. Yang, D. Bowring, A.S. Chou, R. Khatiwada, A. Sonnenschein, W. Wester, G. Carosi, N. Woollett, L.D. Duffy, R. Bradley, C. Boutan, M. Jones, B.H. LaRoque, N.S. Oblath, M.S. Taubman, J. Clarke, A. Dove, A. Eddins, S.R. O'Kelley, S. Nawaz, I. Siddiqi, N. Stevenson, A. Agrawal, A.V. Dixit, J.R. Gleason, S. Jois, P. Sikivie, N.S. Sullivan, D.B. Tanner, E. Lentz, E.J. Daw, J.H. Buckley, P.M. Harrington, E.A. Henriksen, K.W. Murch
- **Posted on arXiv:** 18 October 2019
- **Published in Phys. Rev. Lett. 124, 101303 (2020):** 11 March 2020

2.5.3.8 A quantum-enhanced search for dark matter axions [100]

- **Authors:** K.M. Backes, D.A. Palken, S. Al Kenany, B.M. Brubaker, S.B. Cahn, A. Droster, Gene C. Hilton, Sumita Ghosh, H. Jackson, S.K. Lamoreaux, A.F. Leder, K.W. Lehnert, S.M. Lewis, M. Malnou, R.H. Maruyama, N.M. Rapidis, M. Simanovskaia, Sukhman Singh, D.H. Speller, I. Urdinaran, Leila R. Vale, E.C. van Assendelft, K. van Bibber, H. Wang
- **Posted on arXiv:** 04 August 2020
- **Published in Nature:** 10 February 2021

2.5.3.9 Searching for Dark Matter with a Superconducting Qubit [101]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

2.5.3.10 Cavity Entanglement and State Swapping to Accelerate the Search for Axion Dark Matter [102]

- **Authors:** K. Wurtz, B.M. Brubaker, Y. Jiang, E.P. Ruddy, D.A. Palken, K.W. Lehnert
- **Posted on arXiv:** 08 July 2021
- **Published in PRX Quantum:** 10 December 2021

2.5.3.11 Search for Invisible Axion Dark Matter in the 3.3–4.2 μeV Mass Range [103]

- **Authors:** C. Bartram, T. Braine, E. Burns, R. Cervantes, N. Crisosto, N. Du, H. Korandla, G. Leum, P. Mohapatra, T. Nitta, L.J. Rosenberg, G. Rybka, J. Yang, John Clarke, I. Siddiqi, A. Agrawal, A.V. Dixit, M.H. Awida, A.S. Chou, M. Hollister, S. Knirck, A. Sonnenschein, W. Wester, J.R. Gleason, A.T. Hipp, S. Jois, P. Sikivie, N.S. Sullivan, D.B. Tanner, E. Lentz, R. Khatiwada, G. Carosi, N. Robertson, N. Woollett, L.D. Duffy, C. Boutan, M. Jones, B.H. LaRoque, N.S. Oblath, M.S. Taubman, E.J. Daw, M.G. Perry, J.H. Buckley, C. Gaikwad, J. Hoffman, K.W. Murch, M. Goryachev, B.T. McAllister, A. Quiskamp, C. Thomson, M.E. Tobar
- **Posted on arXiv:** 12 October 2021
- **Published in Phys. Rev. Lett. 127, 261803 (2021):** 23 December 2021

2.5.3.12 Entangled Sensor-Networks for Dark-Matter Searches [104]

- **Authors:** Anthony J. Brady, Christina Gao, Roni Harnik, Zhen Liu, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 10 March 2022
- **Published in PRX Quantum 3, 030333 (2022):** 01 September 2022

2.5.3.13 Detecting Hidden Photon Dark Matter Using the Direct Excitation of Transmon Qubits [105]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 07 December 2022
- **Published in Phys.Rev.Lett.:** 22 November 2023

2.5.3.14 New results from HAYSTAC's phase II operation with a squeezed state receiver [106]

- **Authors:** M.J. Jewell, A.F. Leder, K.M. Backes, Xiran Bai, K. van Bibber, B.M. Brubaker, S.B. Cahn, A. Droster, Maryam H. Esmat, Sumita Ghosh, Eleanor Graham, Gene C. Hilton, H. Jackson, Claire Laffan, S.K. Lamoreaux, K.W. Lehnert, S.M. Lewis, M. Malnou, R.H. Maruyama, D.A. Palken, N.M. Rapidis, E.P. Ruddy, M. Simanovskaia, Sukhman Singh, D.H. Speller, Leila R. Vale, H. Wang, Yuqi Zhu
- **Posted on arXiv:** 23 January 2023
- **Published in Phys.Rev.D:** 01 April 2023

2.5.3.15 Stimulated Emission of Signal Photons from Dark Matter Waves [107]

- **Authors:** Ankur Agrawal, Akash V. Dixit, Tanay Roy, Srivatsan Chakram, Kevin He, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 05 May 2023
- **Published in Phys. Rev. Lett. 132, 140801 (2024):** 03 April 2024

2.5.3.16 Quantum measurements in fundamental physics: a user's manual [108]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023
- **Published in SciPost Phys. Rev. 2 (2026):** 2026

2.5.3.17 Quantum Enhancement in Dark Matter Detection with Quantum Computation [109]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 17 November 2023
- **Published in Phys.Rev.Lett.:** 08 July 2024

2.5.3.18 Quantum-Enhanced Sensing of Axion Dark Matter with a Transmon-Based Single Microwave Photon Counter [110]

- **Authors:** C. Braggio, L. Balembois, R. Di Vora, Z. Wang, J. Travesedo, L. Pallegoix, G. Carugno, A. Ortolan, G. Ruoso, U. Gambardella, D. D'Agostino, P. Bertet, E. Flurin
- **Posted on arXiv:** 04 March 2024
- **Published in Phys.Rev.X:** 01 April 2025

2.5.3.19 Search for QCD axion dark matter with transmon qubits and quantum circuit [111]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 29 July 2024
- **Published in Phys.Rev.D:** 01 December 2024

2.5.3.20 Axion bounds from quantum technology [112]

- **Authors:** Martin Bauer, Sreemanti Chakraborti, Guillaume Rostagni
- **Posted on arXiv:** 12 August 2024
- **Published in JHEP:** 05 May 2025

2.5.3.21 ADMX Axion Dark Matter Bounds around $3.3\,\mu$eV with Dine-Fischler-Srednicki-Zhitnitsky Discovery Ability [113]

- **Authors:** C. Goodman, M. Guzzetti, C. Hanretty, L.J. Rosenberg, G. Rybka, J. Sinnis, D. Zhang, John Clarke, I. Siddiqi, A.S. Chou, M. Hollister, S. Knirck, A. Sonnenschein, T.J. Caligiure, J.R. Gleason, A.T. Hipp, P. Sikivie, M.E. Solano, N.S. Sullivan, D.B. Tanner, R. Khatiwada, G. Carosi, C. Cisneros, N. Du, N. Robertson, N. Woollett, L.D. Duffy, C. Boutan, T. Braine, E. Lentz, N.S. Oblath, M.S. Taubman, E.J. Daw, C. Mostyn, M.G. Perry, C. Bartram, T.A. Dyson, S. Ruppert, M.O. Withers, C.L. Kuo, B.T. McAllister, J.H. Buckley, C. Gaikwad, J. Hoffman, K. Murch, M. Goryachev, E. Hartman, A. Quiskamp, M.E. Tobar
- **Posted on arXiv:** 27 August 2024
- **Published in Phys. Rev. Lett. 134, 111002 (2025):** 21 March 2025

2.5.3.22 Quantum-enhanced dark matter detection with in-cavity control: mitigating the Rayleigh curse [114]

- **Authors:** Haowei Shi, Anthony J. Brady, Wojciech Górecki, Lorenzo Maccone, Roberto Di Candia, Quntao Zhuang
- **Posted on arXiv:** 06 September 2024
- **Published in npj Quantum Information 11, 48 (2025):** 17 March 2025

2.5.3.23 Transmon qubit modeling and characterization for Dark Matter search [115]

- **Authors:** Roberto Moretti, Danilo Labranca, Pietro Campana, Rodolfo Carobene, Marco Gobbo, Manuel A. Castellanos-Beltran, David Olaya, Peter F. Hopkins, Leonardo Banchi, Matteo Borghesi, Alessandro Candido, Stefano Carrazza, Hervè Atsè Corti, Alessandro D'Elia, Marco Faverzani, Elena Ferri, Angelo Nucciotti, Luca Origo, Andrea Pasquale, Alex Stephane Piedjou Komnang, Alessio Rettaroli, Simone Tocci, Claudio Gatti, Andrea Giachero
- **Posted on arXiv:** 09 September 2024
- **Published in IEEE Trans.Quantum Eng.:** 2026

2.5.3.24 Dark Matter Axion Search with HAYSTAC Phase II [116]

- **Authors:** Xiran Bai, M.J. Jewell, J. Echevers, K. van Bibber, A. Droster, Maryam H. Esmat, Sumita Ghosh, Eleanor Graham, H. Jackson, Claire Laffan, S.K. Lamoreaux, A.F. Leder, K.W. Lehnert, S.M. Lewis, R.H. Maruyama, R.D. Nath, N.M. Rapidis, E.P. Ruddy, M. Silva-Feaver, M. Simanovskaia, Sukhman Singh, D.H. Speller, Sabrina Zacarias, Yuqi Zhu
- **Posted on arXiv:** 13 September 2024
- **Published in Phys. Rev. Lett. 134, 151006 (2025):** 18 April 2025

2.5.3.25 Flux-Tunable Cavity for Dark Matter Detection [117]

- **Authors:** Fang Zhao, Ziqian Li, Akash V. Dixit, Tanay Roy, Andrei Vrajitoarea, Riju Banerjee, Alexander Anferov, Kan-Heng Lee, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 12 January 2025
- **Published in Phys.Rev.Lett.:** 13 November 2025

2.5.3.26 Search for Axion Dark Matter from 1.1 to 1.3 GHz with ADMX [118]

- **Authors:** G. Carosi, C. Cisneros, N. Du, S. Durham, N. Robertson, C. Goodman, M. Guzzetti, C. Hanretty, K. Enzian, L.J. Rosenberg, G. Rybka, J. Sinnis, D. Zhang, John Clarke, I. Siddiqi, A.S. Chou, M. Hollister, A. Sonnenschein, S. Knirck, T.J. Caligiure, J.R. Gleason, A.T. Hipp, P. Sikivie, M.E. Solano, N.S. Sullivan, D.B. Tanner, R. Khatiwada, L.D. Duffy, C. Boutan, T. Braine, E. Lentz, N.S. Oblath, M.S. Taubman, E.J. Daw, C. Mostyn, M.G. Perry, C. Bartram, J. Laurel, A. Yi, T.A. Dyson, S. Ruppert, M.O. Withers, C.L. Kuo, B.T. McAllister, J.H. Buckley, C. Gaikwad, J. Hoffman, K. Murch, M. Goryachev, E. Hartman, A. Quiskamp, M.E. Tobar
- **Posted on arXiv:** 09 April 2025
- **Published in Phys. Rev. Lett. 135, 191001 (2025):** 03 November 2025

2.5.3.27 Quantum Enhanced Dark-Matter Search with Entangled Fock States in High-Quality Cavities [119]

- **Authors:** Benjamin Freiman, Xinyuan You, Andy C.Y. Li, Raphael Cervantes, Taeyoon Kim, Anna Grasselino, Roni Harnik, Yao Lu
- **Posted on arXiv:** 30 October 2025

2.5.4 Quantum Sensors - Optomechanical Sensors

2.5.4.1 Sound of Dark Matter: Searching for Light Scalars with Resonant-Mass Detectors [120]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Ken Van Tilburg
- **Posted on arXiv:** 07 August 2015
- **Published in Phys.Rev.Lett.:** 22 January 2016

2.5.4.2 Dark Matter Direct Detection with Accelerometers [121]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

2.5.4.3 Ultralight dark matter detection with mechanical quantum sensors [122]

- **Authors:** Daniel Carney, Anson Hook, Zhen Liu, Jacob M. Taylor, Yue Zhao
- **Posted on arXiv:** 13 August 2019
- **Published in New Journal of Physics, Volume 23, February 2021:** 02 March 2021

2.5.4.4 Searching for scalar dark matter with compact mechanical resonators [123]

- **Authors:** Jack Manley, Russell Stump, Dalziel Wilson, Daniel Grin, Swati Singh
- **Posted on arXiv:** 16 October 2019
- **Published in Phys. Rev. Lett. 124, 151301 (2020):** 17 April 2020

2.5.4.5 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [124]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

2.5.4.6 Searching for vector dark matter with an optomechanical accelerometer [125]

- **Authors:** Jack Manley, Mitul Dey Chowdhury, Daniel Grin, Swati Singh, Dalziel J. Wilson
- **Posted on arXiv:** 09 July 2020
- **Published in Phys. Rev. Lett. 126, 061301 (2021):** 11 February 2021

2.5.4.7 Entanglement-enhanced optomechanical sensor array with application to dark matter searches [126]

- **Authors:** Anthony J. Brady, Xin Chen, Kewen Xiao, Yi Xia, Jack Manley, Mitul Dey Chowdhury, Zhen Liu, Roni Harnik, Dalziel J. Wilson, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 13 October 2022
- **Published in Commun.Phys.:** 01 September 2023

2.5.4.8 Axion detection with optomechanical cavities [127]

- **Authors:** Clara Murgui, Yikun Wang, Kathryn M. Zurek
- **Posted on arXiv:** 15 November 2022
- **Published in Phys.Lett.B:** 27 June 2024

2.5.4.9 Superfluid helium ultralight dark matter detector [128]

- **Authors:** M. Hirschel, V. Vadakkumbatt, N.P. Baker, F.M. Schweizer, J.C. Sankey, S. Singh, J.P. Davis
- **Posted on arXiv:** 14 September 2023
- **Published in Physical Review D 109, 095011 (2024):** 01 May 2024

2.5.4.10 Vector wave dark matter and terrestrial quantum sensors [129]

- **Authors:** Dorian W.P. Amaral, Mudit Jain, Mustafa A. Amin, Christopher Tunnell
- **Posted on arXiv:** 04 March 2024
- **Published in JCAP:** 21 June 2024

2.5.5 Quantum Sensors - Solid State Sensors

2.5.5.1 Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor [130]

- **Authors:** Xing Rong, Mengqi Wang, Jianpei Geng, Xi Qin, Maosen Guo, Man Jiao, Yijin Xie, Pengfei Wang, Pu Huang, Fazhan Shi, Yi-Fu Cai, Chongwen Zou, Jiangfeng Du
- **Posted on arXiv:** 12 June 2017
- **Published in Nature Commun.:** 21 February 2018

2.5.5.2 First Results from ABRACADABRA-10 cm: A Search for Sub-μeV Axion Dark Matter [131]

- **Authors:** Jonathan L. Ouellet, Chiara P. Salemi, Joshua W. Foster, Reyco Henning, Zachary Bogorad, Janet M. Conrad, Joseph A. Formaggio, Yonatan Kahn, Joe Minervini, Alexey Radovinsky, Nicholas L. Rodd, Benjamin R. Safdi, Jesse Thaler, Daniel Winklehner, Lindley Winslow
- **Posted on arXiv:** 29 October 2018
- **Published in Phys. Rev. Lett. 122, 121802 (2019):** 30 March 2019

2.5.5.3 Search for axion-like dark matter with ferromagnets [132]

- **Authors:** Alexander V. Gramolin, Deniz Aybas, Dorian Johnson, Janos Adam, Alexander O. Sushkov
- **Posted on arXiv:** 06 March 2020
- **Published in Nature Physics 17, 79-84 (2021):** 17 August 2020

2.5.5.4 Search for Low-Mass Axion Dark Matter with ABRACADABRA-10 cm [133]

- **Authors:** Chiara P. Salemi, Joshua W. Foster, Jonathan L. Ouellet, Andrew Gavin, Kaliroe M.W. Pappas, Sabrina Cheng, Kate A. Richardson, Reyco Henning, Yonatan Kahn, Rachel Nguyen, Nicholas L. Rodd, Benjamin R. Safdi, Lindley Winslow
- **Posted on arXiv:** 12 February 2021
- **Published in Phys. Rev. Lett. 127, 081801 (2021):** 17 August 2021

2.5.5.5 Axion search with quantum nondemolition detection of magnons [134]

- **Authors:** Tomonori Ikeda, Asuka Ito, Kentaro Miuchi, Jiro Soda, Hisaya Kurashige, Yutaka Shikano
- **Posted on arXiv:** 17 February 2021
- **Published in Phys. Rev. D 105, 102004 (2022):** 15 May 2022

2.5.5.6 Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors [135]

- **Authors:** P.-H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov, V.M. Acosta
- **Posted on arXiv:** 29 December 2021
- **Published in Physical Review Research 4, 023162 (2022):** 31 May 2022

2.5.5.7 Ultimate precision limit of noise sensing and dark matter search [136]

- **Authors:** Haowei Shi, Quntao Zhuang
- **Posted on arXiv:** 29 August 2022
- **Published in npj Quantum Inf.:** 20 March 2023

2.5.5.8 Light dark matter search with nitrogen-vacancy centers in diamonds [137]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 24 February 2023
- **Published in JHEP:** 12 March 2025

2.5.5.9 Axion detection via superfluid ^{3}He ferromagnetic phase and quantum measurement techniques [138]

- **Authors:** So Chigusa, Dan Kondo, Hitoshi Murayama, Risshin Okabe, Hiroyuki Sudo
- **Posted on arXiv:** 17 September 2023
- **Published in JHEP:** 26 September 2024

2.5.5.10 Nuclear spin metrology with nitrogen vacancy center in diamond for axion dark matter detection [139]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 09 July 2024
- **Published in Phys. Rev. D 111 (2025) 075028:** 01 April 2025

2.5.5.11 Listening for new physics with quantum acoustics [140]

- **Authors:** Ryan Linehan, Tanner Trickle, Christopher R. Conner, Sohitri Ghosh, Tongyan Lin, Mukul Sholapurkar, Andrew N. Cleland
- **Posted on arXiv:** 22 October 2024
- **Published in Phys.Rev.D:** 01 December 2025

2.5.5.12 Entanglement-enhanced ac magnetometry in the presence of Markovian noise [141]

- **Authors:** Thanaporn Sichanugrist, Hajime Fukuda, Takeo Moroi, Kazunori Nakayama, So Chigusa, Norikazu Mizuochi, Masashi Hazumi, Yuichiro Matsuzaki
- **Posted on arXiv:** 28 October 2024
- **Published in Phys.Rev.A:** 07 April 2025

2.5.5.13 Probing Sub-eV Dark Photon, Scalar and Axion-like Particle Dark Matters with Transmon Qubits [142]

- **Authors:** Wei Chao, Yu Gao, Ming-Jie Jin, Xiao-Sheng Liu, Xi-Lei Sun
- **Posted on arXiv:** 30 December 2024
- **Published in Phys.Dark Univ. 50 (2025) 102171:** 15 November 2025

2.5.5.14 Piezoelectric bulk acoustic resonators for dark photon detection [143]

- **Authors:** Tanner Trickle
- **Posted on arXiv:** 09 January 2025
- **Published in Phys.Rev.D:** 01 September 2025

2.5.5.15 Background suppression in quantum sensing of dark matter via collective entangled-state projection [144]

- **Authors:** Shion Chen, Hajime Fukuda, Yutaro Iiyama, Yuya Mino, Takeo Moroi, Mikio Nakahara, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 02 October 2025
- **Published in Phys.Rev.D:** 01 March 2026

2.5.5.16 Hybrid-spin decoupling for noise-resilient DC quantum sensing [145]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 20 November 2025

2.5.6 Crosslists

2.5.6.1 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

2.5.6.2 Quantum Sensing for High Energy Physics [12]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

2.5.6.3 Matter-wave Atomic Gradiometer InterferometricSensor (MAGIS-100) at Fermilab [146]

- **Authors:** Jon Coleman
- **Posted on arXiv:** 02 December 2018
- **Published in PoS:** 28 November 2018

2.5.6.4 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar

- **Posted on arXiv:** 13 March 2019

2.5.6.5 ZAIGA: Zhaoshan Long-baseline Atom Interferometer Gravitation Antenna [147]

- **Authors:** Ming-Sheng Zhan, Jin Wang, Wei-Tou Ni, Dong-Feng Gao, Gang Wang, Ling-Xiang He, Run-Bing Li, Lin Zhou, Xi Chen, Jia-Qi Zhong, Biao Tang, Zhan-Wei Yao, Lei Zhu, Zong-Yuan Xiong, Si-Bin Lu, Geng-Hua Yu, Qun-Feng Cheng, Min Liu, Yu-Rong Liang, Peng Xu, Xiao-Dong He, Min Ke, Zheng Tan, Jun Luo

- **Posted on arXiv:** 21 March 2019

- **Published in Int.J.Mod.Phys.D:** 02 July 2019

2.5.6.6 SAGE: A Proposal for a Space Atomic Gravity Explorer [148]

- **Authors:** Guglielmo M. Tino, Angelo Bassi, Giuseppe Bianco, Kai Bongs, Philippe Bouyer, Luigi Cacciapuoti, Salvatore Capozziello, Xuzong Chen, Maria L. Chiofalo, Andrei Derevianko, Wolfgang Ertmer, Naceur Gaaloul, Patrick Gill, Peter W. Graham, Jason M. Hogan, Luciano Iess, Mark A. Kasevich, Hidetoshi Katori, Carsten Klempt, Xuanhui Lu, Long-Sheng Ma, Holger Müller, Nathan R. Newbury, Chris W. Oates, Achim Peters, Nicola Poli, Ernst M. Rasel, Gabriele Rosi, Albert Roura, Christophe Salomon, Stephan Schiller, Wolfgang Schleich, Dennis Schlippert, Florian Schreck, Christian Schubert, Fiodor Sorrentino, Uwe Sterr, Jan W. Thomsen, Giuseppe Vallone, Flavio Vetrano, Paolo Villoresi, Wolf von Klitzing, David Wilkowski, Peter Wolf, Jun Ye, Nan Yu, Mingsheng Zhan

- **Posted on arXiv:** 08 July 2019

- **Published in Eur. Phys. J. D 73, 228 (2019):** 15 November 2019

2.5.6.7 AION: An Atom Interferometer Observatory and Network [149]

- **Authors:** L. Badurina, E. Bentine, D. Blas, K. Bongs, D. Bortoletto, T. Bowcock, K. Bridges, W. Bowden, O. Buchmueller, C. Burrage, J. Coleman, G. Elertas, J. Ellis, C. Foot, V. Gibson, M.G. Haehnelt, T. Harte, S. Hedges, R. Hobson, M. Holynski, T. Jones,

M. Langlois, S. Lellouch, M. Lewicki, R. Maiolino, P. Majewski, S. Malik, J. March-Russell, C. McCabe, D. Newbold, B. Sauer, U. Schneider, I. Shipsey, Y. Singh, M.A. Uchida, T. Valenzuela, M. van der Grinten, V. Vaskonen, J. Vossebeld, D. Weatherill, I. Wilmut

- **Posted on arXiv:** 26 November 2019
- **Published in JCAP:** 06 May 2020

2.5.6.8 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [150]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

2.5.6.9 Mechanical quantum sensing in the search for dark matter [15]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.5.6.10 Matter-wave Atomic Gradiometer Interferometric Sensor (MAGIS-100) [151]

- **Authors:** Mahiro Abe, Philip Adamson, Marcel Borcean, Daniela Bortoletto, Kieran Bridges, Samuel P. Carman, Swapan Chattopadhyay, Jonathon Coleman, Noah M. Curfman, Kenneth DeRose, Tejas Deshpande, Savas Dimopoulos, Christopher J. Foot, Josef C. Frisch, Benjamin E. Garber, Steve Geer, Valerie Gibson, Jonah Glick, Peter W. Graham, Steve R. Hahn, Roni Harnik, Leonie Hawkins, Sam Hindley, Jason M. Hogan, Yijun Jiang (姜一君), Mark A. Kasevich, Ronald J. Kellett, Mandy Kiburg, Tim Kovachy, Joseph D. Lykken, John March-Russell, Jeremiah Mitchell, Martin Murphy, Megan Nantel, Lucy E. Nobrega, Robert K. Plunkett, Surjeet Rajendran, Jan Rudolph, Natasha Sachdeva, Murtaza Safdari, James K. Santucci, Ariel G. Schwartzman, Ian Shipsey, Hunter Swan, Linda R. Valerio, Arvydas Vasonis, Yiping Wang, Thomas Wilkason
- **Posted on arXiv:** 06 April 2021
- **Published in Quantum Sci. Technol. 6, 044003 (2021):** 16 July 2021

2.5.6.11 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [19]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.5.6.12 New Horizons: Scalar and Vector Ultralight Dark Matter [23]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou
- **Posted on arXiv:** 28 March 2022

2.5.6.13 Quantum Networks for High Energy Physics [24]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

2.5.6.14 Electromagnetic cavities as mechanical bars for gravitational waves [152]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D'Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

2.5.6.15 Quantum sensing for particle physics [153]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.5.6.16 Quantum Sensors for High Energy Physics [30]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.5.6.17 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [154]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis
- **Posted on arXiv:** 07 August 2024
- **Published in Phys.Rev.D:** 01 March 2025

2.5.6.18 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [155]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

2.5.6.19 Superradiant interactions for relic detection with entangled nuclear spins [156]

- **Authors:** Marios Galanis, Onur Hosten, Asimina Arvanitaki, Savas Dimopoulos
- **Posted on arXiv:** 28 August 2025
- **Published in Phys.Rev.A:** 10 June 2026

2.5.6.20 A prototype differential atom interferometer for fundamental physics [157]

- **Authors:** C.F.A. Baynham, R. Hobson, O. Buchmüller, D. Evans, L. Hawkins, L. Iannizzotto Venezze, A. Josset, D. Lee, E. Pasatembou, B.E. Sauer, M.R. Tarbutt, T. Walker, O. Ennis, U. Chauhan, A. Brzakalik, S. Dey, S. Hedges, B. Stray, M. Langlois, K. Bongs, T. Hird, S. Lellouch, M. Holynski, B. Bostwick f, J. Chen, Z. Eyler, V. Gibson, T.L. Harte, C.C. Hsu, M. Karzazi, C. Lu, B. Millward, J. Mitchell, N. Mouelle, B. Panchumarthi g, J. Scheper, U. Schneider, X. Su, X. Suh, Y. Tang, K. Tkalčec, M. Zeuneri, S. Zhang, Y. Zhij, Y. Zhi, L. Badurina, A. Beniwal l, D. Blas, J. Carlton, J. Ellis, C. McCabe, G. Parish, D. Pathak Govardhan, V. Vaskonenn, T. Bowcock, K. Bridges, A. Carroll, J. Coleman, G. Elertas, S. Hindley, C. Metelko, H. Throssell, J.N. Tinsley, E. Bentine, M. Booth, D. Bortoletto, N. Callaghan, C. Foot, C. Gómez-Monedero, K. Hughes, A. James, T. Leese, A. Lowe, J. March-Russell, J. Sander, J. Schelfhout, I. Shipsey, D. Weatherill, D. Wood, M.G. Bason, K. Hussain, H. Labiad, A.L. Marchant, T.C. Thornton, T. Valenzuela, S.N. Balashov, P. Majewski, D. Newbold, M.G.D. van der Grinten, Z. Pan, Z. Tam, I. Wilmut, K. Clarke, A. Vick
- **Posted on arXiv:** 12 April 2025
- **Published in Nature:** 17 June 2026

2.6 Detector Technologies and Simulations

2.6.1 Quantum Algorithms - Quantum Walks

2.6.1.1 Quantum walk based Monte Carlo simulation for photon interaction cross sections [158]

- **Authors:** Euimin Lee, Sangmin Lee, Shiho Kim
- **Posted on arXiv:** 20 February 2025
- **Published in Phys.Rev.D:** 01 June 2025

2.6.2 Quantum Computing Paradigms - Continuous Variable Quantum Computing

2.6.2.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [159]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

2.6.3 Quantum Computing Paradigms - Quantum Annealing

2.6.3.1 Unfolding measurement distributions via quantum annealing [160]

- **Authors:** Kyle Cormier, Riccardo Di Sipio, Peter Wittek
- **Posted on arXiv:** 22 August 2019
- **Published in JHEP:** 22 November 2019

2.6.4 Quantum Machine Learning - Unsupervised Methods

2.6.4.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [161]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

2.6.4.2 Running the Dual-PQC GAN on noisy simulators and real quantum hardware [162]

- **Authors:** Su Yeon Chang, Edwin Agnew, Elías F. Combarro, Michele Grossi, Steven Herbert, Sofia Vallecorsa
- **Posted on arXiv:** 30 May 2022
- **Published in nan:** 2023

2.6.4.3 Precise image generation on current noisy quantum computing devices [163]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borras, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

2.6.5 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.6.5.1 Quantum Systems for Enhanced High Energy Particle Physics Detectors [164]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

2.6.5.2 Quantum sensing for particle physics [153]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.6.6 Quantum Sensors - Solid State Sensors

2.6.6.1 Impact of ionizing radiation on superconducting qubit coherence [165]

- **Authors:** Antti P. Vepsäläinen, Amir H. Karamlou, John L. Orrell, Akshunna S. Dogra, Ben Loer, Francisca Vasconcelos, David K. Kim, Alexander J. Melville, Bethany M. Niedzielski, Jonilyn L. Yoder, Simon Gustavsson, Joseph A. Formaggio, Brent A. VanDevender, William D. Oliver
- **Posted on arXiv:** 24 January 2020
- **Published in Nature:** 26 August 2020

2.6.6.2 Correlated charge noise and relaxation errors in superconducting qubits [166]

- **Authors:** C.D. Wilen, S. Abdullah, N.A. Kurinsky, C. Stanford, L. Cardani, G. D'Imperio, C. Tomei, L. Faoro, L.B. Ioffe, C.H. Liu, A. Opremcak, B.G. Christensen, J.L. DuBois, R. McDermott
- **Posted on arXiv:** 10 December 2020
- **Published in Nature 594, 369–373 (2021):** 16 June 2021

2.6.6.3 Reducing the impact of radioactivity on quantum circuits in a deep-underground facility [167]

- **Authors:** Laura Cardani, Francesco Valenti, Nicola Casali, Gianluigi Catelani, Thibault Charpentier, Massimiliano Clemenza, Ivan Colantoni, Angelo Cruciani, G. D'Imperio, Luca Gironi, Lukas Grünhaupt, Daria Gusenkova, Fabio Henriques, Marc Lagoin, Maria Martinez, Giorgio Pettinari, Claudia Rusconi, Oliver Sander, C. Tomei, Alexey V. Ustinov, Marc Weber, Wolfgang Wernsdorfer, Marco Vignati, Stefano Pirro, Ioan M. Pop
- **Posted on arXiv:** 05 May 2020
- **Published in Nature Commun.:** 12 May 2021

2.6.6.4 Synchronous detection of cosmic rays and correlated errors in superconducting qubit arrays [168]

- **Authors:** Patrick M. Harrington, Mingyu Li, Max Hays, Wouter Van De Pontseele, Daniel Mayer, H. Douglas Pinckney, Felipe Contipelli, Michael Gingras, Bethany M. Niedzielski, Hannah Stickler, Jonilyn L. Yoder, Mollie E. Schwartz, Jeffrey A. Grover, Kyle Serniak, William D. Oliver, Joseph A. Formaggio
- **Posted on arXiv:** 05 February 2024
- **Published in Nature Commun.:** 11 July 2025

2.6.6.5 Cosmic-ray-induced correlated errors in superconducting qubit array [169]

- **Authors:** Xue-Gang Li, Jun-Hua Wang, Yao-Yao Jiang, Guang-Ming Xue, Xiao-Xia Cai, Jun Zhou, Ming Gong, Zhao-Feng Liu, Shuang-Yu Zheng, Deng-Ke Ma, Mo Chen, Wei-Jie Sun, Shuang Yang, Fei Yan, Yi-Rong Jin, Xue-Feng Ding, Hai-Feng Yu
- **Posted on arXiv:** 06 February 2024
- **Published in Nature Commun.:** 20 May 2025

2.6.7 Crosslists

2.6.7.1 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

2.6.7.2 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.6.7.3 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao
- **Posted on arXiv:** 28 February 2023

2.6.7.4 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.6.7.5 Quantum-centric Supercomputing for Physics Research [170]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.7 Effective Field Theories

2.7.1 Quantum Algorithms - Quantum Simulations

2.7.1.1 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [171]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

2.7.1.2 Quantum Algorithms for Simulating Nuclear Effective Field Theories [172]

- **Authors:** James D. Watson, Jacob Bringewatt, Alexander F. Shaw, Andrew M. Childs, Alexey V. Gorshkov, Zohreh Davoudi
- **Posted on arXiv:** 08 December 2023

2.7.2 Quantum Entanglement and Bell Inequalities

2.7.2.1 Geometric quantum information structure in quantum fields and their lattice simulation [173]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 August 2020
- **Published in Phys. Rev. D 103, 065007 (2021):** 23 March 2021

2.7.2.2 Entanglement Spheres and a UV-IR Connection in Effective Field Theories [174]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 27 March 2021
- **Published in Phys.Rev.Lett.:** 16 November 2021

2.7.2.3 Entanglement structures in quantum field theories: Negativity cores and bound entanglement in the vacuum [175]

- **Authors:** Natalie Klco, D.H. Beck, Martin J. Savage
- **Posted on arXiv:** 20 October 2021
- **Published in Phys.Rev.A:** 11 January 2023

2.7.2.4 Minimal entanglement and emergent symmetries in low-energy QCD [176]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

2.7.2.5 Entanglement in flavored scalar scattering [177]

- **Authors:** Kamila Kowalska, Enrico Maria Sessolo
- **Posted on arXiv:** 21 April 2024
- **Published in JHEP:** 18 July 2024

2.7.3 Crosslists

2.7.3.1 Cloud Quantum Computing of an Atomic Nucleus [178]

- **Authors:** E.F. Dumitrescu, A.J. McCaskey, G. Hagen, G.R. Jansen, T.D. Morris, T. Papenbrock, R.C. Pooser, D.J. Dean, P. Lougovski
- **Posted on arXiv:** 11 January 2018
- **Published in Phys. Rev. Lett. 120, 210501 (2018):** 24 May 2018

2.7.3.2 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

2.7.3.3 Quantum Computing for Neutrino-Nucleus Scattering [179]

- **Authors:** Alessandro Roggero, Andy C.Y. Li, Joseph Carlson, Rajan Gupta, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019
- **Published in Phys. Rev. D 101, 074038 (2020):** 28 April 2020

2.7.3.4 Improving Hamiltonian encodings with the Gray code [180]

- **Authors:** Olivia Di Matteo, Anna McCoy, Peter Gysbers, Takayuki Miyagi, R.M. Woloshyn, Petr Navrátil
- **Posted on arXiv:** 11 August 2020
- **Published in Phys. Rev. A 103, 042405 (2021):** 03 April 2021

2.7.3.5 Standard model physics and the digital quantum revolution: thoughts about the interface [16]

- **Authors:** Natalie Klco, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 10 July 2021
- **Published in Rept.Prog.Phys.:** 19 May 2022

2.7.3.6 Quantum SMEFT tomography: Top quark pair production at the LHC [181]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.7.3.7 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [18]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.7.3.8 Quantum Simulation for High-Energy Physics [25]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.7.3.9 Subatomic Many-Body Physics Simulations on a Quantum Frequency Processor [182]

- **Authors:** Hsuan-Hao Lu, Natalie Klco, Joseph M. Lukens, Titus D. Morris, Aaina Bansal, Andreas Ekström, Gaute Hagen, Thomas Papenbrock, Andrew M. Weiner, Martin J. Savage, Pavel Lougovski
- **Published in nan:** 2019

2.8 Event Classification

2.8.1 Quantum Algorithms - Grover's Search Algorithm

2.8.1.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [183]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

2.8.2 Quantum Computing Paradigms - Continuous Variable Quantum Computing

2.8.2.1 Continuous-variable photonic quantum extreme learning machines for fast collider-data selection [184]

- **Authors:** Benedikt Maier, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 October 2025
- **Published in EPJ Quant.Technol.:** 27 April 2026

2.8.3 Quantum Computing Paradigms - Quantum Annealing

2.8.3.1 Solving a Higgs optimization problem with quantum annealing for machine learning [185]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

2.8.3.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [186]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

2.8.3.3 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [187]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

2.8.3.4 Quantum algorithm for the classification of supersymmetric top quark events [46]

- **Authors:** Pedrame Bargassa, Timothée Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothée Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.8.3.5 Completely quantum neural networks [188]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

2.8.4 Quantum Machine Learning - Supervised Methods

2.8.4.1 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [189]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevior
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

2.8.4.2 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [190]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

2.8.4.3 Higgs analysis with quantum classifiers [191]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in nan:** 2021

2.8.4.4 Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets [192]

- **Authors:** Zhelun Li, Lento Nagano, Koji Terashi
- **Posted on arXiv:** 17 May 2024
- **Published in Phys.Rev.Res.:** 10 October 2024

2.8.4.5 From Reachability to Learnability: Geometric Design Principles for Quantum Neural Networks [193]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky
- **Posted on arXiv:** 03 March 2026

2.8.5 Quantum Machine Learning - Unsupervised Methods

2.8.5.1 Guided graph compression for quantum graph neural networks [194]

- **Authors:** Mikel Casals, Vasilis Belis, Elias F. Combarro, Eduard Alarcón, Sofia Vallecorsa, Michele Grossi
- **Posted on arXiv:** 11 June 2025
- **Published in Mach.Learn.Sci.Tech.:** 12 September 2025

2.8.6 Quantum Machine Learning - Variational Quantum Algorithms

2.8.6.1 Event Classification with Quantum Machine Learning in High-Energy Physics [43]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.8.6.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [44]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.8.6.3 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [195]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

2.8.6.4 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [196]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

2.8.6.5 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [197]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

2.8.6.6 Quantum Metric Learning for New Physics Searches at the LHC [47]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

2.8.6.7 Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics [198]

- **Authors:** Eyup B. Unlu, Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva
- **Posted on arXiv:** 01 February 2024
- **Published in Axioms v. 13, no 3, (2024) 187:** 13 March 2024

2.8.6.8 Quantum Vision Transformers for Quark–Gluon Classification [199]

- **Authors:** Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva, Eyup B. Unlu
- **Posted on arXiv:** 16 May 2024
- **Published in Axioms 2024, 13(5), 323:** 13 May 2024

2.8.7 Crosslists

2.8.7.1 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.8.7.2 Unsupervised event classification with graphs on classical and photonic quantum computers [31]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.8.7.3 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.8.7.4 Quantum Computing Applications in Future Colliders [5]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.8.7.5 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.8.7.6 Quantum-centric Supercomputing for Physics Research [170]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.9 Event Generation

2.9.1 Quantum Algorithms - Grover's Search Algorithm

2.9.1.1 Quantum algorithm for Feynman loop integrals [200]

- **Authors:** Selomit Ramírez-Uribe, Andrés E. Rentería-Olivo, Germán Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

2.9.1.2 Loop Feynman integration on a quantum computer [201]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

2.9.2 Quantum Algorithms - Quantum Simulations

2.9.2.1 Quantum Algorithm for High Energy Physics Simulations [202]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

2.9.2.2 Towards a quantum computing algorithm for helicity amplitudes and parton showers [203]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

2.9.2.3 Improving quantum simulation efficiency of final state radiation with dynamic quantum circuits [204]

- **Authors:** Plato Deliyannis, James Sud, Diana Chamaki, Zoë Webb-Mack, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 18 March 2022
- **Published in Physical Review D 106, 036007 (2022):** 01 August 2022

2.9.2.4 Collider events on a quantum computer [205]

- **Authors:** Gösta Gustafson, Stefan Prestel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 21 July 2022
- **Published in JHEP:** 07 November 2022

2.9.2.5 Quantum parton shower with kinematics [206]

- **Authors:** Christian W. Bauer, So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 30 October 2023
- **Published in Phys.Rev.A:** 26 March 2024

2.9.3 Quantum Algorithms - Quantum Walks

2.9.3.1 Quantum walk approach to simulating parton showers [207]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

2.9.4 Quantum Machine Learning - Unsupervised Methods

2.9.4.1 Style-based quantum generative adversarial networks for Monte Carlo events [208]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

2.9.4.2 Quantum integration of elementary particle processes [209]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

2.9.4.3 Generative invertible quantum neural networks [210]

- **Authors:** Armand Rousselot, Michael Spannowsky
- **Posted on arXiv:** 24 February 2023
- **Published in SciPost Phys. 16, 146 (2024):** 04 June 2024

2.9.4.4 Quantum-centric Supercomputing for Physics Research [170]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.9.5 Quantum Machine Learning - Variational Quantum Algorithms

2.9.5.1 Partonic collinear structure by quantum computing [211]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.9.5.2 Unsupervised quantum circuit learning in high energy physics [212]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.9.5.3 Quantum Wishlist: Lessons from Parton Showers [213]

- **Authors:** Masahito Yamazaki
- **Posted on arXiv:** 25 February 2025
- **Published in nan:** 24 October 2025

2.9.6 Crosslists

2.9.6.1 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.9.6.2 Quantum Simulation for High-Energy Physics [25]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.9.6.3 Quantum Computing Applications in Future Colliders [5]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.9.6.4 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [27]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

2.9.6.5 Quantum data learning for quantum simulations in high-energy physics [214]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.9.6.6 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.9.6.7 Quantum Algorithms in Particle Physics [215]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.10 Experimental Infrastructure - Detector Technologies

2.11 Experimental Infrastructure - Accelerator Physics

2.12 Gravitational Waves

2.12.1 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.12.1.1 Exploring gravity with the MIGA large scale atom interferometer [216]

- **Authors:** B. Canuel, A. Bertoldi, L. Amand, E. Pozzo di Borgo, B. Fang, R. Geiger, J. Gillot, S. Henry, J. Hinderer, D. Holleville, G. Lefèvre, M. Merzougui, N. Mielec, T. Monfret, S. Pelisson, M. Prevedelli, S. Reynaud, I. Riou, Y. Rogister, S. Rosat, E. Cormier, A. Landragin, W. Chaibi, S. Gaffet, P. Bouyer

- **Posted on arXiv:** 07 March 2017
- **Published in Scientific Reports (2018) 8:14064:** 14 September 2018

2.12.1.2 Mid-band gravitational wave detection with precision atomic sensors [217]

- **Authors:** Peter W. Graham, Jason M. Hogan, Mark A. Kasevich, Surjeet Rajendran, Roger W. Romani
- **Posted on arXiv:** 06 November 2017

2.12.1.3 Matter-wave Atomic Gradiometer InterferometricSensor (MAGIS-100) at Fermilab [146]

- **Authors:** Jon Coleman
- **Posted on arXiv:** 02 December 2018
- **Published in PoS:** 28 November 2018

2.12.1.4 ZAIGA: Zhaoshan Long-baseline Atom Interferometer Gravitation Antenna [147]

- **Authors:** Ming-Sheng Zhan, Jin Wang, Wei-Tou Ni, Dong-Feng Gao, Gang Wang, Ling-Xiang He, Run-Bing Li, Lin Zhou, Xi Chen, Jia-Qi Zhong, Biao Tang, Zhan-Wei Yao, Lei Zhu, Zong-Yuan Xiong, Si-Bin Lu, Geng-Hua Yu, Qun-Feng Cheng, Min Liu, Yu-Rong Liang, Peng Xu, Xiao-Dong He, Min Ke, Zheng Tan, Jun Luo
- **Posted on arXiv:** 21 March 2019
- **Published in Int.J.Mod.Phys.D:** 02 July 2019

2.12.1.5 SAGE: A Proposal for a Space Atomic Gravity Explorer [148]

- **Authors:** Guglielmo M. Tino, Angelo Bassi, Giuseppe Bianco, Kai Bongs, Philippe Bouyer, Luigi Cacciapuoti, Salvatore Capozziello, Xuzong Chen, Maria L. Chiofalo, Andrei Derevianko, Wolfgang Ertmer, Naceur Gaaloul, Patrick Gill, Peter W. Graham, Jason M. Hogan, Luciano Iess, Mark A. Kasevich, Hidetoshi Katori, Carsten Klempt, Xuanhui Lu, Long-Sheng Ma, Holger Müller, Nathan R. Newbury, Chris W. Oates, Achim Peters, Nicola Poli, Ernst M. Rasel, Gabriele Rosi, Albert Roura, Christophe Salomon, Stephan Schiller, Wolfgang Schleich, Dennis Schlippert, Florian Schreck, Christian Schubert, Fiodor Sorrentino, Uwe Sterr, Jan W. Thomsen, Giuseppe Vallone, Flavio Vetrano, Paolo Villoresi, Wolf von Klitzing, David Wilkowski, Peter Wolf, Jun Ye, Nan Yu, Mingsheng Zhan
- **Posted on arXiv:** 08 July 2019
- **Published in Eur. Phys. J. D 73, 228 (2019):** 15 November 2019

2.12.1.6 ELGAR—a European Laboratory for Gravitation and Atom-interferometric Research [218]

- **Authors:** B. Canuel, S. Abend, P. Amaro-Seoane, F Badaracco, Q. Beaufils, A. Bertoldi, K. Bongs, P. Bouyer, C. Braxmaier, W. Chaibi, N. Christensen, F. Fitzek, G. Flouris, N. Gaaloul, S. Gaffet, C.L. Garrido Alzar, R. Geiger, S. Guellati-Khelifa, K. Hammerer, J. Harms, J. Hinderer, J. Junca, S. Katsanevas, C. Klempt, C. Kozanitis, M. Krutzik, A. Landragin, I. Làzaro Roche, B. Leykauf, Y-H. Lien, S. Loriani, S. Merlet, M. Merzougui, M. Nofrarias, P. Papadakos, F. Pereira, A. Peters, D. Plexousakis, M. Prevedelli, E. Rasel, Y. Rogister, S. Rosat, A. Roura, D.O. Sabulsky, V. Schkolnik, D. Schlippert, C. Schubert, L. Sidorenkov, J.-N. Siemß, C.F. Sopuerta, F. Sorrentino, C. Struckmann, G.M. Tino, G. Tsagkatakis, A. Viceré, W. von Klitzing, L. Woerner, X. Zou
- **Posted on arXiv:** 09 November 2019
- **Published in Class.Quant.Grav.:** 31 October 2020

2.12.1.7 AION: An Atom Interferometer Observatory and Network [149]

- **Authors:** L. Badurina, E. Bentine, D. Blas, K. Bongs, D. Bortoletto, T. Bowcock, K. Bridges, W. Bowden, O. Buchmueller, C. Burrage, J. Coleman, G. Elertas, J. Ellis, C. Foot, V. Gibson, M.G. Haehnelt, T. Harte, S. Hedges, R. Hobson, M. Holynski, T. Jones, M. Langlois, S. Lellouch, M. Lewicki, R. Maiolino, P. Majewski, S. Malik, J. March-Russell, C. McCabe, D. Newbold, B. Sauer, U. Schneider, I. Shipsey, Y. Singh, M.A. Uchida, T. Valenzuela, M. van der Grinten, V. Vaskonen, J. Vossebeld, D. Weatherill, I. Wilmut
- **Posted on arXiv:** 26 November 2019
- **Published in JCAP:** 06 May 2020

2.12.1.8 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [150]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

2.12.1.9 Matter-wave Atomic Gradiometer Interferometric Sensor (MAGIS-100) [151]

- **Authors:** Mahiro Abe, Philip Adamson, Marcel Borcean, Daniela Bortoletto, Kieran Bridges, Samuel P. Carman, Swapan Chattopadhyay, Jonathon Coleman, Noah M.

Curfman, Kenneth DeRose, Tejas Deshpande, Savas Dimopoulos, Christopher J. Foot, Josef C. Frisch, Benjamin E. Garber, Steve Geer, Valerie Gibson, Jonah Glick, Peter W. Graham, Steve R. Hahn, Roni Harnik, Leonie Hawkins, Sam Hindley, Jason M. Hogan, Yijun Jiang (姜一君), Mark A. Kasevich, Ronald J. Kellett, Mandy Kiburg, Tim Kovachy, Joseph D. Lykken, John March-Russell, Jeremiah Mitchell, Martin Murphy, Megan Nantel, Lucy E. Nobrega, Robert K. Plunkett, Surjeet Rajendran, Jan Rudolph, Natasha Sachdeva, Murtaza Safdari, James K. Santucci, Ariel G. Schwartzman, Ian Shipsey, Hunter Swan, Linda R. Valerio, Arvydas Vasonis, Yiping Wang, Thomas Wilkason

- **Posted on arXiv:** 06 April 2021
- **Published in Quantum Sci. Technol. 6, 044003 (2021):** 16 July 2021

2.12.1.10 Prospective sensitivities of atom interferometers to gravitational waves and ultralight dark matter [219]

- **Authors:** Leonardo Badurina, Oliver Buchmueller, John Ellis, Marek Lewicki, Christopher McCabe, Ville Vaskonen
- **Posted on arXiv:** 05 August 2021
- **Published in Phil. Trans. R. Soc. A.380 20210060, 2022:** 07 February 2022

2.12.1.11 Letter of Intent: AICE – 100m Atom Interferometer Experiment at CERN [220]

- **Authors:** Charles Baynham, Andrea Bertoldi, Diego Blas, Oliver Buchmueller, Sergio Calatroni, Vassilis Charmandaris, Maria Luisa Marilu Chiofalo, Pierre Cladé, Jonathon Coleman, Fabio Di Pumpo, John Ellis, Naceur Gaaloul, Saïda Guellati-Khelifa, Tiffany Harte, Richard Hobson, Michael Holynski, Samuel Lellouch, Lucas Lombriser, Elias Lopez Asamar, Michele Maggiore, Christopher McCabe, Jeremiah Mitchell, Ernst M. Rasel, Federico Sanchez Nieto, Wolfgang Schleich, Dennis Schlippert, Ulrich Schneider, Steven Schramm, Marcelle Soares-Santos, Guglielmo M. Tino, Jonathan N. Tinsley, Tristan Valenzuela, Maurits van der Grinten, Wolf von Klitzing
- **Posted on arXiv:** 15 September 2025

2.12.1.12 A prototype differential atom interferometer for fundamental physics [157]

- **Authors:** C.F.A. Baynham, R. Hobson, O. Buchmüller, D. Evans, L. Hawkins, L. Iannizzotto Venezze, A. Josset, D. Lee, E. Pasatembou, B.E. Sauer, M.R. Tarbutt, T. Walker, O. Ennis, U. Chauhan, A. Brzakalik, S. Dey, S. Hedges, B. Stray, M. Langlois, K. Bongs, T. Hird, S. Lellouch, M. Holynski, B. Bostwick f, J. Chen, Z. Eyler, V. Gibson, T.L. Harte, C.C. Hsu, M. Karzazi, C. Lu, B. Millward, J. Mitchell, N. Mouelle,

B. Panchumarthi g, J. Scheper, U. Schneider, X. Su, X. Suh, Y. Tang, K. Tkalčec, M. Zeuneri, S. Zhang, Y. Zhij, Y. Zhi, L. Badurina, A. Beniwal l, D. Blas, J. Carlton, J. Ellis, C. McCabe, G. Parish, D. Pathak Govardhan, V. Vaskonenn, T. Bowcock, K. Bridges, A. Carroll, J. Coleman, G. Elertas, S. Hindley, C. Metelko, H. Throssell, J.N. Tinsley, E. Bentine, M. Booth, D. Bortoletto, N. Callaghan, C. Foot, C. Gómez-Monedero, K. Hughes, A. James, T. Leese, A. Lowe, J. March-Russell, J. Sander, J. Schelfhout, I. Shipsey, D. Weatherill, D. Wood, M.G. Bason, K. Hussain, H. Labiad, A.L. Marchant, T.C. Thornton, T. Valenzuela, S.N. Balashov, P. Majewski, D. Newbold, M.G.D. van der Grinten, Z. Pan, Z. Tam, I. Wilmut, K. Clarke, A. Vick

- **Posted on arXiv:** 12 April 2025
- **Published in Nature:** 17 June 2026

2.12.2 Quantum Sensors - Optical/Photonic Sensors

2.12.2.1 Electromagnetic cavities as mechanical bars for gravitational waves [152]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D'Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

2.12.2.2 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [155]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

2.12.3 Quantum Sensors - Optomechanical Sensors

2.12.3.1 Quantum interactions between a laser interferometer and gravitational waves [221]

- **Authors:** Belinda Pang, Yanbei Chen
- **Posted on arXiv:** 28 August 2018
- **Published in Phys. Rev. D 98, 124006 (2018):** 11 December 2018

2.12.3.2 Superconducting Levitated Detector of Gravitational Waves [222]

- **Authors:** Daniel Carney, Gerard Higgins, Giacomo Marocco, Michael Wentzel
- **Posted on arXiv:** 02 August 2024
- **Published in Carney (2025) Phys. Rev. Lett. 134, 181402:** 06 May 2025

2.12.4 Crosslists

2.12.4.1 AEDGE: Atomic Experiment for Dark Matter and Gravity Exploration in Space [14]

- **Authors:** Yousef Abou El-Neaj, Cristiano Alpigiani, Sana Amairi-Pyka, Henrique Araújo, Antun Balaž, Angelo Bassi, Lars Bathe-Peters, Baptiste Battelier, Aleksandar Belić, Elliot Bentine, José Bernabeu, Andrea Bertoldi, Robert Bingham, Diego Blas, Vasiliki Bolpasi, Kai Bongs, Sougato Bose, Philippe Bouyer, Themis Bowcock, William Bowden, Oliver Buchmueller, Clare Burrage, Xavier Calmet, Benjamin Canuel, Laurentiu-Ioan Caramete, Andrew Carroll, Giancarlo Cella, Vassilis Charmandaris, Swapan Chattopadhyay, Xuzong Chen, Maria Luisa Chiofalo, Jonathon Coleman, Joseph Cotter, Yanou Cui, Andrei Derevianko, Albert De Roeck, Goran S. Djordjevic, Peter Dornan, Michael Doser, Ioannis Drougkakis, Jacob Dunningham, Ioana Dutan, Sajan Easo, Gedminas Elertas, John Ellis, Mai El Sawy, Farida Fassi, Daniel Felea, Chen-Hao Feng, Robert Flack, Chris Foot, Ivette Fuentes, Naceur Gaaloul, Alexandre Gauguet, Remi Geiger, Valerie Gibson, Gian Giudice, Jon Goldwin, Oleg Grachov, Peter W. Graham, Dario Grasso, Maurits van der Grinten, Mustafa Gündogan, Martin G. Haehnelt, Tiffany Harte, Aurelien Hees, Richard Hobson, Jason Hogan, Bodil Holst, Michael Holynski, Mark Kasevich, Bradley J. Kavanagh, Wolf von Klitzing, Tim Kovachy, Benjamin Krikler, Markus Krutzik, Marek Lewicki, Yu-Hung Lien, Miaoyuan Liu, Giuseppe Gaetano Luciano, Alain Magnon, Mohammed Attia Mahmoud, Sarah Malik, Christopher McCabe, Jeremiah Mitchell, Julia Pahl, Debapriya Pal, Saurabh Pandey, Dimitris Papazoglou, Mauro Paternostro, Bjoern Penning, Achim Peters, Marco Prevedelli, Vishnupriya Puthiya-Veettil, John Quenby, Ernst Rasel, Sean Ravenhall, Jack Ringwood, Albert Roura, Dylan Sabulsky, Muhammed Sameed, Ben Sauer, Stefan Alaric Schäffer, Stephan Schiller, Vladimir Schkolnik, Dennis Schlippert, Christian Schubert, Haifa Rejeb Sfar, Armin Shayeghi, Ian Shipsey, Carla Signorini, Yeshpal Singh, Marcelle Soares-Santos, Fiodor Sorrentino, Timothy Sumner, Konstantinos Tassis, Silvia Tentindo, Guglielmo Maria Tino, Jonathan N. Tinsley, James Unwin, Tristan Valenzuela, Georgios Vasilakis, Ville Vaskonen, Christian Vogt, Alex Webber-Date, Andŕ e} Wenzlawski, Patrick Windpassinger, Marian Woltmann, Efe Yazgan, Ming-Sheng Zhan, Xinhao Zou, Jure Zupan
- **Posted on arXiv:** 02 August 2019
- **Published in EPJ Quant.Technol.:** 04 March 2020

2.12.4.2 Quantum Sensors with Matter Waves for GW Observation [4]

- **Authors:** Andrea Bertoldi, Philippe Bouyer, Benjamin Canuel
- **Posted on arXiv:** 27 October 2020
- **Published in nan:** 2022

2.12.4.3 Snowmass2021 Cosmic Frontier White Paper: Future Gravitational-Wave Detector Facilities [21]

- **Authors:** Stefan W. Ballmer, Rana Adhikari, Leonardo Badurina, Duncan A. Brown, Swapan Chattopadhyay, Matthew Evans, Peter Fritschel, Evan Hall, Jason M. Hogan, Karan Jani, Tim Kovachy, Kevin Kuns, Ariel Schwartzman, Daniel Sigg, Bram Slagmolen, Salvatore Vitale, Christopher Wipf
- **Posted on arXiv:** 15 March 2022

2.13 Jet Reconstruction

2.13.1 Quantum Algorithms - Grover's Search Algorithm

2.13.1.1 Quantum Algorithms for Jet Clustering [223]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

2.13.1.2 Quantum Algorithms in Particle Physics [215]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.13.2 Quantum Algorithms - Quantum Simulations

2.13.2.1 Medium induced jet broadening in a quantum computer [224]

- **Authors:** João Barata, Xiaojian Du, Meijian Li, Wenyang Qian, Carlos A. Salgado
- **Posted on arXiv:** 13 August 2022
- **Published in Phys. Rev. D 106, 074013 (2022):** 01 October 2022

2.13.3 Quantum Computing Paradigms - Quantum Annealing

2.13.3.1 Adiabatic quantum algorithm for multijet clustering in high energy physics [225]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

2.13.3.2 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [226]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

2.13.3.3 Quantum annealing for jet clustering with thrust [227]

- **Authors:** Andrea Delgado, Jesse Thaler
- **Posted on arXiv:** 05 May 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.13.3.4 Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics [228]

- **Authors:** Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, Jesse Thaler
- **Posted on arXiv:** 20 May 2022
- **Published in Phys. Rev. D 106, 056008 (2022):** 01 September 2022

2.13.4 Quantum Machine Learning - Unsupervised Methods

2.13.4.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [229]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

2.13.4.2 Quantum clustering and jet reconstruction at the LHC [230]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

2.13.5 Quantum Machine Learning - Variational Quantum Algorithms

2.13.5.1 Quantum Machine Learning for b-jet charge identification [231]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

2.13.6 Crosslists

2.13.6.1 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.13.6.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.13.6.3 Efficient quantum simulation of QCD jets on the light front [232]

- **Authors:** Wenyang Qian, Meijian Li, Carlos A. Salgado, Michael Kreshchuk
- **Posted on arXiv:** 14 November 2024
- **Published in Phys.Rev.D 111 (2025) 9, 096001:** 01 May 2025

2.14 Lattice Field Theories - Scalar/Fermion Theories

2.14.1 Quantum Algorithms - Quantum Simulations

2.14.1.1 Quantum Algorithms for Quantum Field Theories [233]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** 2012
- **Published in Science:** 2012

2.14.1.2 Quantum Computation of Scattering in Scalar Quantum Field Theories [234]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

2.14.1.3 Quantum Algorithms for Fermionic Quantum Field Theories [235]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

2.14.1.4 Digitization of scalar fields for quantum computing [236]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

2.14.1.5 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [237]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

2.14.1.6 Real-Time Scattering on Quantum Computers via Hamiltonian Truncation [238]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams, Matthew Wingate
- **Posted on arXiv:** 06 May 2025

2.14.2 Quantum Computing Paradigms - Continuous Variable Quantum Computing

2.14.2.1 Simulating quantum field theories on continuous-variable quantum computers [239]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 March 2024
- **Published in Phys. Rev. A 110 (2024) 1, 012607:** 08 July 2024

2.14.2.2 Real-time scattering processes with continuous-variable quantum computers [240]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 03 February 2025
- **Published in Phys. Rev. A 112 (2025), 012614:** 16 July 2025

2.14.2.3 Qumode Tensor Networks for False Vacuum Decay in Quantum Field Theory [241]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 20 June 2025

2.14.3 Quantum Error Correction and Mitigation

2.14.3.1 Hierarchical qubit maps and hierarchically implemented quantum error correction [242]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 04 September 2021
- **Published in Phys.Rev.A:** 15 December 2021

2.15 Lattice Field Theories - Gauge Theories

2.15.1 Quantum Algorithms - Quantum Simulations

2.15.1.1 Simulating lattice gauge theories on a quantum computer [243]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

2.15.1.2 Cold-Atom Quantum Simulator for SU(2) Yang-Mills Lattice Gauge Theory [244]

- **Authors:** Erez Zohar, J. Ignacio Cirac, Benni Reznik
- **Posted on arXiv:** November 2012
- **Published in Phys.Rev.Lett.:** 21 March 2013

2.15.1.3 Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories [245]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 07 May 2013
- **Published in Annalen Phys.:** 2013

2.15.1.4 Towards Quantum Simulating QCD [246]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

2.15.1.5 Quantum Simulations of Lattice Gauge Theories using Ultracold Atoms in Optical Lattices [247]

- **Authors:** Erez Zohar, J. Ignacio Cirac, Benni Reznik
- **Posted on arXiv:** 08 March 2015
- **Published in Rept.Prog.Phys.:** 18 December 2015

2.15.1.6 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [248]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

2.15.1.7 Gauss's law, duality, and the Hamiltonian formulation of U(1) lattice gauge theory [249]

- **Authors:** David B. Kaplan, Jesse R. Stryker
- **Posted on arXiv:** 22 June 2018
- **Published in Phys. Rev. D 102, 094515 (2020):** 20 November 2020

2.15.1.8 Oracles for Gauss's law on digital quantum computers [250]

- **Authors:** Jesse R. Stryker
- **Posted on arXiv:** 04 December 2018
- **Published in Phys. Rev. A 99, 042301 (2019):** 02 April 2019

2.15.1.9 General Methods for Digital Quantum Simulation of Gauge Theories [251]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

2.15.1.10 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [252]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

2.15.1.11 Simulating Lattice Gauge Theories within Quantum Technologies [253]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

2.15.1.12 Loop, string, and hadron dynamics in SU(2) Hamiltonian lattice gauge theories [254]

- **Authors:** Indrakshi Raychowdhury, Jesse R. Stryker
- **Posted on arXiv:** 12 December 2019
- **Published in Phys. Rev. D 101, 114502 (2020):** 09 June 2020

2.15.1.13 Observation of gauge invariance in a 71-site Bose–Hubbard quantum simulator [255]

- **Authors:** Bing Yang, Hui Sun, Robert Ott, Han-Yi Wang, Torsten V. Zache, Jad C. Halimeh, Zhen-Sheng Yuan, Philipp Hauke, Jian-Wei Pan
- **Posted on arXiv:** 19 March 2020
- **Published in Nature:** 2020

2.15.1.14 Toward scalable simulations of Lattice Gauge Theories on quantum computers [256]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

2.15.1.15 Role of boundary conditions in quantum computations of scattering observables [257]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

2.15.1.16 Quantum simulation of gauge theory via orbifold lattice [258]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

2.15.1.17 Trailhead for quantum simulation of SU(3) Yang-Mills lattice gauge theory in the local multiplet basis [259]

- **Authors:** Anthony Ciavarella, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 25 January 2021
- **Published in Phys. Rev. D 103, 094501 (2021):** 05 May 2021

2.15.1.18 Lattice renormalization of quantum simulations [260]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

2.15.1.19 Lattice Quantum Chromodynamics and Electrodynamics on a Universal Quantum Computer [261]

- **Authors:** Angus Kan, Yunseong Nam
- **Posted on arXiv:** 27 July 2021

2.15.1.20 Thermalization dynamics of a gauge theory on a quantum simulator [262]

- **Authors:** Zhao-Yu Zhou, Guo-Xian Su, Jad C. Halimeh, Robert Ott, Hui Sun, Philipp Hauke, Bing Yang, Zhen-Sheng Yuan, Jürgen Berges, Jian-Wei Pan
- **Posted on arXiv:** 28 July 2021
- **Published in Science 377, 311-314 (2022):** 15 July 2022

2.15.1.21 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [263]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

2.15.1.22 Preparations for quantum simulations of quantum chromodynamics in $1+1$ dimensions. I. Axial gauge [264]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

2.15.1.23 Simulating one-dimensional quantum chromodynamics on a quantum computer: Real-time evolutions of tetra- and pentaquarks [265]

- **Authors:** Yasar Y. Atas, Jan F. Haase, Jinglei Zhang, Victor Wei, Sieglinde M.-L. Pfaendler, Randy Lewis, Christine A. Muschik
- **Posted on arXiv:** 07 July 2022
- **Published in Phys.Rev.Res.:** 13 September 2023

2.15.1.24 Preparations for quantum simulations of quantum chromodynamics in $1+1$ dimensions. II. Single-baryon β-decay in real time [266]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 22 September 2022
- **Published in Phys. Rev. D 107, 054513 (2023):** 01 March 2023

2.15.1.25 Quantum Computation of Dynamical Quantum Phase Transitions and Entanglement Tomography in a Lattice Gauge Theory [267]

- **Authors:** Niklas Mueller, Joseph A. Carolan, Andrew Connelly, Zohreh Davoudi, Eugene F. Dumitrescu, Kübra Yeter-Aydeniz
- **Posted on arXiv:** 06 October 2022
- **Published in PRX Quantum:** 01 August 2023

2.15.1.26 Measurement-based quantum simulation of Abelian lattice gauge theories [268]

- **Authors:** Hiroki Sukeno, Takuya Okuda
- **Posted on arXiv:** 19 October 2022
- **Published in SciPost Phys. 14, 129 (2023):** 25 May 2023

2.15.1.27 Loop-string-hadron formulation of an SU(3) gauge theory with dynamical quarks [269]

- **Authors:** Saurabh V. Kadam, Indrakshi Raychowdhury, Jesse R. Stryker
- **Posted on arXiv:** 08 December 2022
- **Published in Phys. Rev. D 107, 094513 (2023):** 01 May 2023

2.15.1.28 General quantum algorithms for Hamiltonian simulation with applications to a non-Abelian lattice gauge theory [270]

- **Authors:** Zohreh Davoudi, Alexander F. Shaw, Jesse R. Stryker
- **Posted on arXiv:** 28 December 2022
- **Published in Quantum 7, 1213 (2023):** 20 December 2023

2.15.1.29 Simulating $\mathbb{Z}_2$ lattice gauge theory on a quantum computer [271]

- **Authors:** Clement Charles, Erik J. Gustafson, Elizabeth Hardt, Florian Herren, Norman Hogan, Henry Lamm, Sara Starecheski, Ruth S. Van de Water, Michael L. Wagman
- **Posted on arXiv:** 03 May 2023
- **Published in Phys.Rev.E:** 26 January 2024

2.15.1.30 Observation of microscopic confinement dynamics by a tunable topological θ-angle [272]

- **Authors:** Wei-Yong Zhang, Ying Liu, Yanting Cheng, Ming-Gen He, Han-Yi Wang, Tian-Yi Wang, Zi-Hang Zhu, Guo-Xian Su, Zhao-Yu Zhou, Yong-Guang Zheng, Hui Sun, Bing Yang, Philipp Hauke, Wei Zheng, Jad C. Halimeh, Zhen-Sheng Yuan, Jian-Wei Pan
- **Posted on arXiv:** 20 June 2023
- **Published in Nature Phys.:** 20 December 2024

2.15.1.31 Quantum simulation of fundamental particles and forces [273]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 April 2024
- **Published in Nature Rev.Phys.:** 21 June 2023

2.15.1.32 Toward QCD on quantum computer: orbifold lattice approach [274]

- **Authors:** Georg Bergner, Masanori Hanada, Enrico Rinaldi, Andreas Schafer
- **Posted on arXiv:** 22 January 2024
- **Published in JHEP:** 20 May 2024

2.15.1.33 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [275]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

2.15.1.34 Quantum Simulation of SU(3) Lattice Yang-Mills Theory at Leading Order in Large-N_c Expansion [276]

- **Authors:** Anthony N. Ciavarella, Christian W. Bauer
- **Posted on arXiv:** 15 February 2024
- **Published in Phys.Rev.Lett.:** 09 September 2024

2.15.1.35 Loop-string-hadron approach to SU(3) lattice Yang-Mills theory: Hilbert space of a trivalent vertex [277]

- **Authors:** Saurabh V. Kadam, Aahiri Naskar, Indrakshi Raychowdhury, Jesse R. Stryker
- **Posted on arXiv:** 27 July 2024
- **Published in Phys. Rev. D 111, 074516 (2025):** 01 April 2025

2.15.1.36 Quantum computing universal thermalization dynamics in a $(2+1)$D Lattice Gauge Theory [278]

- **Authors:** Niklas Mueller, Tianyi Wang, Or Katz, Zohreh Davoudi, Marko Cetina
- **Posted on arXiv:** 31 July 2024
- **Published in Nature Commun.:** 01 July 2025

2.15.1.37 Visualizing dynamics of charges and strings in (2 + 1)D lattice gauge theories [279]

- **Authors:** Tyler A. Cochran, Bernhard Jobst, Eliott Rosenberg, Yuri D. Lensky, Gaurav Gyawali, Norhan Eassa, Melissa Will, Murphy Yuezhen Niu, William D. Oliver, Aaron Szasz, Dmitry Abanin, Rajeev Acharya, Laleh Aghababaie Beni, Trond I. Andersen, Markus Ansmann, Frank Arute, Kunal Arya, Abraham Asfaw, Juan Atalaya, Ryan Babbush, Brian Ballard, Joseph C. Bardin, Andreas Bengtsson, Alexander Bilmes, Alexandre Bourassa, Jenna Bovaird, Michael Broughton, David A. Browne, Brett Buchea, Bob B. Buckley, Tim Burger, Brian Burkett, Nicholas Bushnell, Anthony Cabrera, Juan Campero, Hung-Shen Chang, Zijun Chen, Ben Chiaro, Jahan Claes, Agnetta Y. Cleland, Josh Cogan, Roberto Collins, Paul Conner, William Courtney, Alexander L. Crook, Ben Curtin, Sayan Das, Sean Demura, Laura De Lorenzo, Agustin Di Paolo, Paul Donohoe, Ilya Drozdov, Andrew Dunsworth, Alec Eickbusch, Aviv Moshe Elbag, Mahmoud Elzouka, Catherine Erickson, Vinicius S. Ferreira, Leslie Flores Burgos, Ebrahim Forati, Austin G. Fowler, Brooks Foxen, Suhas Ganjam, Robert Gasca, Élie Genois, William Giang, Dar Gilboa, Raja Gosula, Alejandro Grajales Dau, Dietrich Graumann, Alex Greene, Jonathan A. Gross, Steve Habegger, Monica Hansen, Matthew P. Harrigan, Sean D. Harrington, Paula Heu, Oscar Higgott, Jeremy Hilton, Hsin-Yuan Huang, Ashley Huff, William J. Huggins, Evan Jeffrey, Zhang Jiang, Cody Jones, Chaitali Joshi, Pavol Juhas, Dvir Kafri, Hui Kang, Amir H. Karamlou, Kostyantyn Kechedzhi, Trupti Khaire, Tanuj Khattar, Mostafa Khezri, Seon Kim, Paul V. Klimov, Bryce Kobrin, Alexander N. Korotkov, Fedor Kostritsa, John Mark Kreikebaum, Vladislav D. Kurilovich, David Landhuis, Tiano Lange-Dei, Brandon W. Langley, Kim-Ming Lau, Justin Ledford, Kenny Lee, Brian J. Lester, Loïck Le Guevel, Wing Yan Li, Alexander T. Lill, William P. Livingston, Aditya Locharla, Daniel Lundahl, Aaron Lunt, Sid

Madhuk, Ashley Maloney, Salvatore Mandrà, Leigh S. Martin, Orion Martin, Cameron Maxfield, Jarrod R. McClean, Matt McEwen, Seneca Meeks, Anthony Megrant, Kevin C. Miao, Reza Molavi, Sebastian Molina, Shirin Montazeri, Ramis Movassagh, Charles Neill, Michael Newman, Anthony Nguyen, Murray Nguyen, Chia-Hung Ni, Kristoffer Ottosson, Alex Pizzuto, Rebecca Potter, Orion Pritchard, Chris Quintana, Ganesh Ramachandran, Matthew J. Reagor, David M. Rhodes, Gabrielle Roberts, Kannan Sankaragomathi, Kevin J. Satzinger, Henry F. Schurkus, Michael J. Shearn, Aaron Shorter, Noah Shutty, Vladimir Shvarts, Volodymyr Sivak, Spencer Small, W. Clarke Smith, Sofia Springer, George Sterling, Jordan Suchard, Alex Sztein, Douglas Thor, M. Mert Torunbalci, Abeer Vaishnav, Justin Vargas, Sergey Vdovichev, Guifre Vidal, Catherine Vollgraff Heidweiller, Steven Waltman, Shannon X. Wang, Brayden Ware, Theodore White, Kristi Wong, Bryan W.K. Woo, Cheng Xing, Z. Jamie Yao, Ping Yeh, Bicheng Ying, Juhwan Yoo, Noureldin Yosri, Grayson Young, Adam Zalcman, Yaxing Zhang, Ningfeng Zhu, Nicholas Zobris, Sergio Boixo, Julian Kelly, Erik Lucero, Yu Chen, Vadim Smelyanskiy, Hartmut Neven, Adam Gammon-Smith, Frank Pollmann, Michael Knap, Pedram Roushan

- **Posted on arXiv:** 25 September 2024
- **Published in Nature:** 04 June 2025

2.15.1.38 Observation of string breaking on a (2 + 1)D Rydberg quantum simulator [280]

- **Authors:** Daniel Gonzalez-Cuadra, Majd Hamdan, Torsten V. Zache, Boris Braverman, Milan Kornjaca, Alexander Lukin, Sergio H. Cantu, Fangli Liu, Sheng-Tao Wang, Alexander Keesling, Mikhail D. Lukin, Peter Zoller, Alexei Bylinskii
- **Posted on arXiv:** 21 October 2024
- **Published in Nature:** 04 June 2025

2.15.1.39 Probing false vacuum decay on a cold-atom gauge-theory quantum simulator [281]

- **Authors:** Zi-Hang Zhu, Ying Liu, Gianluca Lagnese, Federica Maria Surace, Wei-Yong Zhang, Ming-Gen He, Jad C. Halimeh, Marcello Dalmonte, Siddhardh C. Morampudi, Frank Wilczek, Zhen-Sheng Yuan, Jian-Wei Pan
- **Posted on arXiv:** 19 November 2024

2.15.1.40 Efficient truncations of $\mathrm{SU}(N_c)$ lattice gauge theory for quantum simulation [282]

- **Authors:** Anthony N. Ciavarella, Ivan M. Burbano, Christian W. Bauer
- **Posted on arXiv:** 14 March 2025
- **Published in Phys.Rev.D:** 01 September 2025

2.15.1.41 Observation of hadron scattering in a lattice gauge theory on a quantum computer [283]

- **Authors:** Julian Schuhmacher, Guo-Xian Su, Jesse J. Osborne, Anthony Gandon, Jad C. Halimeh, Ivano Tavernelli
- **Posted on arXiv:** 26 May 2025

2.15.1.42 Scalable cold-atom quantum simulator of a $3+1$D U(1) lattice gauge theory with dynamical matter [284]

- **Authors:** Simone Orlando, Guo-Xian Su, Bing Yang, Jad C. Halimeh
- **Posted on arXiv:** 07 January 2026

2.15.2 Quantum Computing Paradigms - Quantum Annealing

2.15.2.1 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [285]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

2.15.2.2 SU(2) lattice gauge theory on a quantum annealer [286]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

2.15.3 Quantum Entanglement and Bell Inequalities

2.15.3.1 Gauge and diffeomorphism invariance from quantum information principles [287]

- **Authors:** Claudia Núñez, Miguel Pardina, Manuel Asorey, José Ignacio Latorre, Alba Cervera-Lierta
- **Posted on arXiv:** 06 November 2025
- **Published in Phys. Rev. D 113, 096019 (2026):** 01 May 2026

2.15.4 Quantum Error Correction and Mitigation

2.15.4.1 Quantum error correction with gauge symmetries [288]

- **Authors:** Abhishek Rajput, Alessandro Roggero, Nathan Wiebe
- **Posted on arXiv:** 09 December 2021
- **Published in npj Quantum Inf.:** 25 April 2023

2.15.4.2 Mitigating depolarizing noise on quantum computers with noise-estimation circuits [289]

- **Authors:** Miroslav Urbanek, Benjamin Nachman, Vincent R. Pascuzzi, Andre He, Christian W. Bauer, Wibe A. de Jong
- **Posted on arXiv:** 15 March 2021
- **Published in Phys.Rev.Lett.:** 2021

2.15.4.3 Robustness of Gauge Digitization to Quantum Noise [290]

- **Authors:** Erik J. Gustafson, Henry Lamm
- **Posted on arXiv:** 24 January 2023

2.15.4.4 Quantum error thresholds for gauge-redundant digitizations of lattice field theories [291]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 26 February 2024
- **Published in Phys.Rev.D:** 01 September 2024

2.15.4.5 Symmetry verification for noisy quantum simulations of non-Abelian lattice gauge theories [292]

- **Authors:** Edoardo Ballini, Julius Mildenberger, Matteo M. Wauters, Philipp Hauke
- **Posted on arXiv:** 10 December 2024
- **Published in Quantum 9, 1802 (2025):** 22 July 2025

2.15.5 Quantum Machine Learning - Supervised Methods

2.15.5.1 Quantum data learning for quantum simulations in high-energy physics [214]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.15.6 Quantum Machine Learning - Variational Quantum Algorithms

2.15.6.1 SU(2) hadrons on a quantum computer via a variational approach [293]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

2.15.7 Crosslists

2.15.7.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [294]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

2.15.7.2 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

2.15.7.3 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

2.15.7.4 Standard model physics and the digital quantum revolution: thoughts about the interface [16]

- **Authors:** Natalie Klco, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 10 July 2021
- **Published in Rept.Prog.Phys.:** 19 May 2022

2.15.7.5 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [17]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

2.15.7.6 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [20]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

2.15.7.7 Quantum Simulation for High-Energy Physics [25]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.15.7.8 Quantum computing hardware for HEP algorithms and sensing [26]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti
- **Posted on arXiv:** 18 April 2022

2.15.7.9 Review on Quantum Computing for Lattice Field Theory [7]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in nan:** 01 February 2023

2.15.7.10 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao
- **Posted on arXiv:** 28 February 2023

2.15.7.11 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico

Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.15.7.12 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [295]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

2.15.7.13 Steps toward quantum simulations of hadronization and energy loss in dense matter [296]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

2.15.7.14 Pathfinding quantum simulations of neutrinoless double-β decay [297]

- **Authors:** Ivan A. Chernyshev, Roland C. Farrell, Marc Illa, Martin J. Savage, Andrii Maksymov, Felix Tripier, Miguel Angel Lopez-Ruiz, Andrew Arrasmith, Yvette de Sereville, Aharon Brodutch, Claudio Girotto, Ananth Kaushik, Martin Roetteler
- **Posted on arXiv:** 06 June 2025
- **Published in Nature Commun.:** 23 January 2026

2.16 Lattice Field Theories - Schwinger Model

2.16.1 Quantum Algorithms - Quantum Simulations

2.16.1.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [294]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

2.16.1.2 Quantum Algorithms for Simulating the Lattice Schwinger Model [298]

- **Authors:** Alexander F. Shaw, Pavel Lougovski, Jesse R. Stryker, Nathan Wiebe
- **Posted on arXiv:** 25 February 2020
- **Published in Quantum 4, 306 (2020):** 10 August 2020

2.16.1.3 Quantum simulation of nonequilibrium dynamics and thermalization in the Schwinger model [299]

- **Authors:** Wibe A. de Jong, Kyle Lee, James Mulligan, Mateusz Płoskoń, Felix Ringer, Xiaojun Yao
- **Posted on arXiv:** 15 June 2021
- **Published in Phys. Rev. D 106, 054508 (2022):** 01 September 2022

2.16.1.4 Digital Quantum Simulation of the Schwinger Model and Symmetry Protection with Trapped Ions [300]

- **Authors:** Nhung H. Nguyen, Minh C. Tran, Yingyue Zhu, Alaina M. Green, C. Huerta Alderete, Zohreh Davoudi, Norbert M. Linke
- **Posted on arXiv:** 28 December 2021
- **Published in PRX Quantum 3, 020324 (2022):** 01 May 2022

2.16.1.5 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [295]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

2.16.1.6 Quantum simulations of hadron dynamics in the Schwinger model using 112 qubits [301]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 15 January 2024
- **Published in Phys.Rev.D 109 (2024) 11, 114510:** 01 June 2024

2.16.1.7 Steps toward quantum simulations of hadronization and energy loss in dense matter [296]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

2.16.1.8 Enhancing quantum field theory simulations on NISQ devices with Hamiltonian truncation [302]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams
- **Posted on arXiv:** 26 July 2024
- **Published in Phys. Rev. D 110 (2024), 096016:** 01 November 2024

2.16.1.9 Digital quantum simulations of scattering in quantum field theories using W states [303]

- **Authors:** Roland C. Farrell, Nikita A. Zemlevskiy, Marc Illa, John Preskill
- **Posted on arXiv:** 05 May 2025

2.16.2 Quantum Error Correction and Mitigation

2.16.2.1 Measurement error mitigation in quantum computers through classical bit-flip correction [304]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn, Paolo Stornati, Xiaoyang Wang
- **Posted on arXiv:** 07 July 2020
- **Published in Phys. Rev. A 105, 062404 (2022):** 03 June 2022

2.16.3 Quantum Machine Learning - Variational Quantum Algorithms

2.16.3.1 Quantum-classical computation of Schwinger model dynamics using quantum computers [305]

- **Authors:** N. Klco, E.F. Dumitrescu, A.J. McCaskey, T.D. Morris, R.C. Pooser, M. Sanz, E. Solano, P. Lougovski, M.J. Savage
- **Posted on arXiv:** 08 March 2018
- **Published in Phys. Rev. A 98, 032331 (2018):** 29 September 2018

2.16.3.2 Quantum-Classical Simulation of Quantum Field Theory by Quantum Circuit Learning [306]

- **Authors:** Kazuki Ikeda
- **Posted on arXiv:** 27 November 2023
- **Published in Annalen Phys.:** 01 June 2025

2.16.4 Crosslists

2.16.4.1 Quantum data learning for quantum simulations in high-energy physics [214]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.17 Neutrino Physics

2.17.1 Quantum Algorithms - Quantum Simulations

2.17.1.1 Simulation of collective neutrino oscillations on a quantum computer [307]

- **Authors:** Benjamin Hall, Alessandro Roggero, Alessandro Baroni, Joseph Carlson
- **Posted on arXiv:** 24 February 2021
- **Published in Phys. Rev. D 104, 063009 (2021):** 03 September 2021

2.17.1.2 Collective neutrino oscillations on a quantum computer [308]

- **Authors:** Kübra Yeter-Aydeniz, Shikha Bangar, George Siopsis, Raphael C. Pooser
- **Posted on arXiv:** 07 April 2021
- **Published in Quant.Inf.Proc.:** 17 February 2022

2.17.1.3 Trapped-ion quantum simulation of collective neutrino oscillations [309]

- **Authors:** Valentina Amitrano, Alessandro Roggero, Piero Luchi, Francesco Turro, Luca Vespucci, Francesco Pederiva
- **Posted on arXiv:** 07 July 2022
- **Published in Phys.Rev.D:** 13 January 2023

2.17.1.4 Multi-Neutrino Entanglement and Correlations in Dense Neutrino Systems [310]

- **Authors:** Marc Illa, Martin J. Savage
- **Posted on arXiv:** 16 October 2022
- **Published in Phys. Rev. Lett. 130, 221003 (2023):** 31 May 2023

2.17.1.5 Qutrit and qubit circuits for three-flavor collective neutrino oscillations [311]

- **Authors:** Francesco Turro, Ivan A. Chernyshev, Ramya Bhaskar, Marc Illa
- **Posted on arXiv:** 18 July 2024
- **Published in Phys. Rev. D 111, 043038 (2025):** 15 February 2025

2.17.1.6 Collective neutrino oscillations in three flavors on qubit and qutrit processors [312]

- **Authors:** Luca Spagnoli, Noah Goss, Alessandro Roggero, Ermal Rrapaj, Michael J. Cervia, Amol V. Patwardhan, Ravi K. Naik, A. Baha Balantekin, David I. Santiago, Irfan Siddiqi, Sheakha Aldaihan
- **Posted on arXiv:** 11 May 2025
- **Published in Phys.Rev.D:** 15 May 2025

2.17.1.7 Quantum simulation of collective neutrino oscillations in dense neutrino environment [313]

- **Authors:** Shvetaank Tripathi, Sandeep Joshi, Garima Rajpoot, Prashant Shukla
- **Posted on arXiv:** 15 August 2025
- **Published in Quant.Inf.Proc.:** 06 May 2026

2.17.1.8 Quantum Simulation of Collective Neutrino Oscillations using Dicke States [314]

- **Authors:** Katarina Bleau, Nikolina Ilic, Joachim Kopp, Ushak Rahaman, Xin Yue Yu
- **Posted on arXiv:** 08 April 2026

2.17.2 Quantum Computing Paradigms - Quantum Annealing

2.17.2.1 Three-flavor Collective Neutrino Oscillations on D-Wave's {\tt Advantage} Quantum Annealer [315]

- **Authors:** Ivan A. Chernyshev
- **Posted on arXiv:** 30 May 2024

2.17.3 Quantum Entanglement and Bell Inequalities

2.17.3.1 Violation of the Leggett-Garg Inequality in Neutrino Oscillations [316]

- **Authors:** J.A. Formaggio, D.I. Kaiser, M.M. Murskyj, T.E. Weiss
- **Posted on arXiv:** 29 January 2016
- **Published in Phys.Rev.Lett.:** 26 July 2016

2.17.3.2 Entanglement and collective flavor oscillations in a dense neutrino gas [317]

- **Authors:** Michael J. Cervia, Amol V. Patwardhan, A.B. Balantekin, ‡ S.N. Coppersmith, Calvin W. Johnson
- **Posted on arXiv:** 09 August 2019
- **Published in Phys.Rev.D:** 03 October 2019

2.17.3.3 Entanglement and many-body effects in collective neutrino oscillations [318]

- **Authors:** Alessandro Roggero
- **Posted on arXiv:** 19 February 2021
- **Published in Phys. Rev. D 104 (2021) 103016:** 10 November 2021

2.17.3.4 Dynamical phase transitions in models of collective neutrino oscillations [319]

- **Authors:** Alessandro Roggero
- **Posted on arXiv:** 21 March 2021
- **Published in Phys.Rev.D:** 14 December 2021

2.17.3.5 Spectral splits and entanglement entropy in collective neutrino oscillations [320]

- **Authors:** Amol V. Patwardhan, Michael J. Cervia, A.B. Balantekin
- **Posted on arXiv:** 18 September 2021
- **Published in Phys. Rev. D 104, 123035 (2021):** 15 December 2021

2.17.3.6 Entanglement and correlations in fast collective neutrino flavor oscillations [321]

- **Authors:** Alessandro Roggero, Ermal Rrapaj, Zewei Xiong
- **Posted on arXiv:** 05 March 2022
- **Published in Phys.Rev.D:** 15 August 2022

2.17.3.7 Role of non-Gaussian quantum fluctuations in neutrino entanglement [322]

- **Authors:** Denis Lacroix, A.B. Balantekin, Michael J. Cervia, Amol V. Patwardhan, Pooja Siwach
- **Posted on arXiv:** 19 May 2022
- **Published in Phys.Rev.D:** 06 December 2022

2.17.3.8 Entanglement in three-flavor collective neutrino oscillations [323]

- **Authors:** Pooja Siwach, Anna M. Suliga, A. Baha Balantekin
- **Posted on arXiv:** 14 November 2022
- **Published in Phys.Rev.D 107 023019 (2023):** 15 January 2023

2.17.3.9 Many-body neutrino flavor entanglement in a simple dynamic model [324]

- **Authors:** Joshua D. Martin, A. Roggero, Huaiyu Duan, J. Carlson
- **Posted on arXiv:** 17 January 2023

2.17.3.10 Neutrino many-body correlations [325]

- **Authors:** Lucas Johns
- **Posted on arXiv:** 08 May 2023
- **Published in Int.J.Mod.Phys.A:** 13 November 2024

2.17.3.11 Equilibration of quantum many-body fast neutrino flavor oscillations [326]

- **Authors:** Joshua D. Martin, Duff Neill, A. Roggero, Huaiyu Duan, J. Carlson
- **Posted on arXiv:** 31 July 2023
- **Published in Phys.Rev.D:** 07 December 2023

2.17.3.12 Time Scales in Many-Body Fast Neutrino Flavor Conversion [327]

- **Authors:** Ramya Bhaskar, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 23 December 2023

2.17.3.13 Quantum magic and computational complexity in the neutrino sector [328]

- **Authors:** Ivan Chernyshev, Caroline E.P. Robin, Martin J. Savage
- **Posted on arXiv:** 06 November 2024
- **Published in Phys.Rev.Res.:** 04 June 2025

2.17.3.14 Exploring entanglement and spectral split correlations in three-flavor collective neutrino oscillations [329]

- **Authors:** Pooja Siwach, A. Baha Balantekin, Amol V. Patwardhan, Anna M. Suliga
- **Posted on arXiv:** 07 November 2024
- **Published in Phys.Rev.D:** 14 March 2025

2.17.3.15 Quantum resource redistribution drives spectral splits in dense neutrino gases [330]

- **Authors:** Michael Hite, Pooja Siwach
- **Posted on arXiv:** 22 May 2026

2.17.3.16 Collective neutrino oscillations: Many-body non-forward effects and non-classicality [331]

- **Authors:** Julien Froustey, Ermal Rrapaj, Yuhao Liu, Gushu Li, Costin Iancu, Vincenzo Cirigliano
- **Posted on arXiv:** 10 June 2026

2.17.4 Quantum Machine Learning - Variational Quantum Algorithms

2.17.4.1 Collective neutrino oscillations on a quantum computer with hybrid quantum-classical algorithm [332]

- **Authors:** Pooja Siwach, Kaytlin Harrison, A. Baha Balantekin
- **Posted on arXiv:** 17 August 2023
- **Published in Phys.Rev.D:** 15 October 2023

2.17.4.2 Three-flavor supernova neutrino simulation using a hybrid quantum-classical algorithm with qutrits [333]

- **Authors:** Daniel J. Heimsoth, Pooja Siwach, A. Baha Balantekin
- **Posted on arXiv:** 01 May 2026
- **Published in Phys. Rev. D 114, 043055 (2026):** 15 August 2026

2.17.5 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.17.5.1 Superradiant interactions for relic detection with entangled nuclear spins [156]

- **Authors:** Marios Galanis, Onur Hosten, Asimina Arvanitaki, Savas Dimopoulos
- **Posted on arXiv:** 28 August 2025
- **Published in Phys.Rev.A:** 10 June 2026

2.17.6 Quantum Sensors - Optical/Photonic Sensors

2.17.6.1 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [154]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis
- **Posted on arXiv:** 07 August 2024
- **Published in Phys.Rev.D:** 01 March 2025

2.17.7 Quantum Sensors - Optomechanical Sensors

2.17.7.1 Requirements on quantum superpositions of macro-objects for sensing neutrinos [334]

- **Authors:** Eva Kilian, Marko Toroš, Frank F. Deppisch, Ruben Saakyan, Sougato Bose
- **Posted on arXiv:** 27 April 2022
- **Published in Phys. Rev. Research 5, 023012 (2023):** 07 April 2023

2.17.7.2 Searches for Massive Neutrinos with Mechanical Quantum Sensors [335]

- **Authors:** Daniel Carney, Kyle G. Leach, David C. Moore
- **Posted on arXiv:** 12 July 2022
- **Published in Phys.Rev.X:** 01 February 2023

2.17.8 Crosslists

2.17.8.1 Standard model physics and the digital quantum revolution: thoughts about the interface [16]

- **Authors:** Natalie Klco, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 10 July 2021
- **Published in Rept.Prog.Phys.:** 19 May 2022

2.17.8.2 Many-Body Collective Neutrino Oscillations: Recent Developments [6]

- **Authors:** Amol V. Patwardhan, Michael J. Cervia, Ermal Rrapaj, Pooja Siwach, A.B. Balantekin
- **Posted on arXiv:** 31 December 2022
- **Published in nan:** 2023

2.17.8.3 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao
- **Posted on arXiv:** 28 February 2023

2.17.8.4 Quantum information and quantum simulation of neutrino physics [8]

- **Authors:** A.B. Balantekin, Michael J. Cervia, Amol V. Patwardhan, Ermal Rrapaj, Pooja Siwach
- **Posted on arXiv:** 01 May 2023
- **Published in Eur.Phys.J.A:** 17 August 2023

2.18 Nuclear Many-Body Physics - Nuclear Structure

2.18.1 Quantum Algorithms - Quantum Simulations

2.18.1.1 Symmetry assisted preparation of entangled many-body states on a quantum computer [336]

- **Authors:** D. Lacroix
- **Posted on arXiv:** 11 June 2020
- **Published in Phys.Rev.Lett.:** 03 December 2020

2.18.1.2 Quantum computation of nuclear observables involving linear combinations of unitary operators [337]

- **Authors:** Pooja Siwach, P. Arumugam
- **Posted on arXiv:** 16 June 2022
- **Published in Phys.Rev.C:** 28 June 2022

2.18.2 Quantum Computing Paradigms - Quantum Annealing

2.18.2.1 Quasiparticle pairing encoding of atomic nuclei for quantum annealing [338]

- **Authors:** Emanuele Costa, Axel Pérez-Obiol, Javier Menéndez, Arnau Rios, Artur García-Sáez, Bruno Juliá-Díaz
- **Posted on arXiv:** 11 October 2025
- **Published in Phys.Lett.B:** 30 November 2025

2.18.3 Quantum Machine Learning - Variational Quantum Algorithms

2.18.3.1 Cloud Quantum Computing of an Atomic Nucleus [178]

- **Authors:** E.F. Dumitrescu, A.J. McCaskey, G. Hagen, G.R. Jansen, T.D. Morris, T. Papenbrock, R.C. Pooser, D.J. Dean, P. Lougovski
- **Posted on arXiv:** 11 January 2018
- **Published in Phys. Rev. Lett. 120, 210501 (2018):** 24 May 2018

2.18.3.2 Improving Hamiltonian encodings with the Gray code [180]

- **Authors:** Olivia Di Matteo, Anna McCoy, Peter Gysbers, Takayuki Miyagi, R.M. Woloshyn, Petr Navrátil
- **Posted on arXiv:** 11 August 2020
- **Published in Phys. Rev. A 103, 042405 (2021):** 03 April 2021

2.18.3.3 Lipkin model on a quantum computer [339]

- **Authors:** Michael J. Cervia, A.B. Balantekin, S.N. Coppersmith, Calvin W. Johnson, Peter J. Love, C. Poole, K. Robbins, M. Saffman
- **Posted on arXiv:** 08 November 2020
- **Published in Phys. Rev. C 104, 024305 (2021):** 03 August 2021

2.18.3.4 Variational approaches to constructing the many-body nuclear ground state for quantum computing [340]

- **Authors:** I. Stetcu, A. Baroni, J. Carlson
- **Posted on arXiv:** 12 October 2021
- **Published in Phys. Rev. C 105, 064308 (2022):** 21 June 2022

2.18.3.5 Simulating excited states of the Lipkin model on a quantum computer [341]

- **Authors:** Manqoba Q. Hlatshwayo, Yinu Zhang, Herlik Wibowo, Ryan LaRose, Denis Lacroix, Elena Litvinova
- **Posted on arXiv:** 02 March 2022
- **Published in Phys.Rev.C:** 18 August 2022

2.18.3.6 Solving nuclear structure problems with the adaptive variational quantum algorithm [342]

- **Authors:** A.M. Romero, J. Engel, Ho Lun Tang, Sophia E. Economou
- **Posted on arXiv:** 03 March 2022
- **Published in Phys. Rev. C 105, 064317 (2022):** 27 June 2022

2.18.3.7 Quantum computing of the ^{6}Li nucleus via ordered unitary coupled cluster [343]

- **Authors:** Oriel Kiss, Michele Grossi, Pavel Lougovski, Federico Sanchez, Sofia Vallecorsa, Thomas Papenbrock
- **Posted on arXiv:** 02 May 2022
- **Published in Phys. Rev. C 106, 034325 (2022):** 29 September 2022

2.18.3.8 QCSH: A full quantum computer nuclear shell-model package [344]

- **Authors:** Peng Lv, Shi-Jie Wei, Hao-Nan Xie, Gui-Lu Long
- **Posted on arXiv:** 24 May 2022
- **Published in Sci.China Phys.Mech.Astron.:** 06 March 2023

2.18.3.9 Subatomic Many-Body Physics Simulations on a Quantum Frequency Processor [182]

- **Authors:** Hsuan-Hao Lu, Natalie Klco, Joseph M. Lukens, Titus D. Morris, Aaina Bansal, Andreas Ekström, Gaute Hagen, Thomas Papenbrock, Andrew M. Weiner, Martin J. Savage, Pavel Lougovski
- **Published in nan:** 2019

2.18.3.10 Nuclear shell-model simulation in digital quantum computers [345]

- **Authors:** A. Pérez-Obiol, A.M. Romero, J. Menéndez, A. Rios, A. García-Sáez, B. Juliá-Díaz
- **Posted on arXiv:** 07 February 2023
- **Published in Sci.Rep.:** 29 July 2023

2.18.3.11 Deep quantum circuit simulations of low-energy nuclear states [346]

- **Authors:** Ang Li, Alessandro Baroni, Ionel Stetcu, Travis S. Humble
- **Posted on arXiv:** 26 October 2023
- **Published in Eur.Phys.J.A:** 14 May 2024

2.18.3.12 Exploiting symmetries in nuclear Hamiltonians for ground state preparation [347]

- **Authors:** Joe Gibbs, Zoë Holmes, Paul Stevenson
- **Posted on arXiv:** 15 February 2024
- **Published in Quantum Machine Intelligence:** 30 January 2025

2.18.3.13 Comparison of variational quantum eigensolvers in light nuclei [348]

- **Authors:** Miquel Carrasco-Codina, Emanuele Costa, Antonio Márquez Romero, Javier Menéndez, Arnau Rios
- **Posted on arXiv:** 18 July 2025
- **Published in Phys.Rev.C:** 23 February 2026

2.18.3.14 Toward scalable quantum computations of atomic nuclei [349]

- **Authors:** Chenyi Gu, Matthias Heinz, Oriel Kiss, Thomas Papenbrock
- **Posted on arXiv:** 19 July 2025
- **Published in Phys. Rev. C 113, 034321 (2026):** 24 March 2026

2.18.3.15 Bridging quantum computing and nuclear structure: Atomic nuclei on a trapped-ion quantum computer [350]

- **Authors:** Sota Yoshida, Takeshi Sato, Takumi Ogata, Masaaki Kimura
- **Posted on arXiv:** 25 September 2025
- **Published in Phys. Rev. Research 8, 013134 (2026):** 06 February 2026

2.18.4 Quantum Measurement and Tomography

2.18.4.1 Shadow-based quantum subspace algorithm for the nuclear shell model [351]

- **Authors:** Ruyu Yang, Tianren Wang, Bing-Nan Lu, Ying Li, Xiaosi Xu
- **Posted on arXiv:** 15 June 2023
- **Published in Phys.Rev.A:** 21 January 2025

2.18.4.2 Quantum complexity fluctuations from nuclear and hypernuclear forces [352]

- **Authors:** Caroline E. P. Robin, Martin J. Savage
- **Posted on arXiv:** 16 May 2024
- **Published in Phys.Rev.C 112, 044004 (2025):** 17 October 2025

2.18.5 Crosslists

2.18.5.1 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

2.18.5.2 Standard model physics and the digital quantum revolution: thoughts about the interface [16]

- **Authors:** Natalie Klco, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 10 July 2021
- **Published in Rept.Prog.Phys.:** 19 May 2022

2.18.5.3 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James

Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao

- **Posted on arXiv:** 28 February 2023

2.18.5.4 Quantum Complexity and New Directions in Nuclear Physics and High-Energy Physics Phenomenology [11]

- **Authors:** Caroline E.P. Robin, Martin J. Savage
- **Posted on arXiv:** 29 April 2026

2.19 Nuclear Many-Body Physics - Nuclear Reactions and Response

2.19.1 Quantum Algorithms - Quantum Phase Estimation

2.19.1.1 Dynamic linear response quantum algorithm [353]

- **Authors:** Alessandro Roggero, Joseph Carlson
- **Posted on arXiv:** 04 April 2018
- **Published in Phys. Rev. C 100, 034610 (2019):** 14 September 2019

2.19.2 Quantum Algorithms - Quantum Simulations

2.19.2.1 Quantum Computing for Neutrino-Nucleus Scattering [179]

- **Authors:** Alessandro Roggero, Andy C.Y. Li, Joseph Carlson, Rajan Gupta, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019
- **Published in Phys. Rev. D 101, 074038 (2020):** 28 April 2020

2.19.2.2 Quantum simulation of nuclear inelastic scattering [354]

- **Authors:** Weijie Du, James P. Vary, Xingbo Zhao, Wei Zuo
- **Posted on arXiv:** 01 June 2020
- **Published in Phys. Rev. A 104, 012611 (2021):** 23 July 2021

2.19.2.3 Imaginary-time propagation on a quantum chip [355]

- **Authors:** Francesco Turro, Alessandro Roggero, Valentina Amitrano, Piero Luchi, Kyle A. Wendt, Jonathan L. DuBois, Sofia Quaglioni, Francesco Pederiva
- **Posted on arXiv:** 24 February 2021
- **Published in Phys.Rev.A:** 28 February 2022

2.19.2.4 Nuclear two point correlation functions on a quantum computer [356]

- **Authors:** Alessandro Baroni, Joseph Carlson, Rajan Gupta, Andy C.Y. Li, Gabriel N. Perdue, Alessandro Roggero
- **Posted on arXiv:** 04 November 2021
- **Published in Phys.Rev.D:** 01 April 2022

2.19.2.5 A quantum algorithm for the linear response of nuclei [357]

- **Authors:** Nifeeya Singh, P. Arumugam
- **Posted on arXiv:** 17 October 2022
- **Published in Indian J.Phys.:** 31 May 2025

2.19.2.6 Faster spectral density calculation using energy moments [358]

- **Authors:** Jeremy Hartse, Alessandro Roggero
- **Posted on arXiv:** 01 November 2022
- **Published in Eur.Phys.J.A:** 09 March 2023

2.19.2.7 Quantum error mitigation for Fourier moment computation [359]

- **Authors:** Oriel Kiss, Michele Grossi, Alessandro Roggero
- **Posted on arXiv:** 23 January 2024
- **Published in Phys. Rev. D 111, 034504 (2025):** 01 February 2025

2.19.2.8 Nuclear scattering via quantum computing [360]

- **Authors:** Peiyan Wang, Weijie Du, Wei Zuo, James P. Vary
- **Posted on arXiv:** 30 January 2024
- **Published in Phys.Rev.C 109 6, 064623 (2024):** 26 June 2024

2.19.2.9 Solving reaction dynamics with quantum computing algorithms [361]

- **Authors:** Ronen Weiss, Alessandro Baroni, Joseph Carlson, Ionel Stetcu
- **Posted on arXiv:** 29 March 2024
- **Published in Physical Review C 111, 064004 (2025):** 25 June 2025

2.19.2.10 Quantum computing for extracting nuclear resonances [362]

- **Authors:** Hantao Zhang, Dong Bai, Zhongzhou Ren
- **Posted on arXiv:** 10 September 2024
- **Published in Phys.Lett.B:** 12 December 2024

2.19.2.11 Pathfinding quantum simulations of neutrinoless double-β decay [297]

- **Authors:** Ivan A. Chernyshev, Roland C. Farrell, Marc Illa, Martin J. Savage, Andrii Maksymov, Felix Tripier, Miguel Angel Lopez-Ruiz, Andrew Arrasmith, Yvette de Sereville, Aharon Brodutch, Claudio Girotto, Ananth Kaushik, Martin Roetteler
- **Posted on arXiv:** 06 June 2025
- **Published in Nature Commun.:** 23 January 2026

2.19.3 Quantum Machine Learning - Variational Quantum Algorithms

2.19.3.1 Quantum simulations of nuclear resonances with variational methods [363]

- **Authors:** Ashutosh Singh, Pooja Siwach, P. Arumugam
- **Posted on arXiv:** 15 April 2025
- **Published in Phys.Rev.C:** 25 August 2025

2.19.4 Quantum Measurement and Tomography

2.19.4.1 Antiflatness and nonlocal nonstabilizerness in two-particle scattering processes [364]

- **Authors:** Caroline E.P. Robin, Martin J. Savage
- **Posted on arXiv:** 27 October 2025
- **Published in Phys. Rev. D 114, 014007 (2026):** 01 July 2026

2.19.5 Crosslists

2.19.5.1 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

2.19.5.2 Standard model physics and the digital quantum revolution: thoughts about the interface [16]

- **Authors:** Natalie Klco, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 10 July 2021
- **Published in Rept.Prog.Phys.:** 19 May 2022

2.19.5.3 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao
- **Posted on arXiv:** 28 February 2023

2.20 Quantum Chromodynamics

2.20.1 Quantum Algorithms - Quantum Simulations

2.20.1.1 Quantum simulation of quantum field theory in the light-front formulation [365]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

2.20.1.2 Quantum simulation of open quantum systems in heavy-ion collisions [366]

- **Authors:** Wibe A. De Jong, Mekena Metcalf, James Mulligan, Mateusz Płoskoń, Felix Ringer, Xiaojun Yao
- **Posted on arXiv:** 07 October 2020
- **Published in Phys. Rev. D 104, 051501 (2021):** 01 September 2021

2.20.1.3 Quantum Simulation of Light-Front QCD for Jet Quenching in Nuclear Environments [367]

- **Authors:** Xiaojun Yao
- **Posted on arXiv:** 16 May 2022

2.20.1.4 Quantum simulation of colour in perturbative quantum chromodynamics [368]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen
- **Posted on arXiv:** 08 March 2023
- **Published in SciPost Phys. 15, 205 (2023):** 24 November 2023

2.20.1.5 Quantum simulation of in-medium QCD jets: Momentum broadening, gluon production, and entropy growth [369]

- **Authors:** João Barata, Xiaojian Du, Meijian Li, Wenyang Qian, Carlos A. Salgado
- **Posted on arXiv:** 04 July 2023
- **Published in Phys.Rev.D:** 01 September 2023

2.20.1.6 Efficient quantum simulation of QCD jets on the light front [232]

- **Authors:** Wenyang Qian, Meijian Li, Carlos A. Salgado, Michael Kreshchuk
- **Posted on arXiv:** 14 November 2024
- **Published in Phys.Rev.D 111 (2025) 9, 096001:** 01 May 2025

2.20.1.7 Quantum simulation of scattering amplitudes and interferences in perturbative QCD [370]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen, Simon Williams
- **Posted on arXiv:** 09 July 2025
- **Published in Eur.Phys.J.C:** 11 August 2026

2.20.1.8 Quantum simulating multi-particle processes in high energy nuclear physics: dijet production and color (de)coherence [371]

- **Authors:** João Barata, Meijian Li, Wenyang Qian, Carlos A. Salgado, João M. Silva
- **Posted on arXiv:** 13 April 2026
- **Published in JHEP:** 08 July 2026

2.20.2 Quantum Entanglement and Bell Inequalities

2.20.2.1 Deep inelastic scattering as a probe of entanglement [372]

- **Authors:** Dmitri E. Kharzeev, Eugene M. Levin
- **Posted on arXiv:** 12 February 2017
- **Published in Phys.Rev.D:** 13 June 2017

2.20.2.2 Einstein-Podolsky-Rosen Paradox and Quantum Entanglement at Subnucleonic Scales [373]

- **Authors:** Zhoudunming Tu, Dmitri E. Kharzeev, Thomas Ullrich
- **Posted on arXiv:** 26 April 2019
- **Published in Phys. Rev. Lett. 124, 062001 (2020):** 15 February 2020

2.20.2.3 Measurement of charged particle multiplicity distributions in DIS at HERA and its implication to entanglement entropy of partons [374]

- **Authors:** V. Andreev, A. Baghdasaryan, A. Baty, K. Begzsuren, A. Belousov, A. Bolz, V. Boudry, G. Brandt, D. Britzger, A. Buniatyan, L. Bystritskaya, A.J. Campbell, K.B. Cantun Avila, K. Cerny, V. Chekelian, Z. Chen, J.G. Contreras, J. Cvach, J.B. Dainton, K. Daum, A. Deshpande, C. Diaconu, G. Eckerlin, S. Egli, E. Elsen, L. Favart, A. Fedotov, J. Feltesse, M. Fleischer, A. Fomenko, C. Gal, J. Gayler, L. Goerlich, N. Gogitidze, M. Gouzevitch, C. Grab, A. Grebenyuk, T. Greenshaw, G. Grindhammer, D. Haidt, R.C.W. Henderson, J. Hladkỳ, D. Hoffmann, R. Horisberger, T. Hreus, F. Huber, M. Jacquet, X. Janssen, A.W. Jung, H. Jung, M. Kapichine, J. Katzy, C. Kiesling, M. Klein, C. Kleinwort, R. Kogler, P. Kostka, J. Kretzschmar, D. Krücker, K. Krüger, M.P.J. Landon, W. Lange, P. Laycock, A. Lebedev, S. Levonian, K. Lipka, B. List, J. List, W. Li, B. Lobodzinski, E. Malinovski, H.U. Martyn, S.J. Maxfield, A. Mehta, A.B. Meyer, H. Meyer, J. Meyer, S. Mikocki, M.M. Mondal, A. Morozov, K. Müller, Th. Naumann, P.R. Newman, C. Niebuhr, G. Nowak, J.E. Olsson, D. Ozerov, S. Park, C. Pascaud, G.D. Patel, E. Perez, A. Petrukhin, I. Picuric, D. Pitzl, R. Polifka, V. Radescu, N. Raicevic, T. Ravdandorj, P. Reimer, E. Rizvi, P. Robmann, R. Roosen, A. Rostovtsev, M. Rotaru, D.P.C. Sankey, M. Sauter, E. Sauvan, S. Schmitt, B.A. Schmookler, L. Schoeffel, A. Schöning, F. Sefkow, S. Shushkevich, Y. Soloviev, P. Sopicki, D. South, V. Spaskov, A. Specka, M. Steder, B. Stella, U. Straumann, T. Sykora, P.D. Thompson, D. Traynor, P. Truöl, B. Tseepeldorj, Z. Tu, T. Ullrich, A. Valkárová, C. Vallée, P.Van Mechelen, D. Wegener, E. Wünsch, J. Žáček, J. Zhang, Z. Zhang, R. Žlebčík, H. Zohrabyan, F. Zomer
- **Posted on arXiv:** 03 November 2020
- **Published in Eur.Phys.J.C:** 03 March 2021

2.20.2.4 Deep inelastic scattering as a probe of entanglement: Confronting experimental data [375]

- **Authors:** Dmitri E. Kharzeev, Eugene Levin
- **Posted on arXiv:** 19 February 2021
- **Published in Phys. Rev. D 104, 031503 (2021):** 01 August 2021

2.20.2.5 A quantum strategy to compute the jet quenching parameter $\hat{q}$ [376]

- **Authors:** João Barata, Carlos A. Salgado
- **Posted on arXiv:** 09 April 2021
- **Published in Eur.Phys.J.C:** 01 October 2021

2.20.2.6 Symmetry from entanglement suppression [377]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

2.20.2.7 Entanglement entropy production in deep inelastic scattering [378]

- **Authors:** Kun Zhang, Kun Hao, Dmitri Kharzeev, Vladimir Korepin
- **Posted on arXiv:** 10 October 2021
- **Published in Phys. Rev. D 105, 014002 (2022):** 01 January 2022

2.20.2.8 Real-Time Nonperturbative Dynamics of Jet Production in Schwinger Model: Quantum Entanglement and Vacuum Modification [379]

- **Authors:** Adrien Florio, David Frenklakh, Kazuki Ikeda, Dmitri Kharzeev, Vladimir Korepin, Shuzhe Shi, Kwangmin Yu
- **Posted on arXiv:** 27 January 2023
- **Published in Phys. Rev. Lett. 131 (2023) 021902:** 13 July 2023

2.20.2.9 Probing the Onset of Maximal Entanglement inside the Proton in Diffractive Deep Inelastic Scattering [380]

- **Authors:** Martin Hentschinski, Dmitri E. Kharzeev, Krzysztof Kutak, Zhoudunming Tu
- **Posted on arXiv:** 04 May 2023
- **Published in Phys.Rev.Lett.:** 13 December 2023

2.20.2.10 Universal rapidity scaling of entanglement entropy inside hadrons from conformal invariance [381]

- **Authors:** Umut Gürsoy, Dmitri E. Kharzeev, Juan F. Pedraza
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

2.20.2.11 Entanglement as a Probe of Hadronization [382]

- **Authors:** Jaydeep Datta, Abhay Deshpande, Dmitri E. Kharzeev, Charles Joseph Naïm, Zhoudunming Tu
- **Posted on arXiv:** 29 October 2024
- **Published in Phys.Rev.Lett.:** 19 March 2025

2.20.2.12 Quantum information at the electron-ion collider [383]

- **Authors:** Kun Cheng, Tao Han, Sokratis Trifinopoulos
- **Posted on arXiv:** 27 October 2025
- **Published in Phys.Rev.D:** 01 September 2026

2.20.3 Crosslists

2.20.3.1 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

2.20.3.2 Partonic collinear structure by quantum computing [211]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.20.3.3 Minimal entanglement and emergent symmetries in low-energy QCD [176]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

2.20.3.4 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay

Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao

- **Posted on arXiv:** 28 February 2023

2.20.3.5 Quantum Complexity and New Directions in Nuclear Physics and High-Energy Physics Phenomenology [11]

- **Authors:** Caroline E.P. Robin, Martin J. Savage
- **Posted on arXiv:** 29 April 2026

2.21 Quantum Correlations at Colliders

2.21.1 Quantum Entanglement and Bell Inequalities

2.21.1.1 Maximal Entanglement in High Energy Physics [384]

- **Authors:** Alba Cervera-Lierta, José I. Latorre, Juan Rojo, Luca Rottoli
- **Posted on arXiv:** 08 March 2017
- **Published in SciPost Phys. 3, 036 (2017):** 24 November 2017

2.21.1.2 Polarization and Entanglement in Baryon-Antibaryon Pair Production in Electron-Positron Annihilation [385]

- **Authors:** M. Ablikim, M.N. Achasov, S. Ahmed, M. Albrecht, M. Alekseev, A. Amoroso, F.F. An, Q. An, Y. Bai, O. Bakina, R. Baldini Ferroli, Y. Ban, K. Begzsuren, D.W. Bennett, J.V. Bennett, N. Berger, M. Bertani, D. Bettoni, F. Bianchi, E. Boger, I. Boyko, R.A. Briere, H. Cai, X. Cai, A. Calcaterra, G.F. Cao, S.A. Cetin, J. Chai, J.F. Chang, W.L. Chang, G. Chelkov, G. Chen, H.S. Chen, J.C. Chen, M.L. Chen, P.L. Chen, S.J. Chen, X.R. Chen, Y.B. Chen, W. Cheng, X.K. Chu, G. Cibinetto, F. Cossio, H.L. Dai, J.P. Dai, A. Dbeyssi, D. Dedovich, Z.Y. Deng, A. Denig, I. Denysenko, M. Destefanis, F. De Mori, Y. Ding, C. Dong, J. Dong, L.Y. Dong, M.Y. Dong, Z.L. Dou,

S.X. Du, P.F. Duan, J.Z. Fan, J. Fang, S.S. Fang, Y. Fang, R. Farinelli, L. Fava, S. Fegan, F. Feldbauer, G. Felici, C.Q. Feng, E. Fioravanti, M. Fritsch, C.D. Fu, Q. Gao, X.L. Gao, Y. Gao, Y.G. Gao, Z. Gao, B. Garillon, I. Garzia, A. Gilman, K. Goetzen, L. Gong, W.X. Gong, W. Gradl, M. Greco, L.M. Gu, M.H. Gu, Y.T. Gu, A.Q. Guo, L.B. Guo, R.P. Guo, Y.P. Guo, A. Guskov, Z. Haddadi, S. Han, X.Q. Hao, F.A. Harris, K.L. He, F.H. Heinsius, T. Held, Y.K. Heng, Z.L. Hou, H.M. Hu, J.F. Hu, T. Hu, Y. Hu, G.S. Huang, J.S. Huang, X.T. Huang, X.Z. Huang, Z.L. Huang, T. Hussain, W. Ikegami Andersson, M. Irshad, Q. Ji, Q.P. Ji, X.B. Ji, X.L. Ji, H.L. Jiang, X.S. Jiang, X.Y. Jiang, J.B. Jiao, Z. Jiao, D.P. Jin, S. Jin, Y. Jin, T. Johansson, A. Julin, N. Kalantar-Nayestanaki, X.S. Kang, M. Kavatsyuk, B.C. Ke, I.K. Keshk, T. Khan, A. Khoukaz, P. Kiese, R. Kiuchi, R. Kliemt, L. Koch, O.B. Kolcu, B. Kopf, M. Kornicer, M. Kuemmel, M. Kuessner, A. Kupsc, M. Kurth, W. Kühn, J.S. Lange, P. Larin, L. Lavezzi, S. Leiber, H. Leithoff, C. Li, Cheng Li, D.M. Li, F. Li, F.Y. Li, G. Li, H.B. Li, H.J. Li, J.C. Li, J.W. Li, K.J. Li, Kang Li, Ke Li, Lei Li, P.L. Li, P.R. Li, Q.Y. Li, T. Li, W.D. Li, W.G. Li, X.L. Li, X.N. Li, X.Q. Li, Z.B. Li, H. Liang, Y.F. Liang, Y.T. Liang, G.R. Liao, L.Z. Liao, J. Libby, C.X. Lin, D.X. Lin, B. Liu, B.J. Liu, C.X. Liu, D. Liu, D.Y. Liu, F.H. Liu, Fang Liu, Feng Liu, H.B. Liu, H.L. Liu, H.M. Liu, Huanhuan Liu, Huihui Liu, J.J. Liu, J.B. Liu, J.Y. Liu, K.Y. Liu, Ke Liu, L.D. Liu, Q. Liu, S.B. Liu, X. Liu, Y.B. Liu, Z.A. Liu, Zhiqing Liu, Y.F. Long, X.C. Lou, H.J. Lu, J.G. Lu, Y. Lu, Y.P. Lu, C.L. Luo, M.X. Luo, P.W. Luo, T. Luo, X.L. Luo, S. Lusso, X.R. Lyu, F.C. Ma, H.L. Ma, L.L. Ma, M.M. Ma, Q.M. Ma, X.N. Ma, X.Y. Ma, Y.M. Ma, F.E. Maas, M. Maggiora, S. Maldaner, Q.A. Malik, A. Mangoni, Y.J. Mao, Z.P. Mao, S. Marcello, Z.X. Meng, J.G. Messchendorp, G. Mezzadri, J. Min, T.J. Min, R.E. Mitchell, X.H. Mo, Y.J. Mo, C. Morales Morales, N.Yu. Muchnoi, H. Muramatsu, A. Mustafa, S. Nakhoul, Y. Nefedov, F. Nerling, I.B. Nikolaev, Z. Ning, S. Nisar, S.L. Niu, X.Y. Niu, S.L. Olsen, Q. Ouyang, S. Pacetti, Y. Pan, M. Papenbrock, P. Patteri, M. Pelizaeus, J. Pellegrino, H.P. Peng, Z.Y. Peng, K. Peters, J. Pettersson, J.L. Ping, R.G. Ping, A. Pitka, R. Poling, V. Prasad, H.R. Qi, M. Qi, T.Y. Qi, S. Qian, C.F. Qiao, N. Qin, X.S. Qin, Z.H. Qin, J.F. Qiu, S.Q. Qu, K.H. Rashid, C.F. Redmer, M. Richter, M. Ripka, A. Rivetti, M. Rolo, G. Rong, Ch. Rosner, M. Rump, A. Sarantsev, M. Savrié, K. Schoenning, W. Shan, X.Y. Shan, M. Shao, C.P. Shen, P.X. Shen, X.Y. Shen, H.Y. Sheng, X. Shi, J.J. Song, W.M. Song, X.Y. Song, S. Sosio, C. Sowa, S. Spataro, F.F. Sui, G.X. Sun, J.F. Sun, L. Sun, S.S. Sun, X.H. Sun, Y.J. Sun, Y.K. Sun, Y.Z. Sun, Z.J. Sun, Z.T. Sun, Y.T. Tan, C.J. Tang, G.Y. Tang, X. Tang, M. Tiemens, B. Tsednee, I. Uman, B. Wang, B.L. Wang, C.W. Wang, D. Wang, D.Y. Wang, Dan Wang, H.H. Wang, K. Wang, L.L. Wang, L.S. Wang, M. Wang, Meng Wang, P. Wang, P.L. Wang, W.P. Wang, X.F. Wang, Y. Wang, Y.F. Wang, Z. Wang, Z.G. Wang, Z.Y. Wang, Zongyuan Wang, T. Weber, D.H. Wei, P. Weidenkaff, S.P. Wen, U. Wiedner, M. Wolke, L.H. Wu, L.J. Wu, Z. Wu, L. Xia, X. Xia, Y. Xia, D. Xiao, Y.J. Xiao, Z.J. Xiao, Y.G. Xie, Y.H. Xie, X.A. Xiong, Q.L. Xiu, G.F. Xu, J.J. Xu, L. Xu, Q.J. Xu, X.P. Xu, F. Yan, L. Yan, W.B. Yan, W.C. Yan, Y.H. Yan, H.J. Yang, H.X. Yang, L. Yang,

R.X. Yang, S.L. Yang, Y.H. Yang, Y.X. Yang, Yifan Yang, Z.Q. Yang, M. Ye, M.H. Ye, J.H. Yin, Z.Y. You, B.X. Yu, C.X. Yu, J.S. Yu, C.Z. Yuan, Y. Yuan, A. Yuncu, A.A. Zafar, Y. Zeng, B.X. Zhang, B.Y. Zhang, C.C. Zhang, D.H. Zhang, H.H. Zhang, H.Y. Zhang, J. Zhang, J.L. Zhang, J.Q. Zhang, J.W. Zhang, J.Y. Zhang, J.Z. Zhang, K. Zhang, L. Zhang, S.F. Zhang, T.J. Zhang, X.Y. Zhang, Y. Zhang, Y.H. Zhang, Y.T. Zhang, Yang Zhang, Yao Zhang, Yu Zhang, Z.H. Zhang, Z.P. Zhang, Z.Y. Zhang, G. Zhao, J.W. Zhao, J.Y. Zhao, J.Z. Zhao, Lei Zhao, Ling Zhao, M.G. Zhao, Q. Zhao, S.J. Zhao, T.C. Zhao, Y.B. Zhao, Z.G. Zhao, A. Zhemchugov, B. Zheng, J.P. Zheng, W.J. Zheng, Y.H. Zheng, B. Zhong, L. Zhou, Q. Zhou, X. Zhou, X.K. Zhou, X.R. Zhou, X.Y. Zhou, Xiaoyu Zhou, Xu Zhou, A.N. Zhu, J. Zhu, K. Zhu, K.J. Zhu, S. Zhu, S.H. Zhu, X.L. Zhu, Y.C. Zhu, Y.S. Zhu, Z.A. Zhu, J. Zhuang, B.S. Zou, J.H. Zou

- **Posted on arXiv:** 27 August 2018
- **Published in Nature Phys.:** 06 May 2019

2.21.1.3 Entanglement and quantum tomography with top quarks at the LHC [386]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

2.21.1.4 Testing Bell Inequalities at the LHC with Top-Quark Pairs [387]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

2.21.1.5 Testing Bell inequalities in Higgs boson decays [388]

- **Authors:** Alan J. Barr
- **Posted on arXiv:** 02 June 2021
- **Published in Physics Letters B 825 (2022) 136866:** 04 January 2022

2.21.1.6 Measurement of Bell-type inequalities and quantum entanglement from Λ-hyperon spin correlations at high energy colliders [389]

- **Authors:** Wenjie Gong, Ganesh Parida, Zhoudunming Tu, Raju Venugopalan
- **Posted on arXiv:** 27 July 2021
- **Published in Phys.Rev.D:** 01 August 2022

2.21.1.7 Quantum tops at the LHC: from entanglement to Bell inequalities [390]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

2.21.1.8 General analysis for observing quantum interference at colliders [391]

- **Authors:** Andrew J. Larkoski
- **Posted on arXiv:** 09 January 2022
- **Published in Phys.Rev.D:** 01 May 2022

2.21.1.9 Quantum SMEFT tomography: Top quark pair production at the LHC [181]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.21.1.10 Quantum information with top quarks in QCD [392]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

2.21.1.11 Improved tests of entanglement and Bell inequalities with LHC tops [393]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
- **Posted on arXiv:** 01 May 2022
- **Published in Eur.Phys.J.C:** 03 August 2022

2.21.1.12 Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs [394]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli
- **Posted on arXiv:** 24 August 2022
- **Published in Eur.Phys.J.C:** 19 February 2023

2.21.1.13 Quantum Discord and Steering in Top Quarks at the LHC [395]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 08 September 2022
- **Published in Phys. Rev. Lett. 130, 221801 (2023):** 31 May 2023

2.21.1.14 Testing entanglement and Bell inequalities in $H \to ZZ$ [396]

- **Authors:** J.A. Aguilar-Saavedra, A. Bernal, J.A. Casas, J.M. Moreno
- **Posted on arXiv:** 27 September 2022
- **Published in Phys.Rev.D:** 01 January 2023

2.21.1.15 Laboratory-frame tests of quantum entanglement in $H \to WW$ [397]

- **Authors:** J.A. Aguilar-Saavedra
- **Posted on arXiv:** 28 September 2022
- **Published in Phys.Rev.D:** 01 April 2023

2.21.1.16 Quantum state tomography, entanglement detection and Bell violation prospects in weak decays of massive particles [398]

- **Authors:** Rachel Ashby-Pickering, Alan J. Barr, Agnieszka Wierzchucka
- **Posted on arXiv:** 28 September 2022
- **Published in JHEP:** 02 May 2023

2.21.1.17 Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders [399]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 01 February 2023
- **Published in Eur.Phys.J.C:** 15 September 2023

2.21.1.18 Bell inequality is violated in $B^0 \to J/\psi K^*(892)^0$ decays [400]

- **Authors:** M. Fabbrichesi, R. Floreanini, E. Gabrielli, L. Marzola
- **Posted on arXiv:** 08 May 2023
- **Published in Phys.Rev.D:** 01 February 2024

2.21.1.19 Probing new physics through entanglement in diboson production [401]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 18 July 2023
- **Published in JHEP:** 01 December 2023

2.21.1.20 Entanglement and Bell inequalities violation in $H \to ZZ$ with anomalous coupling [402]

- **Authors:** Alexander Bernal, Paweł Caban, Jakub Rembieliński
- **Posted on arXiv:** 25 July 2023
- **Published in Eur.Phys.J.C:** 17 November 2023

2.21.1.21 Isolating semi-leptonic $H \to WW^*$decays for Bell inequality tests [403]

- **Authors:** Federica Fabbri, James Howarth, Theo Maurin
- **Posted on arXiv:** 25 July 2023
- **Published in Eur.Phys.J.C:** 09 January 2024

2.21.1.22 New observables for testing Bell inequalities in W boson pair production [404]

- **Authors:** Qi Bi, Qing-Hong Cao, Kun Cheng, Hao Zhang
- **Posted on arXiv:** 27 July 2023
- **Published in Phys.Rev.D:** 01 February 2024

2.21.1.23 Three-Body Entanglement in Particle Decays [405]

- **Authors:** Kazuki Sakurai, Michael Spannowsky
- **Posted on arXiv:** 02 October 2023
- **Published in Phys.Rev.Lett.:** 09 April 2024

2.21.1.24 Quantum entanglement and Bell inequality violation in semi-leptonic top decays [406]

- **Authors:** Tao Han, Matthew Low, Tong Arthur Wu
- **Posted on arXiv:** 26 October 2023
- **Published in JHEP:** 22 July 2024

2.21.1.25 Observation of quantum entanglement with top quarks at the ATLAS detector [407]

- **Authors:** Georges Aad, Braden Keim Abbott, Kira Abeling, Nils Julius Abicht, Haider Abidi, Asmaa Aboulhorma, Halina Abramowicz, Henso Abreu, Yiming Abulaiti, Bobby Samir Acharya, Claire Adam Bourdarios, Leszek Adamczyk, Sagar Addepalli, Matt Addison, Jahred Adelman, Aytul Adiguzel, Tim Adye, A.A. Affolder, Tony Affolder, Yoav Afik, Merve Nazlim Agaras, Jinky Agarwala, Anamika Aggarwal, Catalin Agheorghiesei, Ammara Ahmad, Faig Ahmadov, Waleed Syed Ahmed, Sudha Ahuja, Xiaocong Ai, Giulio Aielli, Arya Aikot, Malak Ait Tamlihat, Brahim Aitbenchikh, Iakov Aizenberg, Melike Akbiyik, Torsten Akesson, Andrei Akimov, Daiya Akiyama, Nilima Nilesh Akolkar, Sina Aktas, Konie Al Khoury, Gian Luigi Alberghi, Justin Albert, Pietro Albicocco, Guillaume Lucas Albouy, Sara Alderweireldt, Zackary Lee Alegria, Martin Aleksa, I.N. Aleksandrov, Calin Alexa, Theodoros Alexopoulos, Fabrizio Alfonsi, Malte Algren, Muhammad Alhroob, Babar Ali, H.M. J. Ali, Shahzad Ali, Samuel William Alibocus, Malik Aliev, Gianluca Alimonti, Wael Alkakhi, Corentin Allaire, Benedict Allbrooke, Julia Frances Allen, Cristian Andres Allendes Flores, Philip Patrick Allport, Alberto Aloisio, Francisco Alonso, Cristiano Alpigiani, Manuel Alvarez Estevez, Adrian Alvarez Fernandez, Mario Alves Cardoso, Mariagrazia Alviggi, Mohamed Aly, Yara Do Amaral Coutinho, Alessandro Ambler, Christoph Amelung, Maximilian Amerl, Christoph Ames, Dante Amidei, Susana Patricia Amor Dos Santos, Kieran Robert Amos, Viktor Ananiev, Christos Anastopoulos, Timothy Robert Andeen, John Kenneth Anders, Stefio Yosse Andrean, Attilio Andreazza, Stylianos Angelidakis, Aaron Angerami, Alexey Anisenkov, Alberto Annovi, Claire Antel, Matthew Thomas Anthony, Egor Antipov, Mario Antonelli, Fabio Anulli, Masato Aoki, Takumi Aoki, Javier Alberto Aparisi Pozo, Marco Aparo, Ludovica Aperio Bella, Christian Appelt, Aram Apyan, Sergio Javier Arbiol Val, Chiara Arcangeletti, Ayana Tamu Arce, Eloisa Arena, Jean-Francois Arguin, Spyros Argyropoulos, Jan-Hendrik Arling, Olivier Arnaez, Hannah Arnold, Giacomo Artoni, Haruka Asada, Kanae Asai, Shoji Asai, Nedaa Alexandra Asbah, Ketevi Adikle Assamagan, Robert Astalos, S. Atashi, Rose Atashi, Ryan Justin Atkin, Markus Julian Atkinson, Hicham Atmani, Prachi Atmasiddha, Kamil Augsten, Silvia Auricchio, Adrien Auriol, Volker Andreas Austrup, Giuseppe Avolio, Konstantinos Axiotis, Georges Azuelos, Dominik Babal, Henri Bachacou, Konstantinos Bachas, Alexander Bachiu, Karl Filip Backman, Anthony Badea, Tamas Marton Baer, Paolo Bagnaia, Marzieh Bahmani, Daniel Bahner, Adam Bailey, Virginia Bailey, John Baines, Luke Baines, Keith Baker, Evelin Bakos, Debottam Bakshi Gupta, Veena Balakrishnan, Rahul Balasubramanian, Evgenii Baldin, Petr Balek, Eric Ballabene, Fabrice Balli, Lisa Marie Baltes, William Keaton Balunas, Johannes Balz, Elzbieta Banas, Marilena Bandieramonte, Anjishnu Bandyopadhyay, Shubham Bansal, Liron Barak, Marawan Barakat, Elisabetta Barberio, Dario Barberis, Marlon Benoit Barbero, Marten Zefanja Barel, Kevin Nicholas Barends, Teresa Barillari, Martin Barisits, Tim Barklow, Petr

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2.21.1.26 Optimizing fictitious states for Bell inequality violation in bipartite qubit systems with applications to the tt^- system [408]

- **Authors:** Kun Cheng, Tao Han, Matthew Low
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2.21.1.27 Probing entanglement and testing Bell inequality violation with $e^+e^-\to\tau^+\tau^-$ at Belle II [409]

- **Authors:** Karl Ehatäht, Marco Fabbrichesi, Luca Marzola, Christian Veelken
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2.21.1.28 Can Bell inequalities be tested via scattering cross-section at colliders ? [410]

- **Authors:** Song Li, Wei Shen, Jin Min Yang
- **Posted on arXiv:** 02 January 2024
- **Published in Eur.Phys.J.C:** 21 November 2024

2.21.1.29 Entanglement Autodistillation from Particle Decays [411]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
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- **Published in Phys.Rev.Lett.:** 10 September 2024

2.21.1.30 Quantum detection of new physics in top-quark pair production at the LHC [412]

- **Authors:** Fabio Maltoni, Claudio Severi, Simone Tentori, Eleni Vryonidou
- **Posted on arXiv:** 16 January 2024
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2.21.1.31 Full quantum tomography of top quark decays [413]

- **Authors:** J.A. Aguilar-Saavedra
- **Posted on arXiv:** 22 February 2024
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2.21.1.32 New physics in spin entanglement [414]

- **Authors:** Mateusz Duch, Alessandro Strumia, Arsenii Titov
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2.21.1.33 On bipartite and tripartite entanglement at present and future particle colliders [415]

- **Authors:** Amir Subba, Rafiqul Rahaman
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2.21.1.34 Observation of quantum entanglement in top quark pair production in proton–proton collisions at $\sqrt{s} = 13$ TeV [416]

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2.21.1.35 Entanglement and Bell nonlocality with bottom-quark pairs at hadron colliders [417]

- **Authors:** Yoav Afik, Yevgeny Kats, Juan Ramón Muñoz de Nova, Abner Soffer, David Uzan
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2.21.1.36 Magic states of top quarks [418]

- **Authors:** Chris D. White, Martin J. White
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2.21.1.37 Bell nonlocality and entanglement in $e^+e^-\to YY^-$ at BESIII [419]

- **Authors:** Sihao Wu, Chen Qian, Qun Wang, Xiao-Rong Zhou
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2.21.1.38 Bell inequality is violated in charmonium decays [420]

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2.21.1.39 Measurements of polarization and spin correlation and observation of entanglement in top quark pairs using lepton + jets events from proton-proton collisions at $\sqrt{s} = 13\,\text{TeV}$ [421]

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2.21.1.40 An improved Bell-CHSH observable for gauge boson pairs [422]

- **Authors:** Radosław Grabarczyk
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2.21.1.41 Measuring quantum discord at the LHC [423]

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2.21.1.42 Bell Inequality Violation of Light Quarks in Dihadron Pair Production at Lepton Colliders [424]

- **Authors:** Kun Cheng, Bin Yan
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2.21.1.43 Quantum information meets high-energy physics: input to the update of the European strategy for particle physics [425]

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2.21.1.44 Quantum Entanglement between gauge boson pairs at a Muon Collider [426]

- **Authors:** Ran Ding, Alim Ruzi, Sitian Qian, Andrew Levin, Youpeng Wu, Qiang Li
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2.21.1.45 Tests of quantum contextuality in particle physics [427]

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2.21.1.46 Test of local realism via entangled $\Lambda\bar{\Lambda}$ system [428]

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2.21.1.47 Precise Measurement of the Λ Electric Dipole Moment through the Entangled Strange Baryon-Antibaryon System [429]

- **Authors:** Medina Ablikim, Mikhail N. Achasov, Patrik Adlarsson Adlarson, Xiaocong Ai, Riccardo Aliberti, Antonio Amoroso, Qi An, Yu Bai, Olga Bakina, Yong Ban, Hao-Ran Bao, Varvara Batozskaya, Khurelbaatar Begzsuren, Niklaus Berger, Marcin Berlowski, Monica Bertani, Diego Bettoni, Fabrizio Bianchi, Edoardo Bianco, Alberto Bortone, Igor Boyko, Roy Briere, Anja Brueggemann, Hao Cai, Minghang Cai, Xiao Cai, Alessandro Calcaterra, Guofu Cao, Ning Cao, Serkant Ali Cetin, Xinyu Chai, Jinfan Chang, Guorong Che, Yuzhi Che, Chunhui Chen, Chao Chen, Gang Chen, Hesheng Chen, Hongyang Chen, Mali Chen, Shenjian Chen, Silu Chen, Shaomin Chen, Tong Chen, Xurong Chen, Xiaoting Chen, Xinyu Chen, Yuanbo Chen, Yongqing Chen, Yongqing Chen, Zhuo Chen, Zhuojun Chen, Zikang Chen, Sookyung Choi, Xiao Chu, Gianluigi Cibinetto, Fabio Cossio, Jozie Cottee-Meldrum, Jiajia Cui, Hongliang Dai, JianPing Dai, Alaa Dbeyssi, Remco Emiel de Boer, Dmitry Dedovich, Chuangqi Deng, Ziyan Deng, Achim Denig, Igor Denysenko, Marco Destefanis, Francesca De Mori, Biao

Ding, Xiaoxuan Ding, Yong Ding, Yi Ding, Yuxin Ding, Jing Dong, Liaoyuan Dong, Mingyi Dong, Xiang Dong, Mengchuan Du, Shuxian Du, Shaoxu Du, Yaoyu Duan, Prokhor Egorov, Gaofeng Fan, Junjie Fan, Yuhan Fan, Jian Fang, Jin Fang, Shuangshi Fang, Wenxing Fang, Yaquan Fang, Riccardo Farinelli, Luciano Fava, Florian Feldbauer, Giulietto Felici, Changqing Feng, Junhua Feng, Lin Feng, Qianxi Feng, Yutong Feng, Miriam Fritsch, Chengdong Fu, Jin-Lin Fu, Yiwei Fu, Han Gao, Xinbo Gao, Yang Gao, Yuanning Gao, Yunong Gao, Yangyang Gao, Sara Garbolino, Isabella Garzia, Panting Ge, Zhenwu Ge, Cong Geng, Evelina Mihova Gersabeck, Alexander Gilman, Klaus Goetzen, Jiading Gong, Li Gong, Wenxuan Gong, Wolfgang Gradl, Stefano Gramigna, Michela Greco, Minhao Gu, Yunting Gu, Chunyi Guan, Aiqiang Guo, Libo Guo, Mengjiao Guo, Rupan Guo, Yuping Guo, Alexey Guskov, Jonathan Gutierrez, Kun-Lin Han, Tingting Han, Frederike Hanisch, Kedi Hao, Xiqing Hao, Frederick A. Harris, Kaikai He, Kanglin He, Fritz-Herbert Heinsius, Christopher Heinz, Yuekun Heng, Christoph Herold, Pengcheng Hong, Guoyi Hou, Xiantao Hou, Yingrui Hou, Zhilong Hou, Haiming Hu, Jifeng Hu, Qipeng Hu, Shengliang Hu, Tao Hu, Yu Hu, Zhongmin Hu, Guangshun Huang, Kaixuan Huang, Linqin Huang, Pan Huang, Xingtao Huang, Yanping Huang, Yongsheng Huang, Talab Hussain, Nils H 眉 sken H\"usken, Nikolai in der Wiesche, Joshua Jackson, Quan Ji, Qingping Ji, Wang Ji, Xiaobin Ji, Xiaolu Ji, Yuyao Ji, Zekun Jia, Di Jiang, Houbing Jiang, Peicheng Jiang, Sijing Jiang, Tingjun Jiang, Xiaoshan Jiang, Yi Jiang, Jianbin Jiao, Junkun Jiao, Zheng Jiao, Shan Jin, Yi Jin, Maoqiang Jing, Xin-Mei Jing, Tord Johansson, Sonia Kabana, Nasser Kalantar-Nayestanaki, Xiaolin Kang, Xiaoshen Kang, Myroslav Kavatsyuk, Bai-Cian Ke, Vladimir Khachatryan, Alfons Khoukaz, Ryuta Kiuchi, Onur Bugra Kolcu, Bertram Kopf, Meike Kuessner, Xian Kui, Neeraj Kumar, Andrzej Kupsc, Wolfgang Kuehn K\"uhn, Qiang Lan, Wenning Lan, Tiantian Lei, Max Lellmann, Thomas Lenz, Cheng Li, Cui Li, Cong Li, Chunhua Li, Chunkai Li, Demin Li, Fei Li, Gang Li, Haibo Li, Huijing Li, Hengne Li, Hui Li, Jiarong Li, Jingshu Li, Ke Li, Kailu Li, Kailu Li, njian Li, Lei Li, Minghao Li, Mingrun Li, Pei-an Li, Pei-Rong Li, Qiming Li, Qixin Li, Ran Li, Suxian Li, Teng Li, Tianyou Li, Weidong Li, Weiguo Li, Xu Li, Xuhong Li, Xiaoling Li, Xiaoyu Li, Xuze Li, Yang Li, Yangu Li, Yanpeng Li, Zhijun Li, Ziyang Li, Hao Liang, Yongfei Liang, Yutie Liang, Guangrui Liao, Libo Liao, Minghua Liao, Yipu Liao, Jim Libby, Ayut Limphirat, Changcheng Lin, Dexu Lin, quan Shao Lin, Tao Lin, Beijiang Liu, Baoxin Liu, Cheng Liu, Chunxiu Liu, Fang Liu, Fuhu Liu, Feng Liu, Guoming Liu, Hao Liu, Hongbang Liu, Huanhuan Liu, Huaimin Liu, Huihui Liu, Jianbei Liu, Jiajia Liu, Kai Liu, Kun Liu, Kuiyong Liu, Ke Liu, Liangchen Liu, Lu Liu, Meihong Liu, Peilian Liu, Qian Liu, Shubin Liu, Tong Liu, Weike Liu, Weimin Liu, Wanting Liu, Xiang Liu, Xin Liu, Xiaokang Liu, Xueyin Liu, Ying Liu, Yi Liu, Yuan Liu, Yu-Bin Liu, Zhenan Liu, Zongde Liu, Zhiqing Liu, Xinchou Lou, Feixiang Lu, Hai-jiang Lu, Junguang Lu, Xiaoling Lu, Yu Lu, Yanghong Lu, Yunpeng Lu, Zehui Lu, Chenglin Luo, Jiarui Luo, Jiashun Luo, Mingxing Luo, Tao Luo, Xiaolan Luo, Zaiyu Lv, Xiao-Rui Lyu, Yifeng Lyu, YunHe Lyu, Fengcai Ma, Hailong Ma, Junli Ma, Lianliang

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- **Posted on arXiv:** 23 June 2025

2.21.1.48 Colliders are not testing locality via Bell's inequality nor providing an unconditional proof of entanglement [430]

- **Authors:** Steven A. Abel, Herbi K. Dreiner, Rhitaja Sengupta, Lorenzo Ubaldi
- **Posted on arXiv:** 21 July 2025
- **Published in JHEP:** 07 August 2026

2.21.2 Quantum Measurement and Tomography

2.21.2.1 The magic of top quarks [431]

- **Authors:** Chris D. White, Martin J. White
- **Posted on arXiv:** 10 December 2024
- **Published in nan:** 29 January 2026

2.21.2.2 Probing new physics in the top sector using quantum information [432]

- **Authors:** Rafael Aoude, Hannah Banks, Chris D. White, Martin J. White
- **Posted on arXiv:** 18 May 2025
- **Published in Phys.Rev.D 113 (2026) 11, 115066:** 01 June 2026

2.21.2.3 Spin versus nonstabilizerness in gluon and graviton scattering [433]

- **Authors:** John Gargalionis, Nathan Moynihan, Sokratis Trifinopoulos, Ewan N.V. Wallace, Chris D. White, Martin J. White
- **Posted on arXiv:** 20 August 2025
- **Published in Phys.Rev.D:** 01 January 2026

2.21.2.4 Z-boson quantum tomography at next-to-leading order [434]

- **Authors:** Morgan Del Gratta, Federica Fabbri, Michele Grossi, Fabio Maltoni, Davide Pagani, Giovanni Pelliccioli, Alessandro Vicini
- **Posted on arXiv:** 24 September 2025
- **Published in JHEP:** 04 February 2026

2.21.2.5 Automated computation of spin-density matrices and quantum observables for collider physics [435]

- **Authors:** Valentin Durupt, Fabio Maltoni, Olivier Mattelaer
- **Posted on arXiv:** 20 October 2025
- **Published in JHEP:** 14 April 2026

2.21.3 Crosslists

2.21.3.1 Quantum entanglement and Bell inequality violation at colliders [10]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

2.22 Quantum Correlations in Scattering Amplitudes

2.22.1 Quantum Entanglement and Bell Inequalities

2.22.1.1 Entanglement Suppression and Emergent Symmetries of Strong Interactions [436]

- **Authors:** Silas R. Beane, David B. Kaplan, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 07 December 2018
- **Published in Phys. Rev. Lett. 122, 102001 (2019):** 16 March 2019

2.22.1.2 Geometry and entanglement in the scattering matrix [437]

- **Authors:** Silas R. Beane, Roland C. Farrell
- **Posted on arXiv:** 02 November 2020
- **Published in Annals Phys.:** 16 August 2021

2.22.1.3 Entanglement minimization in hadronic scattering with pions [438]

- **Authors:** Silas R. Beane, Roland C. Farrell, Mira Varma
- **Posted on arXiv:** 02 August 2021
- **Published in Int.J.Mod.Phys.A:** 11 November 2021

2.22.1.4 Tree-level entanglement in quantum electrodynamics [439]

- **Authors:** Samuel Fedida, Alessio Serafini
- **Posted on arXiv:** 03 September 2022
- **Published in Phys. Rev. D. 107:116007, Jun 2023:** 01 June 2023

2.22.1.5 Positivity in Amplitudes and Quantum Entanglement [440]

- **Authors:** Rafael Aoude, Gilly Elor, Grant N. Remmen, Olcyr Sumensari
- **Posted on arXiv:** 26 February 2024
- **Published in Fortschr. Phys. 74 (2026) e70113:** 01 May 2026

2.22.1.6 Area law for entanglement entropy in particle scattering [441]

- **Authors:** Ian Low, Zhewei Yin
- **Posted on arXiv:** 13 May 2024
- **Published in Phys.Rev.D:** 06 March 2026

2.22.1.7 Qubit entanglement from forward scattering [442]

- **Authors:** Kamila Kowalska, Enrico Maria Sessolo
- **Posted on arXiv:** 05 October 2025
- **Published in JHEP:** 01 April 2026

2.22.2 Quantum Measurement and Tomography

2.22.2.1 Quantum magic in quantum electrodynamics [443]

- **Authors:** Qiaofeng Liu, Ian Low, Zhewei Yin
- **Posted on arXiv:** 04 March 2025
- **Published in Phys.Rev.D:** 01 March 2026

2.23 Track Reconstruction

2.23.1 Quantum Algorithms - Grover's Search Algorithm

2.23.1.1 Quantum Associative Memory in HEP Track Pattern Recognition [444]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

2.23.1.2 Quantum speedup for track reconstruction in particle accelerators [445]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

2.23.1.3 Quantum pathways for charged track finding in high-energy collisions [446]

- **Authors:** Christopher Brown, Michael Spannowsky, Alexander Tapper, Simon Williams, Ioannis Xiotidis
- **Posted on arXiv:** 01 November 2023
- **Published in Front. Artif. Intell. 7:1339785 (2024):** 07 May 2024

2.23.2 Quantum Algorithms - Harrow-Hassidim-Lloyd Algorithm

2.23.2.1 A quantum algorithm for track reconstruction in the LHCb vertex detector [447]

- **Authors:** Davide Nicotra, Miriam Lucio Martinez, Jacco Andreas de Vries, Marcel Merk, Kurt Driessens, Ronald Leonard Westra, Domenica Dibenedetto, Daniel Hugo Cámpora Pérez
- **Posted on arXiv:** 01 August 2023
- **Published in JINST:** 29 November 2023

2.23.3 Quantum Computing Paradigms - Quantum Annealing

2.23.3.1 A pattern recognition algorithm for quantum annealers [448]

- **Authors:** Frédéric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
- **Published in Comput.Softw.Big Sci.:** 09 December 2019

2.23.3.2 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [449]

- **Authors:** Souvik Das, Andrew J. Wildridge, Andreas Jung
- **Posted on arXiv:** 21 March 2019

2.23.3.3 Charged particle tracking with quantum annealing optimization [450]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

2.23.3.4 Quantum annealing algorithms for track pattern recognition [451]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

2.23.3.5 Particle track classification using quantum associative memory [452]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

2.23.3.6 Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders [453]

- **Authors:** Hideki Okawa, Qing-Guo Zeng, Xian-Zhe Tao, Man-Hong Yung
- **Posted on arXiv:** 22 February 2024
- **Published in Comput.Softw.Big Sci.:** 28 August 2024

2.23.4 Quantum Machine Learning - Supervised Methods

2.23.4.1 Reconstructing charged particle track segments with a quantum-enhanced support vector machine [454]

- **Authors:** Philippa Duckett, Gabriel Facini, Marcin Jastrzebski, Sarah Malik, Tim Scanlon, Sebastien Rettie
- **Posted on arXiv:** 14 December 2022
- **Published in Phys.Rev.D:** 01 March 2024

2.23.5 Quantum Machine Learning - Variational Quantum Algorithms

2.23.5.1 Quantum convolutional neural networks for high energy physics data analysis [45]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.23.5.2 Hybrid quantum classical graph neural networks for particle track reconstruction [455]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

2.23.5.3 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [456]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in nan:** 2023

2.23.5.4 Particle track reconstruction with noisy intermediate-scale quantum computers [457]

- **Authors:** Tim Schwägerl, Cigdem Issever, Karl Jansen, Teng Jian Khoo, Stefan Kühn, Cenk Tüysüz, Hannsjörg Weber
- **Posted on arXiv:** 23 March 2023
- **Published in J.Phys.Conf.Ser.:** 2026

2.23.5.5 Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm [458]

- **Authors:** Hideki Okawa
- **Posted on arXiv:** 16 October 2023

2.23.6 Crosslists

2.23.6.1 Hybrid Quantum-Classical Graph Convolutional Network [41]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

2.23.6.2 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.23.6.3 Quantum Computing Applications in Future Colliders [5]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.23.6.4 Quantum Systems for Enhanced High Energy Particle Physics Detectors [164]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

2.23.6.5 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3 Quantum Information Science in Nuclear and Particle Physics

3.1 Reviews and Whitepapers and Proceedings

3.1.1 Reviews

3.1.1.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

3.1.1.2 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

3.1.1.3 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.1.1.4 Quantum Sensors with Matter Waves for GW Observation [4]

- **Authors:** Andrea Bertoldi, Philippe Bouyer, Benjamin Canuel
- **Posted on arXiv:** 27 October 2020
- **Published in nan:** 2022

3.1.1.5 Quantum Computing Applications in Future Colliders [5]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.1.1.6 Many-Body Collective Neutrino Oscillations: Recent Developments [6]

- **Authors:** Amol V. Patwardhan, Michael J. Cervia, Ermal Rrapaj, Pooja Siwach, A.B. Balantekin
- **Posted on arXiv:** 31 December 2022
- **Published in nan:** 2023

3.1.1.7 Review on Quantum Computing for Lattice Field Theory [7]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in nan:** 01 February 2023

3.1.1.8 Quantum information and quantum simulation of neutrino physics [8]

- **Authors:** A.B. Balantekin, Michael J. Cervia, Amol V. Patwardhan, Ermal Rrapaj, Pooja Siwach
- **Posted on arXiv:** 01 May 2023
- **Published in Eur.Phys.J.A:** 17 August 2023

3.1.1.9 Machine learning for anomaly detection in particle physics [9]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klaeboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

3.1.1.10 Quantum entanglement and Bell inequality violation at colliders [10]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

3.1.1.11 Quantum Complexity and New Directions in Nuclear Physics and High-Energy Physics Phenomenology [11]

- **Authors:** Caroline E.P. Robin, Martin J. Savage
- **Posted on arXiv:** 29 April 2026

3.1.2 Whitepapers and Proceedings

3.1.2.1 Quantum Sensing for High Energy Physics [12]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs,

Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.1.2.2 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar

- **Posted on arXiv:** 13 March 2019

3.1.2.3 AEDGE: Atomic Experiment for Dark Matter and Gravity Exploration in Space [14]

- **Authors:** Yousef Abou El-Neaj, Cristiano Alpigiani, Sana Amairi-Pyka, Henrique Araújo, Antun Balaž, Angelo Bassi, Lars Bathe-Peters, Baptiste Battelier, Aleksandar Belić, Elliot Bentine, José Bernabeu, Andrea Bertoldi, Robert Bingham, Diego Blas, Vasiliki Bolpasi, Kai Bongs, Sougato Bose, Philippe Bouyer, Themis Bowcock, William Bowden, Oliver Buchmueller, Clare Burrage, Xavier Calmet, Benjamin Canuel, Laurentiu-Ioan Caramete, Andrew Carroll, Giancarlo Cella, Vassilis Charmandaris, Swapan Chattopadhyay, Xuzong Chen, Maria Luisa Chiofalo, Jonathon Coleman, Joseph Cotter, Yanou Cui, Andrei Derevianko, Albert De Roeck, Goran S. Djordjevic, Peter Dornan, Michael Doser, Ioannis Drougkakis, Jacob Dunningham, Ioana Dutan, Sajan Easo, Gedminas Elertas, John Ellis, Mai El Sawy, Farida Fassi, Daniel Felea, Chen-Hao Feng, Robert Flack, Chris Foot, Ivette Fuentes, Naceur Gaaloul, Alexandre Gauguet, Remi Geiger, Valerie Gibson, Gian Giudice, Jon Goldwin, Oleg Grachov, Peter W. Graham, Dario Grasso, Maurits van der Grinten, Mustafa Gündogan, Martin G. Haehnelt, Tiffany Harte, Aurelien Hees, Richard Hobson, Jason Hogan, Bodil Holst, Michael

Holynski, Mark Kasevich, Bradley J. Kavanagh, Wolf von Klitzing, Tim Kovachy, Benjamin Krikler, Markus Krutzik, Marek Lewicki, Yu-Hung Lien, Miaoyuan Liu, Giuseppe Gaetano Luciano, Alain Magnon, Mohammed Attia Mahmoud, Sarah Malik, Christopher McCabe, Jeremiah Mitchell, Julia Pahl, Debapriya Pal, Saurabh Pandey, Dimitris Papazoglou, Mauro Paternostro, Bjoern Penning, Achim Peters, Marco Prevedelli, Vishnupriya Puthiya-Veettil, John Quenby, Ernst Rasel, Sean Ravenhall, Jack Ringwood, Albert Roura, Dylan Sabulsky, Muhammed Sameed, Ben Sauer, Stefan Alaric Schäffer, Stephan Schiller, Vladimir Schkolnik, Dennis Schlippert, Christian Schubert, Haifa Rejeb Sfar, Armin Shayeghi, Ian Shipsey, Carla Signorini, Yeshpal Singh, Marcelle Soares-Santos, Fiodor Sorrentino, Timothy Sumner, Konstantinos Tassis, Silvia Tentindo, Guglielmo Maria Tino, Jonathan N. Tinsley, James Unwin, Tristan Valenzuela, Georgios Vasilakis, Ville Vaskonen, Christian Vogt, Alex Webber-Date, Andŕ e} Wenzlawski, Patrick Windpassinger, Marian Woltmann, Efe Yazgan, Ming-Sheng Zhan, Xinhao Zou, Jure Zupan

- **Posted on arXiv:** 02 August 2019
- **Published in EPJ Quant.Technol.:** 04 March 2020

3.1.2.4 Mechanical quantum sensing in the search for dark matter [15]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

3.1.2.5 Standard model physics and the digital quantum revolution: thoughts about the interface [16]

- **Authors:** Natalie Klco, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 10 July 2021
- **Published in Rept.Prog.Phys.:** 19 May 2022

3.1.2.6 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [17]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

3.1.2.7 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [18]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.1.2.8 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [19]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.1.2.9 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [20]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.1.2.10 Snowmass2021 Cosmic Frontier White Paper: Future Gravitational-Wave Detector Facilities [21]

- **Authors:** Stefan W. Ballmer, Rana Adhikari, Leonardo Badurina, Duncan A. Brown, Swapan Chattopadhyay, Matthew Evans, Peter Fritschel, Evan Hall, Jason M. Hogan, Karan Jani, Tim Kovachy, Kevin Kuns, Ariel Schwartzman, Daniel Sigg, Bram Slagmolen, Salvatore Vitale, Christopher Wipf
- **Posted on arXiv:** 15 March 2022

3.1.2.11 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo

- **Posted on arXiv:** 15 March 2022

3.1.2.12 New Horizons: Scalar and Vector Ultralight Dark Matter [23]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.1.2.13 Quantum Networks for High Energy Physics [24]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.1.2.14 Quantum Simulation for High-Energy Physics [25]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti

- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

3.1.2.15 Quantum computing hardware for HEP algorithms and sensing [26]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti
- **Posted on arXiv:** 18 April 2022

3.1.2.16 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [27]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.1.2.17 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne,

Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao

- **Posted on arXiv:** 28 February 2023

3.1.2.18 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023

- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.1.2.19 Quantum Sensors for High Energy Physics [30]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

3.2 Quantum Algorithms - Grover's Search Algorithm

3.2.1 Beyond the Standard Model

3.2.1.1 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [37]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker

- **Posted on arXiv:** 01 October 2020

3.2.2 Event Classification

3.2.2.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [183]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

3.2.3 Event Generation

3.2.3.1 Quantum algorithm for Feynman loop integrals [200]

- **Authors:** Selomit Ramírez-Uribe, Andrés E. Rentería-Olivo, Germán Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

3.2.3.2 Loop Feynman integration on a quantum computer [201]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

3.2.4 Jet Reconstruction

3.2.4.1 Quantum Algorithms for Jet Clustering [223]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.2.4.2 Quantum Algorithms in Particle Physics [215]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.2.5 Track Reconstruction

3.2.5.1 Quantum Associative Memory in HEP Track Pattern Recognition [444]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

3.2.5.2 Quantum speedup for track reconstruction in particle accelerators [445]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

3.2.5.3 Quantum pathways for charged track finding in high-energy collisions [446]

- **Authors:** Christopher Brown, Michael Spannowsky, Alexander Tapper, Simon Williams, Ioannis Xiotidis
- **Posted on arXiv:** 01 November 2023
- **Published in Front. Artif. Intell. 7:1339785 (2024):** 07 May 2024

3.2.6 Crosslists

3.2.6.1 Quantum clustering and jet reconstruction at the LHC [230]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

3.3 Quantum Algorithms - Harrow-Hassidim-Lloyd Algorithm

3.3.1 Track Reconstruction

3.3.1.1 A quantum algorithm for track reconstruction in the LHCb vertex detector [447]

- **Authors:** Davide Nicotra, Miriam Lucio Martinez, Jacco Andreas de Vries, Marcel Merk, Kurt Driessens, Ronald Leonard Westra, Domenica Dibenedetto, Daniel Hugo Cámpora Pérez
- **Posted on arXiv:** 01 August 2023
- **Published in JINST:** 29 November 2023

3.3.2 Crosslists

3.3.2.1 Quantum-centric Supercomputing for Physics Research [170]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.4 Quantum Algorithms - Quantum Phase Estimation

3.4.1 Nuclear Many-Body Physics - Nuclear Reactions and Response

3.4.1.1 Dynamic linear response quantum algorithm [353]

- **Authors:** Alessandro Roggero, Joseph Carlson
- **Posted on arXiv:** 04 April 2018
- **Published in Phys. Rev. C 100, 034610 (2019):** 14 September 2019

3.4.2 Crosslists

3.4.2.1 Quantum Computing for Neutrino-Nucleus Scattering [179]

- **Authors:** Alessandro Roggero, Andy C.Y. Li, Joseph Carlson, Rajan Gupta, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019
- **Published in Phys. Rev. D 101, 074038 (2020):** 28 April 2020

3.5 Quantum Algorithms - Quantum Simulations

3.5.1 Beyond the Standard Model

3.5.1.1 A Quantum Computational Determination of the Weak Mixing Angle in the Standard Model [38]

- **Authors:** Qiaofeng Liu, Ian Low, Zhewei Yin
- **Posted on arXiv:** 22 September 2025

3.5.2 Dark Matter - Particle-like Dark Matter

3.5.2.1 Quantum simulations of dark sector showers [48]

- **Authors:** So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 26 April 2022
- **Published in Phys.Lett.B:** 26 September 2022

3.5.3 Effective Field Theories

3.5.3.1 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [171]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

3.5.3.2 Quantum Algorithms for Simulating Nuclear Effective Field Theories [172]

- **Authors:** James D. Watson, Jacob Bringewatt, Alexander F. Shaw, Andrew M. Childs, Alexey V. Gorshkov, Zohreh Davoudi
- **Posted on arXiv:** 08 December 2023

3.5.4 Event Generation

3.5.4.1 Quantum Algorithm for High Energy Physics Simulations [202]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

3.5.4.2 Towards a quantum computing algorithm for helicity amplitudes and parton showers [203]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

3.5.4.3 Improving quantum simulation efficiency of final state radiation with dynamic quantum circuits [204]

- **Authors:** Plato Deliyannis, James Sud, Diana Chamaki, Zoë Webb-Mack, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 18 March 2022
- **Published in Physical Review D 106, 036007 (2022):** 01 August 2022

3.5.4.4 Collider events on a quantum computer [205]

- **Authors:** Gösta Gustafson, Stefan Prestel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 21 July 2022
- **Published in JHEP:** 07 November 2022

3.5.4.5 Quantum parton shower with kinematics [206]

- **Authors:** Christian W. Bauer, So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 30 October 2023
- **Published in Phys.Rev.A:** 26 March 2024

3.5.5 Jet Reconstruction

3.5.5.1 Medium induced jet broadening in a quantum computer [224]

- **Authors:** João Barata, Xiaojian Du, Meijian Li, Wenyang Qian, Carlos A. Salgado
- **Posted on arXiv:** 13 August 2022
- **Published in Phys. Rev. D 106, 074013 (2022):** 01 October 2022

3.5.6 Lattice Field Theories - Scalar/Fermion Theories

3.5.6.1 Quantum Algorithms for Quantum Field Theories [233]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** 2012
- **Published in Science:** 2012

3.5.6.2 Quantum Computation of Scattering in Scalar Quantum Field Theories [234]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

3.5.6.3 Quantum Algorithms for Fermionic Quantum Field Theories [235]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

3.5.6.4 Digitization of scalar fields for quantum computing [236]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

3.5.6.5 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [237]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

3.5.6.6 Real-Time Scattering on Quantum Computers via Hamiltonian Truncation [238]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams, Matthew Wingate
- **Posted on arXiv:** 06 May 2025

3.5.7 Lattice Field Theories - Gauge Theories

3.5.7.1 Simulating lattice gauge theories on a quantum computer [243]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

3.5.7.2 Cold-Atom Quantum Simulator for SU(2) Yang-Mills Lattice Gauge Theory [244]

- **Authors:** Erez Zohar, J. Ignacio Cirac, Benni Reznik
- **Posted on arXiv:** November 2012
- **Published in Phys.Rev.Lett.:** 21 March 2013

3.5.7.3 Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories [245]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 07 May 2013
- **Published in Annalen Phys.:** 2013

3.5.7.4 Towards Quantum Simulating QCD [246]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

3.5.7.5 Quantum Simulations of Lattice Gauge Theories using Ultracold Atoms in Optical Lattices [247]

- **Authors:** Erez Zohar, J. Ignacio Cirac, Benni Reznik
- **Posted on arXiv:** 08 March 2015
- **Published in Rept.Prog.Phys.:** 18 December 2015

3.5.7.6 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [248]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

3.5.7.7 Gauss's law, duality, and the Hamiltonian formulation of U(1) lattice gauge theory [249]

- **Authors:** David B. Kaplan, Jesse R. Stryker
- **Posted on arXiv:** 22 June 2018
- **Published in Phys. Rev. D 102, 094515 (2020):** 20 November 2020

3.5.7.8 Oracles for Gauss's law on digital quantum computers [250]

- **Authors:** Jesse R. Stryker
- **Posted on arXiv:** 04 December 2018
- **Published in Phys. Rev. A 99, 042301 (2019):** 02 April 2019

3.5.7.9 General Methods for Digital Quantum Simulation of Gauge Theories [251]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

3.5.7.10 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [252]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

3.5.7.11 Simulating Lattice Gauge Theories within Quantum Technologies [253]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.5.7.12 Loop, string, and hadron dynamics in SU(2) Hamiltonian lattice gauge theories [254]

- **Authors:** Indrakshi Raychowdhury, Jesse R. Stryker
- **Posted on arXiv:** 12 December 2019
- **Published in Phys. Rev. D 101, 114502 (2020):** 09 June 2020

3.5.7.13 Observation of gauge invariance in a 71-site Bose–Hubbard quantum simulator [255]

- **Authors:** Bing Yang, Hui Sun, Robert Ott, Han-Yi Wang, Torsten V. Zache, Jad C. Halimeh, Zhen-Sheng Yuan, Philipp Hauke, Jian-Wei Pan
- **Posted on arXiv:** 19 March 2020
- **Published in Nature:** 2020

3.5.7.14 Toward scalable simulations of Lattice Gauge Theories on quantum computers [256]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

3.5.7.15 Role of boundary conditions in quantum computations of scattering observables [257]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

3.5.7.16 Quantum simulation of gauge theory via orbifold lattice [258]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

3.5.7.17 Trailhead for quantum simulation of SU(3) Yang-Mills lattice gauge theory in the local multiplet basis [259]

- **Authors:** Anthony Ciavarella, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 25 January 2021
- **Published in Phys. Rev. D 103, 094501 (2021):** 05 May 2021

3.5.7.18 Lattice renormalization of quantum simulations [260]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

3.5.7.19 Lattice Quantum Chromodynamics and Electrodynamics on a Universal Quantum Computer [261]

- **Authors:** Angus Kan, Yunseong Nam
- **Posted on arXiv:** 27 July 2021

3.5.7.20 Thermalization dynamics of a gauge theory on a quantum simulator [262]

- **Authors:** Zhao-Yu Zhou, Guo-Xian Su, Jad C. Halimeh, Robert Ott, Hui Sun, Philipp Hauke, Bing Yang, Zhen-Sheng Yuan, Jürgen Berges, Jian-Wei Pan
- **Posted on arXiv:** 28 July 2021
- **Published in Science 377, 311-314 (2022):** 15 July 2022

3.5.7.21 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [263]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

3.5.7.22 Preparations for quantum simulations of quantum chromodynamics in $1+1$ dimensions. I. Axial gauge [264]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

3.5.7.23 Simulating one-dimensional quantum chromodynamics on a quantum computer: Real-time evolutions of tetra- and pentaquarks [265]

- **Authors:** Yasar Y. Atas, Jan F. Haase, Jinglei Zhang, Victor Wei, Sieglinde M.-L. Pfaendler, Randy Lewis, Christine A. Muschik
- **Posted on arXiv:** 07 July 2022
- **Published in Phys.Rev.Res.:** 13 September 2023

3.5.7.24 Preparations for quantum simulations of quantum chromodynamics in $1+1$ dimensions. II. Single-baryon β-decay in real time [266]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 22 September 2022
- **Published in Phys. Rev. D 107, 054513 (2023):** 01 March 2023

3.5.7.25 Quantum Computation of Dynamical Quantum Phase Transitions and Entanglement Tomography in a Lattice Gauge Theory [267]

- **Authors:** Niklas Mueller, Joseph A. Carolan, Andrew Connelly, Zohreh Davoudi, Eugene F. Dumitrescu, Kübra Yeter-Aydeniz
- **Posted on arXiv:** 06 October 2022
- **Published in PRX Quantum:** 01 August 2023

3.5.7.26 Measurement-based quantum simulation of Abelian lattice gauge theories [268]

- **Authors:** Hiroki Sukeno, Takuya Okuda
- **Posted on arXiv:** 19 October 2022
- **Published in SciPost Phys. 14, 129 (2023):** 25 May 2023

3.5.7.27 Loop-string-hadron formulation of an SU(3) gauge theory with dynamical quarks [269]

- **Authors:** Saurabh V. Kadam, Indrakshi Raychowdhury, Jesse R. Stryker
- **Posted on arXiv:** 08 December 2022
- **Published in Phys. Rev. D 107, 094513 (2023):** 01 May 2023

3.5.7.28 General quantum algorithms for Hamiltonian simulation with applications to a non-Abelian lattice gauge theory [270]

- **Authors:** Zohreh Davoudi, Alexander F. Shaw, Jesse R. Stryker
- **Posted on arXiv:** 28 December 2022
- **Published in Quantum 7, 1213 (2023):** 20 December 2023

3.5.7.29 Simulating $\mathbb{Z}_2$ lattice gauge theory on a quantum computer [271]

- **Authors:** Clement Charles, Erik J. Gustafson, Elizabeth Hardt, Florian Herren, Norman Hogan, Henry Lamm, Sara Starecheski, Ruth S. Van de Water, Michael L. Wagman
- **Posted on arXiv:** 03 May 2023
- **Published in Phys.Rev.E:** 26 January 2024

3.5.7.30 Observation of microscopic confinement dynamics by a tunable topological θ-angle [272]

- **Authors:** Wei-Yong Zhang, Ying Liu, Yanting Cheng, Ming-Gen He, Han-Yi Wang, Tian-Yi Wang, Zi-Hang Zhu, Guo-Xian Su, Zhao-Yu Zhou, Yong-Guang Zheng, Hui Sun, Bing Yang, Philipp Hauke, Wei Zheng, Jad C. Halimeh, Zhen-Sheng Yuan, Jian-Wei Pan
- **Posted on arXiv:** 20 June 2023
- **Published in Nature Phys.:** 20 December 2024

3.5.7.31 Quantum simulation of fundamental particles and forces [273]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 April 2024
- **Published in Nature Rev.Phys.:** 21 June 2023

3.5.7.32 Toward QCD on quantum computer: orbifold lattice approach [274]

- **Authors:** Georg Bergner, Masanori Hanada, Enrico Rinaldi, Andreas Schafer
- **Posted on arXiv:** 22 January 2024
- **Published in JHEP:** 20 May 2024

3.5.7.33 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [275]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

3.5.7.34 Quantum Simulation of SU(3) Lattice Yang-Mills Theory at Leading Order in Large-N_c Expansion [276]

- **Authors:** Anthony N. Ciavarella, Christian W. Bauer
- **Posted on arXiv:** 15 February 2024
- **Published in Phys.Rev.Lett.:** 09 September 2024

3.5.7.35 Loop-string-hadron approach to SU(3) lattice Yang-Mills theory: Hilbert space of a trivalent vertex [277]

- **Authors:** Saurabh V. Kadam, Aahiri Naskar, Indrakshi Raychowdhury, Jesse R. Stryker
- **Posted on arXiv:** 27 July 2024
- **Published in Phys. Rev. D 111, 074516 (2025):** 01 April 2025

3.5.7.36 Quantum computing universal thermalization dynamics in a $(2+1)$D Lattice Gauge Theory [278]

- **Authors:** Niklas Mueller, Tianyi Wang, Or Katz, Zohreh Davoudi, Marko Cetina
- **Posted on arXiv:** 31 July 2024
- **Published in Nature Commun.:** 01 July 2025

3.5.7.37 Visualizing dynamics of charges and strings in (2 + 1)D lattice gauge theories [279]

- **Authors:** Tyler A. Cochran, Bernhard Jobst, Eliott Rosenberg, Yuri D. Lensky, Gaurav Gyawali, Norhan Eassa, Melissa Will, Murphy Yuezhen Niu, William D. Oliver, Aaron Szasz, Dmitry Abanin, Rajeev Acharya, Laleh Aghababaie Beni, Trond I. Andersen, Markus Ansmann, Frank Arute, Kunal Arya, Abraham Asfaw, Juan Atalaya, Ryan Babbush, Brian Ballard, Joseph C. Bardin, Andreas Bengtsson, Alexander Bilmes, Alexandre Bourassa, Jenna Bovaird, Michael Broughton, David A. Browne, Brett Buchea, Bob B. Buckley, Tim Burger, Brian Burkett, Nicholas Bushnell, Anthony Cabrera, Juan Campero, Hung-Shen Chang, Zijun Chen, Ben Chiaro, Jahan Claes, Agnetta Y. Cleland, Josh Cogan, Roberto Collins, Paul Conner, William Courtney, Alexander L. Crook, Ben Curtin, Sayan Das, Sean Demura, Laura De Lorenzo, Agustin Di Paolo, Paul Donohoe, Ilya Drozdov, Andrew Dunsworth, Alec Eickbusch, Aviv Moshe Elbag, Mahmoud Elzouka, Catherine Erickson, Vinicius S. Ferreira, Leslie Flores Burgos, Ebrahim Forati, Austin G. Fowler, Brooks Foxen, Suhas Ganjam, Robert Gasca, Élie Genois, William Giang, Dar Gilboa, Raja Gosula, Alejandro Grajales Dau, Dietrich Graumann, Alex Greene, Jonathan A. Gross, Steve Habegger, Monica Hansen, Matthew P. Harrigan, Sean D. Harrington, Paula Heu, Oscar Higgott, Jeremy Hilton, Hsin-Yuan Huang, Ashley Huff, William J. Huggins, Evan Jeffrey, Zhang Jiang, Cody Jones, Chaitali Joshi, Pavol Juhas, Dvir Kafri, Hui Kang, Amir H. Karamlou, Kostyantyn Kechedzhi, Trupti Khaire, Tanuj Khattar, Mostafa Khezri, Seon Kim, Paul V. Klimov, Bryce Kobrin, Alexander N. Korotkov, Fedor Kostritsa, John Mark Kreikebaum, Vladislav D. Kurilovich, David Landhuis, Tiano Lange-Dei, Brandon W. Langley, Kim-Ming Lau, Justin Ledford, Kenny Lee, Brian J. Lester, Loïck Le Guevel, Wing Yan Li, Alexander T. Lill, William P. Livingston, Aditya Locharla, Daniel Lundahl, Aaron Lunt, Sid Madhuk, Ashley Maloney, Salvatore Mandrà, Leigh S. Martin, Orion Martin, Cameron Maxfield, Jarrod R. McClean, Matt McEwen, Seneca Meeks, Anthony Megrant, Kevin C. Miao, Reza Molavi, Sebastian Molina, Shirin Montazeri, Ramis Movassagh, Charles Neill, Michael Newman, Anthony Nguyen, Murray Nguyen, Chia-Hung Ni, Kristoffer Ottosson, Alex Pizzuto, Rebecca Potter, Orion Pritchard, Chris Quintana, Ganesh Ramachandran, Matthew J. Reagor, David M. Rhodes, Gabrielle Roberts, Kannan Sankaragomathi, Kevin J. Satzinger, Henry F. Schurkus, Michael J. Shearn, Aaron Shorter, Noah Shutty, Vladimir Shvarts, Volodymyr Sivak, Spencer Small, W. Clarke Smith, Sofia Springer, George Sterling, Jordan Suchard, Alex Sztein, Douglas Thor, M. Mert Torunbalci, Abeer Vaishnav, Justin Vargas, Sergey Vdovichev, Guifre Vidal, Catherine Vollgraff Heidweiller, Steven Waltman, Shannon X. Wang, Brayden Ware, Theodore White, Kristi Wong, Bryan W.K. Woo, Cheng Xing, Z. Jamie Yao, Ping Yeh, Bicheng Ying, Juhwan Yoo, Noureldin Yosri, Grayson Young, Adam Zalcman, Yaxing Zhang, Ningfeng Zhu, Nicholas Zobris, Sergio Boixo, Julian Kelly, Erik Lucero, Yu Chen, Vadim Smelyanskiy, Hartmut Neven, Adam Gammon-Smith, Frank Pollmann,

Michael Knap, Pedram Roushan

- **Posted on arXiv:** 25 September 2024
- **Published in Nature:** 04 June 2025

3.5.7.38 Observation of string breaking on a (2 + 1)D Rydberg quantum simulator [280]

- **Authors:** Daniel Gonzalez-Cuadra, Majd Hamdan, Torsten V. Zache, Boris Braverman, Milan Kornjaca, Alexander Lukin, Sergio H. Cantu, Fangli Liu, Sheng-Tao Wang, Alexander Keesling, Mikhail D. Lukin, Peter Zoller, Alexei Bylinskii
- **Posted on arXiv:** 21 October 2024
- **Published in Nature:** 04 June 2025

3.5.7.39 Probing false vacuum decay on a cold-atom gauge-theory quantum simulator [281]

- **Authors:** Zi-Hang Zhu, Ying Liu, Gianluca Lagnese, Federica Maria Surace, Wei-Yong Zhang, Ming-Gen He, Jad C. Halimeh, Marcello Dalmonte, Siddhardh C. Morampudi, Frank Wilczek, Zhen-Sheng Yuan, Jian-Wei Pan
- **Posted on arXiv:** 19 November 2024

3.5.7.40 Efficient truncations of $\mathrm{SU}(N_c)$ lattice gauge theory for quantum simulation [282]

- **Authors:** Anthony N. Ciavarella, Ivan M. Burbano, Christian W. Bauer
- **Posted on arXiv:** 14 March 2025
- **Published in Phys.Rev.D:** 01 September 2025

3.5.7.41 Observation of hadron scattering in a lattice gauge theory on a quantum computer [283]

- **Authors:** Julian Schuhmacher, Guo-Xian Su, Jesse J. Osborne, Anthony Gandon, Jad C. Halimeh, Ivano Tavernelli
- **Posted on arXiv:** 26 May 2025

3.5.7.42 Scalable cold-atom quantum simulator of a $3 + 1$D $\mathbf{U}(1)$ lattice gauge theory with dynamical matter [284]

- **Authors:** Simone Orlando, Guo-Xian Su, Bing Yang, Jad C. Halimeh
- **Posted on arXiv:** 07 January 2026

3.5.8 Lattice Field Theories - Schwinger Model

3.5.8.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [294]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

3.5.8.2 Quantum Algorithms for Simulating the Lattice Schwinger Model [298]

- **Authors:** Alexander F. Shaw, Pavel Lougovski, Jesse R. Stryker, Nathan Wiebe
- **Posted on arXiv:** 25 February 2020
- **Published in Quantum 4, 306 (2020):** 10 August 2020

3.5.8.3 Quantum simulation of nonequilibrium dynamics and thermalization in the Schwinger model [299]

- **Authors:** Wibe A. de Jong, Kyle Lee, James Mulligan, Mateusz Płoskoń, Felix Ringer, Xiaojun Yao
- **Posted on arXiv:** 15 June 2021
- **Published in Phys. Rev. D 106, 054508 (2022):** 01 September 2022

3.5.8.4 Digital Quantum Simulation of the Schwinger Model and Symmetry Protection with Trapped Ions [300]

- **Authors:** Nhung H. Nguyen, Minh C. Tran, Yingyue Zhu, Alaina M. Green, C. Huerta Alderete, Zohreh Davoudi, Norbert M. Linke
- **Posted on arXiv:** 28 December 2021
- **Published in PRX Quantum 3, 020324 (2022):** 01 May 2022

3.5.8.5 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [295]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

3.5.8.6 Quantum simulations of hadron dynamics in the Schwinger model using 112 qubits [301]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 15 January 2024
- **Published in Phys.Rev.D 109 (2024) 11, 114510:** 01 June 2024

3.5.8.7 Steps toward quantum simulations of hadronization and energy loss in dense matter [296]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

3.5.8.8 Enhancing quantum field theory simulations on NISQ devices with Hamiltonian truncation [302]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams
- **Posted on arXiv:** 26 July 2024
- **Published in Phys. Rev. D 110 (2024), 096016:** 01 November 2024

3.5.8.9 Digital quantum simulations of scattering in quantum field theories using W states [303]

- **Authors:** Roland C. Farrell, Nikita A. Zemlevskiy, Marc Illa, John Preskill
- **Posted on arXiv:** 05 May 2025

3.5.9 Neutrino Physics

3.5.9.1 Simulation of collective neutrino oscillations on a quantum computer [307]

- **Authors:** Benjamin Hall, Alessandro Roggero, Alessandro Baroni, Joseph Carlson
- **Posted on arXiv:** 24 February 2021
- **Published in Phys. Rev. D 104, 063009 (2021):** 03 September 2021

3.5.9.2 Collective neutrino oscillations on a quantum computer [308]

- **Authors:** Kübra Yeter-Aydeniz, Shikha Bangar, George Siopsis, Raphael C. Pooser
- **Posted on arXiv:** 07 April 2021
- **Published in Quant.Inf.Proc.:** 17 February 2022

3.5.9.3 Trapped-ion quantum simulation of collective neutrino oscillations [309]

- **Authors:** Valentina Amitrano, Alessandro Roggero, Piero Luchi, Francesco Turro, Luca Vespucci, Francesco Pederiva
- **Posted on arXiv:** 07 July 2022
- **Published in Phys.Rev.D:** 13 January 2023

3.5.9.4 Multi-Neutrino Entanglement and Correlations in Dense Neutrino Systems [310]

- **Authors:** Marc Illa, Martin J. Savage
- **Posted on arXiv:** 16 October 2022
- **Published in Phys. Rev. Lett. 130, 221003 (2023):** 31 May 2023

3.5.9.5 Qutrit and qubit circuits for three-flavor collective neutrino oscillations [311]

- **Authors:** Francesco Turro, Ivan A. Chernyshev, Ramya Bhaskar, Marc Illa
- **Posted on arXiv:** 18 July 2024
- **Published in Phys. Rev. D 111, 043038 (2025):** 15 February 2025

3.5.9.6 Collective neutrino oscillations in three flavors on qubit and qutrit processors [312]

- **Authors:** Luca Spagnoli, Noah Goss, Alessandro Roggero, Ermal Rrapaj, Michael J. Cervia, Amol V. Patwardhan, Ravi K. Naik, A. Baha Balantekin, David I. Santiago, Irfan Siddiqi, Sheakha Aldaihan
- **Posted on arXiv:** 11 May 2025
- **Published in Phys.Rev.D:** 15 May 2025

3.5.9.7 Quantum simulation of collective neutrino oscillations in dense neutrino environment [313]

- **Authors:** Shvetaank Tripathi, Sandeep Joshi, Garima Rajpoot, Prashant Shukla
- **Posted on arXiv:** 15 August 2025
- **Published in Quant.Inf.Proc.:** 06 May 2026

3.5.9.8 Quantum Simulation of Collective Neutrino Oscillations using Dicke States [314]

- **Authors:** Katarina Bleau, Nikolina Ilic, Joachim Kopp, Ushak Rahaman, Xin Yue Yu
- **Posted on arXiv:** 08 April 2026

3.5.10 Nuclear Many-Body Physics - Nuclear Structure

3.5.10.1 Symmetry assisted preparation of entangled many-body states on a quantum computer [336]

- **Authors:** D. Lacroix
- **Posted on arXiv:** 11 June 2020
- **Published in Phys.Rev.Lett.:** 03 December 2020

3.5.10.2 Quantum computation of nuclear observables involving linear combinations of unitary operators [337]

- **Authors:** Pooja Siwach, P. Arumugam
- **Posted on arXiv:** 16 June 2022
- **Published in Phys.Rev.C:** 28 June 2022

3.5.11 Nuclear Many-Body Physics - Nuclear Reactions and Response

3.5.11.1 Quantum Computing for Neutrino-Nucleus Scattering [179]

- **Authors:** Alessandro Roggero, Andy C.Y. Li, Joseph Carlson, Rajan Gupta, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019
- **Published in Phys. Rev. D 101, 074038 (2020):** 28 April 2020

3.5.11.2 Quantum simulation of nuclear inelastic scattering [354]

- **Authors:** Weijie Du, James P. Vary, Xingbo Zhao, Wei Zuo
- **Posted on arXiv:** 01 June 2020
- **Published in Phys. Rev. A 104, 012611 (2021):** 23 July 2021

3.5.11.3 Imaginary-time propagation on a quantum chip [355]

- **Authors:** Francesco Turro, Alessandro Roggero, Valentina Amitrano, Piero Luchi, Kyle A. Wendt, Jonathan L. DuBois, Sofia Quaglioni, Francesco Pederiva
- **Posted on arXiv:** 24 February 2021
- **Published in Phys.Rev.A:** 28 February 2022

3.5.11.4 Nuclear two point correlation functions on a quantum computer [356]

- **Authors:** Alessandro Baroni, Joseph Carlson, Rajan Gupta, Andy C.Y. Li, Gabriel N. Perdue, Alessandro Roggero
- **Posted on arXiv:** 04 November 2021
- **Published in Phys.Rev.D:** 01 April 2022

3.5.11.5 A quantum algorithm for the linear response of nuclei [357]

- **Authors:** Nifeeya Singh, P. Arumugam
- **Posted on arXiv:** 17 October 2022
- **Published in Indian J.Phys.:** 31 May 2025

3.5.11.6 Faster spectral density calculation using energy moments [358]

- **Authors:** Jeremy Hartse, Alessandro Roggero
- **Posted on arXiv:** 01 November 2022
- **Published in Eur.Phys.J.A:** 09 March 2023

3.5.11.7 Quantum error mitigation for Fourier moment computation [359]

- **Authors:** Oriel Kiss, Michele Grossi, Alessandro Roggero
- **Posted on arXiv:** 23 January 2024
- **Published in Phys. Rev. D 111, 034504 (2025):** 01 February 2025

3.5.11.8 Nuclear scattering via quantum computing [360]

- **Authors:** Peiyan Wang, Weijie Du, Wei Zuo, James P. Vary
- **Posted on arXiv:** 30 January 2024
- **Published in Phys.Rev.C 109 6, 064623 (2024):** 26 June 2024

3.5.11.9 Solving reaction dynamics with quantum computing algorithms [361]

- **Authors:** Ronen Weiss, Alessandro Baroni, Joseph Carlson, Ionel Stetcu
- **Posted on arXiv:** 29 March 2024
- **Published in Physical Review C 111, 064004 (2025):** 25 June 2025

3.5.11.10 Quantum computing for extracting nuclear resonances [362]

- **Authors:** Hantao Zhang, Dong Bai, Zhongzhou Ren
- **Posted on arXiv:** 10 September 2024
- **Published in Phys.Lett.B:** 12 December 2024

3.5.11.11 Pathfinding quantum simulations of neutrinoless double-β decay [297]

- **Authors:** Ivan A. Chernyshev, Roland C. Farrell, Marc Illa, Martin J. Savage, Andrii Maksymov, Felix Tripier, Miguel Angel Lopez-Ruiz, Andrew Arrasmith, Yvette de Sereville, Aharon Brodutch, Claudio Girotto, Ananth Kaushik, Martin Roetteler
- **Posted on arXiv:** 06 June 2025
- **Published in Nature Commun.:** 23 January 2026

3.5.12 Quantum Chromodynamics

3.5.12.1 Quantum simulation of quantum field theory in the light-front formulation [365]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

3.5.12.2 Quantum simulation of open quantum systems in heavy-ion collisions [366]

- **Authors:** Wibe A. De Jong, Mekena Metcalf, James Mulligan, Mateusz Płoskoń, Felix Ringer, Xiaojun Yao
- **Posted on arXiv:** 07 October 2020
- **Published in Phys. Rev. D 104, 051501 (2021):** 01 September 2021

3.5.12.3 Quantum Simulation of Light-Front QCD for Jet Quenching in Nuclear Environments [367]

- **Authors:** Xiaojun Yao
- **Posted on arXiv:** 16 May 2022

3.5.12.4 Quantum simulation of colour in perturbative quantum chromodynamics [368]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen
- **Posted on arXiv:** 08 March 2023
- **Published in SciPost Phys. 15, 205 (2023):** 24 November 2023

3.5.12.5 Quantum simulation of in-medium QCD jets: Momentum broadening, gluon production, and entropy growth [369]

- **Authors:** João Barata, Xiaojian Du, Meijian Li, Wenyang Qian, Carlos A. Salgado
- **Posted on arXiv:** 04 July 2023
- **Published in Phys.Rev.D:** 01 September 2023

3.5.12.6 Efficient quantum simulation of QCD jets on the light front [232]

- **Authors:** Wenyang Qian, Meijian Li, Carlos A. Salgado, Michael Kreshchuk
- **Posted on arXiv:** 14 November 2024
- **Published in Phys.Rev.D 111 (2025) 9, 096001:** 01 May 2025

3.5.12.7 Quantum simulation of scattering amplitudes and interferences in perturbative QCD [370]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen, Simon Williams
- **Posted on arXiv:** 09 July 2025
- **Published in Eur.Phys.J.C:** 11 August 2026

3.5.12.8 Quantum simulating multi-particle processes in high energy nuclear physics: dijet production and color (de)coherence [371]

- **Authors:** João Barata, Meijian Li, Wenyang Qian, Carlos A. Salgado, João M. Silva
- **Posted on arXiv:** 13 April 2026
- **Published in JHEP:** 08 July 2026

3.5.13 Crosslists

3.5.13.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

3.5.13.2 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

3.5.13.3 Standard model physics and the digital quantum revolution: thoughts about the interface [16]

- **Authors:** Natalie Klco, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 10 July 2021
- **Published in Rept.Prog.Phys.:** 19 May 2022

3.5.13.4 Quantum error correction with gauge symmetries [288]

- **Authors:** Abhishek Rajput, Alessandro Roggero, Nathan Wiebe
- **Posted on arXiv:** 09 December 2021
- **Published in npj Quantum Inf.:** 25 April 2023

3.5.13.5 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [17]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

3.5.13.6 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [18]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.5.13.7 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [20]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.5.13.8 Quantum Simulation for High-Energy Physics [25]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

3.5.13.9 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [27]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.5.13.10 Many-Body Collective Neutrino Oscillations: Recent Developments [6]

- **Authors:** Amol V. Patwardhan, Michael J. Cervia, Ermal Rrapaj, Pooja Siwach, A.B. Balantekin
- **Posted on arXiv:** 31 December 2022
- **Published in nan:** 2023

3.5.13.11 Review on Quantum Computing for Lattice Field Theory [7]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in nan:** 01 February 2023

3.5.13.12 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay

Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao

- **Posted on arXiv:** 28 February 2023

3.5.13.13 Quantum information and quantum simulation of neutrino physics [8]

- **Authors:** A.B. Balantekin, Michael J. Cervia, Amol V. Patwardhan, Ermal Rrapaj, Pooja Siwach
- **Posted on arXiv:** 01 May 2023
- **Published in Eur.Phys.J.A:** 17 August 2023

3.5.13.14 Quantum data learning for quantum simulations in high-energy physics [214]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

3.5.13.15 Collective neutrino oscillations on a quantum computer with hybrid quantum-classical algorithm [332]

- **Authors:** Pooja Siwach, Kaytlin Harrison, A. Baha Balantekin
- **Posted on arXiv:** 17 August 2023
- **Published in Phys.Rev.D:** 15 October 2023

3.5.13.16 Simulating quantum field theories on continuous-variable quantum computers [239]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 March 2024
- **Published in Phys. Rev. A 110 (2024) 1, 012607:** 08 July 2024

3.5.13.17 Real-time scattering processes with continuous-variable quantum computers [240]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 03 February 2025
- **Published in Phys. Rev. A 112 (2025), 012614:** 16 July 2025

3.5.13.18 Three-flavor supernova neutrino simulation using a hybrid quantum-classical algorithm with qutrits [333]

- **Authors:** Daniel J. Heimsoth, Pooja Siwach, A. Baha Balantekin
- **Posted on arXiv:** 01 May 2026
- **Published in Phys. Rev. D 114, 043055 (2026):** 15 August 2026

3.6 Quantum Algorithms - Quantum Walks

3.6.1 Detector Technologies and Simulations

3.6.1.1 Quantum walk based Monte Carlo simulation for photon interaction cross sections [158]

- **Authors:** Euimin Lee, Sangmin Lee, Shiho Kim
- **Posted on arXiv:** 20 February 2025
- **Published in Phys.Rev.D:** 01 June 2025

3.6.2 Event Generation

3.6.2.1 Quantum walk approach to simulating parton showers [207]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

3.7 Quantum Computing Paradigms - Continuous Variable Quantum Computing

3.7.1 Anomaly Detection

3.7.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [31]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

3.7.2 Detector Technologies and Simulations

3.7.2.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [159]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.7.3 Event Classification

3.7.3.1 Continuous-variable photonic quantum extreme learning machines for fast collider-data selection [184]

- **Authors:** Benedikt Maier, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 October 2025
- **Published in EPJ Quant.Technol.:** 27 April 2026

3.7.4 Lattice Field Theories - Scalar/Fermion Theories

3.7.4.1 Simulating quantum field theories on continuous-variable quantum computers [239]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 March 2024
- **Published in Phys. Rev. A 110 (2024) 1, 012607:** 08 July 2024

3.7.4.2 Real-time scattering processes with continuous-variable quantum computers [240]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 03 February 2025
- **Published in Phys. Rev. A 112 (2025), 012614:** 16 July 2025

3.7.4.3 Qumode Tensor Networks for False Vacuum Decay in Quantum Field Theory [241]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 20 June 2025

3.7.5 Crosslists

3.7.5.1 Subatomic Many-Body Physics Simulations on a Quantum Frequency Processor [182]

- **Authors:** Hsuan-Hao Lu, Natalie Klco, Joseph M. Lukens, Titus D. Morris, Aaina Bansal, Andreas Ekström, Gaute Hagen, Thomas Papenbrock, Andrew M. Weiner, Martin J. Savage, Pavel Lougovski
- **Published in nan:** 2019

3.8 Quantum Computing Paradigms - Quantum Annealing

3.8.1 Anomaly Detection

3.8.1.1 A Quantum Algorithm for Model-Independent Searches for New Physics [32]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

3.8.2 Detector Technologies and Simulations

3.8.2.1 Unfolding measurement distributions via quantum annealing [160]

- **Authors:** Kyle Cormier, Riccardo Di Sipio, Peter Wittek
- **Posted on arXiv:** 22 August 2019
- **Published in JHEP:** 22 November 2019

3.8.3 Event Classification

3.8.3.1 Solving a Higgs optimization problem with quantum annealing for machine learning [185]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

3.8.3.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [186]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

3.8.3.3 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [187]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

3.8.3.4 Quantum algorithm for the classification of supersymmetric top quark events [46]

- **Authors:** Pedrame Bargassa, Timothée Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothée Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

3.8.3.5 Completely quantum neural networks [188]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

3.8.4 Jet Reconstruction

3.8.4.1 Adiabatic quantum algorithm for multijet clustering in high energy physics [225]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

3.8.4.2 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [226]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

3.8.4.3 Quantum annealing for jet clustering with thrust [227]

- **Authors:** Andrea Delgado, Jesse Thaler
- **Posted on arXiv:** 05 May 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.8.4.4 Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics [228]

- **Authors:** Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, Jesse Thaler
- **Posted on arXiv:** 20 May 2022
- **Published in Phys. Rev. D 106, 056008 (2022):** 01 September 2022

3.8.5 Lattice Field Theories - Gauge Theories

3.8.5.1 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [285]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

3.8.5.2 SU(2) lattice gauge theory on a quantum annealer [286]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

3.8.6 Neutrino Physics

3.8.6.1 Three-flavor Collective Neutrino Oscillations on D-Wave's {\tt Advantage} Quantum Annealer [315]

- **Authors:** Ivan A. Chernyshev
- **Posted on arXiv:** 30 May 2024

3.8.7 Nuclear Many-Body Physics - Nuclear Structure

3.8.7.1 Quasiparticle pairing encoding of atomic nuclei for quantum annealing [338]

- **Authors:** Emanuele Costa, Axel Pérez-Obiol, Javier Menéndez, Arnau Rios, Artur García-Sáez, Bruno Juliá-Díaz
- **Posted on arXiv:** 11 October 2025
- **Published in Phys.Lett.B:** 30 November 2025

3.8.8 Track Reconstruction

3.8.8.1 A pattern recognition algorithm for quantum annealers [448]

- **Authors:** Frédéric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
- **Published in Comput.Softw.Big Sci.:** 09 December 2019

3.8.8.2 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [449]

- **Authors:** Souvik Das, Andrew J. Wildridge, Andreas Jung
- **Posted on arXiv:** 21 March 2019

3.8.8.3 Charged particle tracking with quantum annealing optimization [450]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

3.8.8.4 Quantum annealing algorithms for track pattern recognition [451]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

3.8.8.5 Particle track classification using quantum associative memory [452]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

3.8.8.6 Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders [453]

- **Authors:** Hideki Okawa, Qing-Guo Zeng, Xian-Zhe Tao, Man-Hong Yung
- **Posted on arXiv:** 22 February 2024
- **Published in Comput.Softw.Big Sci.:** 28 August 2024

3.8.9 Crosslists

3.8.9.1 Quantum Algorithms for Jet Clustering [223]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.8.9.2 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.8.9.3 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.8.9.4 Quantum Computing Applications in Future Colliders [5]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.8.9.5 Preparations for quantum simulations of quantum chromodynamics in $1+1$ dimensions. I. Axial gauge [264]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

3.8.9.6 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.9 Quantum Entanglement and Bell Inequalities

3.9.1 Beyond the Standard Model

3.9.1.1 Entanglement suppression, enhanced symmetry, and a standard-model-like Higgs boson [39]

- **Authors:** Marcela Carena, Ian Low, Carlos E.M. Wagner, Ming-Lei Xiao
- **Posted on arXiv:** 16 July 2023
- **Published in Phys.Rev.D:** 01 March 2024

3.9.1.2 Flavor patterns of fundamental particles from quantum entanglement? [40]

- **Authors:** Jesse Thaler, Sokratis Trifinopoulos
- **Posted on arXiv:** 30 October 2024
- **Published in Phys.Rev.D:** 01 March 2025

3.9.2 Effective Field Theories

3.9.2.1 Geometric quantum information structure in quantum fields and their lattice simulation [173]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 August 2020
- **Published in Phys. Rev. D 103, 065007 (2021):** 23 March 2021

3.9.2.2 Entanglement Spheres and a UV-IR Connection in Effective Field Theories [174]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 27 March 2021
- **Published in Phys.Rev.Lett.:** 16 November 2021

3.9.2.3 Entanglement structures in quantum field theories: Negativity cores and bound entanglement in the vacuum [175]

- **Authors:** Natalie Klco, D.H. Beck, Martin J. Savage
- **Posted on arXiv:** 20 October 2021
- **Published in Phys.Rev.A:** 11 January 2023

3.9.2.4 Minimal entanglement and emergent symmetries in low-energy QCD [176]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

3.9.2.5 Entanglement in flavored scalar scattering [177]

- **Authors:** Kamila Kowalska, Enrico Maria Sessolo
- **Posted on arXiv:** 21 April 2024
- **Published in JHEP:** 18 July 2024

3.9.3 Lattice Field Theories - Gauge Theories

3.9.3.1 Gauge and diffeomorphism invariance from quantum information principles [287]

- **Authors:** Claudia Núñez, Miguel Pardina, Manuel Asorey, José Ignacio Latorre, Alba Cervera-Lierta
- **Posted on arXiv:** 06 November 2025
- **Published in Phys. Rev. D 113, 096019 (2026):** 01 May 2026

3.9.4 Neutrino Physics

3.9.4.1 Violation of the Leggett-Garg Inequality in Neutrino Oscillations [316]

- **Authors:** J.A. Formaggio, D.I. Kaiser, M.M. Murskyj, T.E. Weiss
- **Posted on arXiv:** 29 January 2016
- **Published in Phys.Rev.Lett.:** 26 July 2016

3.9.4.2 Entanglement and collective flavor oscillations in a dense neutrino gas [317]

- **Authors:** Michael J. Cervia, Amol V. Patwardhan, A.B. Balantekin, ‡ S.N. Coppersmith, Calvin W. Johnson
- **Posted on arXiv:** 09 August 2019
- **Published in Phys.Rev.D:** 03 October 2019

3.9.4.3 Entanglement and many-body effects in collective neutrino oscillations [318]

- **Authors:** Alessandro Roggero
- **Posted on arXiv:** 19 February 2021
- **Published in Phys. Rev. D 104 (2021) 103016:** 10 November 2021

3.9.4.4 Dynamical phase transitions in models of collective neutrino oscillations [319]

- **Authors:** Alessandro Roggero
- **Posted on arXiv:** 21 March 2021
- **Published in Phys.Rev.D:** 14 December 2021

3.9.4.5 Spectral splits and entanglement entropy in collective neutrino oscillations [320]

- **Authors:** Amol V. Patwardhan, Michael J. Cervia, A.B. Balantekin
- **Posted on arXiv:** 18 September 2021
- **Published in Phys. Rev. D 104, 123035 (2021):** 15 December 2021

3.9.4.6 Entanglement and correlations in fast collective neutrino flavor oscillations [321]

- **Authors:** Alessandro Roggero, Ermal Rrapaj, Zewei Xiong
- **Posted on arXiv:** 05 March 2022
- **Published in Phys.Rev.D:** 15 August 2022

3.9.4.7 Role of non-Gaussian quantum fluctuations in neutrino entanglement [322]

- **Authors:** Denis Lacroix, A.B. Balantekin, Michael J. Cervia, Amol V. Patwardhan, Pooja Siwach
- **Posted on arXiv:** 19 May 2022
- **Published in Phys.Rev.D:** 06 December 2022

3.9.4.8 Entanglement in three-flavor collective neutrino oscillations [323]

- **Authors:** Pooja Siwach, Anna M. Suliga, A. Baha Balantekin
- **Posted on arXiv:** 14 November 2022
- **Published in Phys.Rev.D 107 023019 (2023):** 15 January 2023

3.9.4.9 Many-body neutrino flavor entanglement in a simple dynamic model [324]

- **Authors:** Joshua D. Martin, A. Roggero, Huaiyu Duan, J. Carlson
- **Posted on arXiv:** 17 January 2023

3.9.4.10 Neutrino many-body correlations [325]

- **Authors:** Lucas Johns
- **Posted on arXiv:** 08 May 2023
- **Published in Int.J.Mod.Phys.A:** 13 November 2024

3.9.4.11 Equilibration of quantum many-body fast neutrino flavor oscillations [326]

- **Authors:** Joshua D. Martin, Duff Neill, A. Roggero, Huaiyu Duan, J. Carlson
- **Posted on arXiv:** 31 July 2023
- **Published in Phys.Rev.D:** 07 December 2023

3.9.4.12 Time Scales in Many-Body Fast Neutrino Flavor Conversion [327]

- **Authors:** Ramya Bhaskar, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 23 December 2023

3.9.4.13 Quantum magic and computational complexity in the neutrino sector [328]

- **Authors:** Ivan Chernyshev, Caroline E.P. Robin, Martin J. Savage
- **Posted on arXiv:** 06 November 2024
- **Published in Phys.Rev.Res.:** 04 June 2025

3.9.4.14 Exploring entanglement and spectral split correlations in three-flavor collective neutrino oscillations [329]

- **Authors:** Pooja Siwach, A. Baha Balantekin, Amol V. Patwardhan, Anna M. Suliga
- **Posted on arXiv:** 07 November 2024
- **Published in Phys.Rev.D:** 14 March 2025

3.9.4.15 Quantum resource redistribution drives spectral splits in dense neutrino gases [330]

- **Authors:** Michael Hite, Pooja Siwach
- **Posted on arXiv:** 22 May 2026

3.9.4.16 Collective neutrino oscillations: Many-body non-forward effects and non-classicality [331]

- **Authors:** Julien Froustey, Ermal Rrapaj, Yuhao Liu, Gushu Li, Costin Iancu, Vincenzo Cirigliano
- **Posted on arXiv:** 10 June 2026

3.9.5 Quantum Chromodynamics

3.9.5.1 Deep inelastic scattering as a probe of entanglement [372]

- **Authors:** Dmitri E. Kharzeev, Eugene M. Levin
- **Posted on arXiv:** 12 February 2017
- **Published in Phys.Rev.D:** 13 June 2017

3.9.5.2 Einstein-Podolsky-Rosen Paradox and Quantum Entanglement at Sub-nucleonic Scales [373]

- **Authors:** Zhoudunming Tu, Dmitri E. Kharzeev, Thomas Ullrich
- **Posted on arXiv:** 26 April 2019
- **Published in Phys. Rev. Lett. 124, 062001 (2020):** 15 February 2020

3.9.5.3 Measurement of charged particle multiplicity distributions in DIS at HERA and its implication to entanglement entropy of partons [374]

- **Authors:** V. Andreev, A. Baghdasaryan, A. Baty, K. Begzsuren, A. Belousov, A. Bolz, V. Boudry, G. Brandt, D. Britzger, A. Buniatyan, L. Bystritskaya, A.J. Campbell, K.B. Cantun Avila, K. Cerny, V. Chekelian, Z. Chen, J.G. Contreras, J. Cvach, J.B. Dainton, K. Daum, A. Deshpande, C. Diaconu, G. Eckerlin, S. Egli, E. Elsen, L. Favart, A. Fedotov, J. Feltesse, M. Fleischer, A. Fomenko, C. Gal, J. Gayler, L. Goerlich, N. Gogitidze, M. Gouzevitch, C. Grab, A. Grebenyuk, T. Greenshaw, G. Grindhammer, D. Haidt, R.C.W. Henderson, J. Hladkỳ, D. Hoffmann, R. Horisberger, T. Hreus, F. Huber, M. Jacquet, X. Janssen, A.W. Jung, H. Jung, M. Kapichine, J. Katzy, C. Kiesling, M. Klein, C. Kleinwort, R. Kogler, P. Kostka, J. Kretzschmar, D. Krücker, K. Krüger, M.P.J. Landon, W. Lange, P. Laycock, A. Lebedev, S. Levonian, K. Lipka, B. List, J. List, W. Li, B. Lobodzinski, E. Malinovski, H.U. Martyn, S.J. Maxfield,

A. Mehta, A.B. Meyer, H. Meyer, J. Meyer, S. Mikocki, M.M. Mondal, A. Morozov, K. Müller, Th. Naumann, P.R. Newman, C. Niebuhr, G. Nowak, J.E. Olsson, D. Ozerov, S. Park, C. Pascaud, G.D. Patel, E. Perez, A. Petrukhin, I. Picuric, D. Pitzl, R. Polifka, V. Radescu, N. Raicevic, T. Ravdandorj, P. Reimer, E. Rizvi, P. Robmann, R. Roosen, A. Rostovtsev, M. Rotaru, D.P.C. Sankey, M. Sauter, E. Sauvan, S. Schmitt, B.A. Schmookler, L. Schoeffel, A. Schöning, F. Sefkow, S. Shushkevich, Y. Soloviev, P. Sopicki, D. South, V. Spaskov, A. Specka, M. Steder, B. Stella, U. Straumann, T. Sykora, P.D. Thompson, D. Traynor, P. Truöl, B. Tseepeldorj, Z. Tu, T. Ullrich, A. Valkárová, C. Vallée, P.Van Mechelen, D. Wegener, E. Wünsch, J. Žáček, J. Zhang, Z. Zhang, R. Žlebčík, H. Zohrabyan, F. Zomer

- **Posted on arXiv:** 03 November 2020
- **Published in Eur.Phys.J.C:** 03 March 2021

3.9.5.4 Deep inelastic scattering as a probe of entanglement: Confronting experimental data [375]

- **Authors:** Dmitri E. Kharzeev, Eugene Levin
- **Posted on arXiv:** 19 February 2021
- **Published in Phys. Rev. D 104, 031503 (2021):** 01 August 2021

3.9.5.5 A quantum strategy to compute the jet quenching parameter $\hat{q}$ [376]

- **Authors:** João Barata, Carlos A. Salgado
- **Posted on arXiv:** 09 April 2021
- **Published in Eur.Phys.J.C:** 01 October 2021

3.9.5.6 Symmetry from entanglement suppression [377]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

3.9.5.7 Entanglement entropy production in deep inelastic scattering [378]

- **Authors:** Kun Zhang, Kun Hao, Dmitri Kharzeev, Vladimir Korepin
- **Posted on arXiv:** 10 October 2021
- **Published in Phys. Rev. D 105, 014002 (2022):** 01 January 2022

3.9.5.8 Real-Time Nonperturbative Dynamics of Jet Production in Schwinger Model: Quantum Entanglement and Vacuum Modification [379]

- **Authors:** Adrien Florio, David Frenklakh, Kazuki Ikeda, Dmitri Kharzeev, Vladimir Korepin, Shuzhe Shi, Kwangmin Yu
- **Posted on arXiv:** 27 January 2023
- **Published in Phys. Rev. Lett. 131 (2023) 021902:** 13 July 2023

3.9.5.9 Probing the Onset of Maximal Entanglement inside the Proton in Diffractive Deep Inelastic Scattering [380]

- **Authors:** Martin Hentschinski, Dmitri E. Kharzeev, Krzysztof Kutak, Zhoudunming Tu
- **Posted on arXiv:** 04 May 2023
- **Published in Phys.Rev.Lett.:** 13 December 2023

3.9.5.10 Universal rapidity scaling of entanglement entropy inside hadrons from conformal invariance [381]

- **Authors:** Umut Gürsoy, Dmitri E. Kharzeev, Juan F. Pedraza
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

3.9.5.11 Entanglement as a Probe of Hadronization [382]

- **Authors:** Jaydeep Datta, Abhay Deshpande, Dmitri E. Kharzeev, Charles Joseph Naïm, Zhoudunming Tu
- **Posted on arXiv:** 29 October 2024
- **Published in Phys.Rev.Lett.:** 19 March 2025

3.9.5.12 Quantum information at the electron-ion collider [383]

- **Authors:** Kun Cheng, Tao Han, Sokratis Trifinopoulos
- **Posted on arXiv:** 27 October 2025
- **Published in Phys.Rev.D:** 01 September 2026

3.9.6 Quantum Correlations at Colliders

3.9.6.1 Maximal Entanglement in High Energy Physics [384]

- **Authors:** Alba Cervera-Lierta, José I. Latorre, Juan Rojo, Luca Rottoli
- **Posted on arXiv:** 08 March 2017
- **Published in SciPost Phys. 3, 036 (2017):** 24 November 2017

3.9.6.2 Polarization and Entanglement in Baryon-Antibaryon Pair Production in Electron-Positron Annihilation [385]

- **Authors:** M. Ablikim, M.N. Achasov, S. Ahmed, M. Albrecht, M. Alekseev, A. Amoroso, F.F. An, Q. An, Y. Bai, O. Bakina, R. Baldini Ferroli, Y. Ban, K. Begzsuren, D.W. Bennett, J.V. Bennett, N. Berger, M. Bertani, D. Bettoni, F. Bianchi, E. Boger, I. Boyko, R.A. Briere, H. Cai, X. Cai, A. Calcaterra, G.F. Cao, S.A. Cetin, J. Chai, J.F. Chang, W.L. Chang, G. Chelkov, G. Chen, H.S. Chen, J.C. Chen, M.L. Chen, P.L. Chen, S.J. Chen, X.R. Chen, Y.B. Chen, W. Cheng, X.K. Chu, G. Cibinetto, F. Cossio, H.L. Dai, J.P. Dai, A. Dbeyssi, D. Dedovich, Z.Y. Deng, A. Denig, I. Denysenko, M. Destefanis, F. De Mori, Y. Ding, C. Dong, J. Dong, L.Y. Dong, M.Y. Dong, Z.L. Dou, S.X. Du, P.F. Duan, J.Z. Fan, J. Fang, S.S. Fang, Y. Fang, R. Farinelli, L. Fava, S. Fegan, F. Feldbauer, G. Felici, C.Q. Feng, E. Fioravanti, M. Fritsch, C.D. Fu, Q. Gao, X.L. Gao, Y. Gao, Y.G. Gao, Z. Gao, B. Garillon, I. Garzia, A. Gilman, K. Goetzen, L. Gong, W.X. Gong, W. Gradl, M. Greco, L.M. Gu, M.H. Gu, Y.T. Gu, A.Q. Guo, L.B. Guo, R.P. Guo, Y.P. Guo, A. Guskov, Z. Haddadi, S. Han, X.Q. Hao, F.A. Harris, K.L. He, F.H. Heinsius, T. Held, Y.K. Heng, Z.L. Hou, H.M. Hu, J.F. Hu, T. Hu, Y. Hu, G.S. Huang, J.S. Huang, X.T. Huang, X.Z. Huang, Z.L. Huang, T. Hussain, W. Ikegami Andersson, M. Irshad, Q. Ji, Q.P. Ji, X.B. Ji, X.L. Ji, H.L. Jiang, X.S. Jiang, X.Y. Jiang, J.B. Jiao, Z. Jiao, D.P. Jin, S. Jin, Y. Jin, T. Johansson, A. Julin, N. Kalantar-Nayestanaki, X.S. Kang, M. Kavatsyuk, B.C. Ke, I.K. Keshk, T. Khan, A. Khoukaz, P. Kiese, R. Kiuchi, R. Kliemt, L. Koch, O.B. Kolcu, B. Kopf, M. Kornicer, M. Kuemmel, M. Kuessner, A. Kupsc, M. Kurth, W. Kühn, J.S. Lange, P. Larin, L. Lavezzi, S. Leiber, H. Leithoff, C. Li, Cheng Li, D.M. Li, F. Li, F.Y. Li, G. Li, H.B. Li, H.J. Li, J.C. Li, J.W. Li, K.J. Li, Kang Li, Ke Li, Lei Li, P.L. Li, P.R. Li, Q.Y. Li, T. Li, W.D. Li, W.G. Li, X.L. Li, X.N. Li, X.Q. Li, Z.B. Li, H. Liang, Y.F. Liang, Y.T. Liang, G.R. Liao, L.Z. Liao, J. Libby, C.X. Lin, D.X. Lin, B. Liu, B.J. Liu, C.X. Liu, D. Liu, D.Y. Liu, F.H. Liu, Fang Liu, Feng Liu, H.B. Liu, H.L. Liu, H.M. Liu, Huanhuan Liu, Huihui Liu, J.J. Liu, J.B. Liu, J.Y. Liu, K.Y. Liu, Ke Liu, L.D. Liu, Q. Liu, S.B. Liu, X. Liu, Y.B. Liu, Z.A. Liu, Zhiqing Liu, Y.F. Long, X.C. Lou, H.J. Lu, J.G. Lu, Y. Lu, Y.P. Lu, C.L. Luo, M.X. Luo, P.W. Luo, T. Luo, X.L. Luo, S. Lusso, X.R. Lyu, F.C. Ma, H.L. Ma, L.L. Ma, M.M. Ma, Q.M. Ma, X.N. Ma, X.Y. Ma, Y.M. Ma, F.E. Maas, M. Maggiora, S. Maldaner, Q.A. Malik, A. Mangoni, Y.J. Mao, Z.P. Mao, S. Marcello, Z.X. Meng, J.G. Messchendorp, G. Mezzadri, J. Min, T.J. Min,

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- **Posted on arXiv:** 27 August 2018
- **Published in Nature Phys.:** 06 May 2019

3.9.6.3 Entanglement and quantum tomography with top quarks at the LHC [386]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

3.9.6.4 Testing Bell Inequalities at the LHC with Top-Quark Pairs [387]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

3.9.6.5 Testing Bell inequalities in Higgs boson decays [388]

- **Authors:** Alan J. Barr
- **Posted on arXiv:** 02 June 2021
- **Published in Physics Letters B 825 (2022) 136866:** 04 January 2022

3.9.6.6 Measurement of Bell-type inequalities and quantum entanglement from Λ-hyperon spin correlations at high energy colliders [389]

- **Authors:** Wenjie Gong, Ganesh Parida, Zhoudunming Tu, Raju Venugopalan
- **Posted on arXiv:** 27 July 2021
- **Published in Phys.Rev.D:** 01 August 2022

3.9.6.7 Quantum tops at the LHC: from entanglement to Bell inequalities [390]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

3.9.6.8 General analysis for observing quantum interference at colliders [391]

- **Authors:** Andrew J. Larkoski
- **Posted on arXiv:** 09 January 2022
- **Published in Phys.Rev.D:** 01 May 2022

3.9.6.9 Quantum SMEFT tomography: Top quark pair production at the LHC [181]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

3.9.6.10 Quantum information with top quarks in QCD [392]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

3.9.6.11 Improved tests of entanglement and Bell inequalities with LHC tops [393]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
- **Posted on arXiv:** 01 May 2022
- **Published in Eur.Phys.J.C:** 03 August 2022

3.9.6.12 Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs [394]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli
- **Posted on arXiv:** 24 August 2022
- **Published in Eur.Phys.J.C:** 19 February 2023

3.9.6.13 Quantum Discord and Steering in Top Quarks at the LHC [395]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 08 September 2022
- **Published in Phys. Rev. Lett. 130, 221801 (2023):** 31 May 2023

3.9.6.14 Testing entanglement and Bell inequalities in $H \to ZZ$ [396]

- **Authors:** J.A. Aguilar-Saavedra, A. Bernal, J.A. Casas, J.M. Moreno
- **Posted on arXiv:** 27 September 2022
- **Published in Phys.Rev.D:** 01 January 2023

3.9.6.15 Laboratory-frame tests of quantum entanglement in $H \to WW$ [397]

- **Authors:** J.A. Aguilar-Saavedra
- **Posted on arXiv:** 28 September 2022
- **Published in Phys.Rev.D:** 01 April 2023

3.9.6.16 Quantum state tomography, entanglement detection and Bell violation prospects in weak decays of massive particles [398]

- **Authors:** Rachel Ashby-Pickering, Alan J. Barr, Agnieszka Wierzchucka
- **Posted on arXiv:** 28 September 2022
- **Published in JHEP:** 02 May 2023

3.9.6.17 Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders [399]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 01 February 2023
- **Published in Eur.Phys.J.C:** 15 September 2023

3.9.6.18 Bell inequality is violated in $B^0 \to J/\psi K^*(892)^0$ decays [400]

- **Authors:** M. Fabbrichesi, R. Floreanini, E. Gabrielli, L. Marzola
- **Posted on arXiv:** 08 May 2023
- **Published in Phys.Rev.D:** 01 February 2024

3.9.6.19 Probing new physics through entanglement in diboson production [401]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 18 July 2023
- **Published in JHEP:** 01 December 2023

3.9.6.20 Entanglement and Bell inequalities violation in $H \to ZZ$ with anomalous coupling [402]

- **Authors:** Alexander Bernal, Paweł Caban, Jakub Rembieliński
- **Posted on arXiv:** 25 July 2023
- **Published in Eur.Phys.J.C:** 17 November 2023

3.9.6.21 Isolating semi-leptonic $H \to WW^*$ decays for Bell inequality tests [403]

- **Authors:** Federica Fabbri, James Howarth, Theo Maurin
- **Posted on arXiv:** 25 July 2023
- **Published in Eur.Phys.J.C:** 09 January 2024

3.9.6.22 New observables for testing Bell inequalities in W boson pair production [404]

- **Authors:** Qi Bi, Qing-Hong Cao, Kun Cheng, Hao Zhang
- **Posted on arXiv:** 27 July 2023
- **Published in Phys.Rev.D:** 01 February 2024

3.9.6.23 Three-Body Entanglement in Particle Decays [405]

- **Authors:** Kazuki Sakurai, Michael Spannowsky
- **Posted on arXiv:** 02 October 2023
- **Published in Phys.Rev.Lett.:** 09 April 2024

3.9.6.24 Quantum entanglement and Bell inequality violation in semi-leptonic top decays [406]

- **Authors:** Tao Han, Matthew Low, Tong Arthur Wu
- **Posted on arXiv:** 26 October 2023
- **Published in JHEP:** 22 July 2024

3.9.6.25 Observation of quantum entanglement with top quarks at the ATLAS detector [407]

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3.9.6.26 Optimizing fictitious states for Bell inequality violation in bipartite qubit systems with applications to the tt^- system [408]

- **Authors:** Kun Cheng, Tao Han, Matthew Low
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3.9.6.27 Probing entanglement and testing Bell inequality violation with $e^+e^-\to\tau^+\tau^-$ at Belle II [409]

- **Authors:** Karl Ehatäht, Marco Fabbrichesi, Luca Marzola, Christian Veelken
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3.9.6.28 Can Bell inequalities be tested via scattering cross-section at colliders ? [410]

- **Authors:** Song Li, Wei Shen, Jin Min Yang
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3.9.6.29 Entanglement Autodistillation from Particle Decays [411]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
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3.9.6.30 Quantum detection of new physics in top-quark pair production at the LHC [412]

- **Authors:** Fabio Maltoni, Claudio Severi, Simone Tentori, Eleni Vryonidou
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3.9.6.31 Full quantum tomography of top quark decays [413]

- **Authors:** J.A. Aguilar-Saavedra
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3.9.6.32 New physics in spin entanglement [414]

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3.9.6.33 On bipartite and tripartite entanglement at present and future particle colliders [415]

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3.9.6.34 Observation of quantum entanglement in top quark pair production in proton–proton collisions at $\sqrt{s} = 13$ TeV [416]

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3.9.6.35 Entanglement and Bell nonlocality with bottom-quark pairs at hadron colliders [417]

- **Authors:** Yoav Afik, Yevgeny Kats, Juan Ramón Muñoz de Nova, Abner Soffer, David Uzan
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3.9.6.36 Magic states of top quarks [418]

- **Authors:** Chris D. White, Martin J. White
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3.9.6.37 Bell nonlocality and entanglement in $e^+e^-\to YY^-$ at BESIII [419]

- **Authors:** Sihao Wu, Chen Qian, Qun Wang, Xiao-Rong Zhou
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3.9.6.38 Bell inequality is violated in charmonium decays [420]

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3.9.6.39 Measurements of polarization and spin correlation and observation of entanglement in top quark pairs using lepton + jets events from proton-proton collisions at $\sqrt{s} = 13\,\text{TeV}$ [421]

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3.9.6.40 An improved Bell-CHSH observable for gauge boson pairs [422]

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3.9.6.41 Measuring quantum discord at the LHC [423]

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3.9.6.42 Bell Inequality Violation of Light Quarks in Dihadron Pair Production at Lepton Colliders [424]

- **Authors:** Kun Cheng, Bin Yan
- **Posted on arXiv:** 06 January 2025
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3.9.6.43 Quantum information meets high-energy physics: input to the update of the European strategy for particle physics [425]

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3.9.6.44 Quantum Entanglement between gauge boson pairs at a Muon Collider [426]

- **Authors:** Ran Ding, Alim Ruzi, Sitian Qian, Andrew Levin, Youpeng Wu, Qiang Li
- **Posted on arXiv:** 13 April 2025

3.9.6.45 Tests of quantum contextuality in particle physics [427]

- **Authors:** M. Fabbrichesi, R. Floreanini, E. Gabrielli, L. Marzola
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3.9.6.46 Test of local realism via entangled $\Lambda\bar{\Lambda}$ system [428]

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3.9.6.47 Precise Measurement of the Λ Electric Dipole Moment through the Entangled Strange Baryon-Antibaryon System [429]

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- **Posted on arXiv:** 23 June 2025

3.9.6.48 Colliders are not testing locality via Bell's inequality nor providing an unconditional proof of entanglement [430]

- **Authors:** Steven A. Abel, Herbi K. Dreiner, Rhitaja Sengupta, Lorenzo Ubaldi
- **Posted on arXiv:** 21 July 2025
- **Published in JHEP:** 07 August 2026

3.9.7 Quantum Correlations in Scattering Amplitudes

3.9.7.1 Entanglement Suppression and Emergent Symmetries of Strong Interactions [436]

- **Authors:** Silas R. Beane, David B. Kaplan, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 07 December 2018
- **Published in Phys. Rev. Lett. 122, 102001 (2019):** 16 March 2019

3.9.7.2 Geometry and entanglement in the scattering matrix [437]

- **Authors:** Silas R. Beane, Roland C. Farrell
- **Posted on arXiv:** 02 November 2020
- **Published in Annals Phys.:** 16 August 2021

3.9.7.3 Entanglement minimization in hadronic scattering with pions [438]

- **Authors:** Silas R. Beane, Roland C. Farrell, Mira Varma
- **Posted on arXiv:** 02 August 2021
- **Published in Int.J.Mod.Phys.A:** 11 November 2021

3.9.7.4 Tree-level entanglement in quantum electrodynamics [439]

- **Authors:** Samuel Fedida, Alessio Serafini
- **Posted on arXiv:** 03 September 2022
- **Published in Phys. Rev. D. 107:116007, Jun 2023:** 01 June 2023

3.9.7.5 Positivity in Amplitudes and Quantum Entanglement [440]

- **Authors:** Rafael Aoude, Gilly Elor, Grant N. Remmen, Olcyr Sumensari
- **Posted on arXiv:** 26 February 2024
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3.9.7.6 Area law for entanglement entropy in particle scattering [441]

- **Authors:** Ian Low, Zhewei Yin
- **Posted on arXiv:** 13 May 2024
- **Published in Phys.Rev.D:** 06 March 2026

3.9.7.7 Qubit entanglement from forward scattering [442]

- **Authors:** Kamila Kowalska, Enrico Maria Sessolo
- **Posted on arXiv:** 05 October 2025
- **Published in JHEP:** 01 April 2026

3.9.8 Crosslists

3.9.8.1 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

3.9.8.2 Standard model physics and the digital quantum revolution: thoughts about the interface [16]

- **Authors:** Natalie Klco, Alessandro Roggero, Martin J. Savage
- **Posted on arXiv:** 10 July 2021
- **Published in Rept.Prog.Phys.:** 19 May 2022

3.9.8.3 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [18]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.9.8.4 Multi-Neutrino Entanglement and Correlations in Dense Neutrino Systems [310]

- **Authors:** Marc Illa, Martin J. Savage
- **Posted on arXiv:** 16 October 2022
- **Published in Phys. Rev. Lett. 130, 221003 (2023):** 31 May 2023

3.9.8.5 Many-Body Collective Neutrino Oscillations: Recent Developments [6]

- **Authors:** Amol V. Patwardhan, Michael J. Cervia, Ermal Rrapaj, Pooja Siwach, A.B. Balantekin
- **Posted on arXiv:** 31 December 2022
- **Published in nan:** 2023

3.9.8.6 Quantum information and quantum simulation of neutrino physics [8]

- **Authors:** A.B. Balantekin, Michael J. Cervia, Amol V. Patwardhan, Ermal Rrapaj, Pooja Siwach
- **Posted on arXiv:** 01 May 2023
- **Published in Eur.Phys.J.A:** 17 August 2023

3.9.8.7 Quantum entanglement and Bell inequality violation at colliders [10]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

3.9.8.8 Quantum Complexity and New Directions in Nuclear Physics and High-Energy Physics Phenomenology [11]

- **Authors:** Caroline E.P. Robin, Martin J. Savage
- **Posted on arXiv:** 29 April 2026

3.10 Quantum Error Correction and Mitigation

3.10.1 Dark Matter - Wave-like Dark Matter

3.10.1.1 Quantum-Error-Correction-Inspired Noise Mitigation for Wavelike Dark Matter Searches with Quantum Sensors [65]

- **Authors:** Hajime Fukuda, Takeo Moroi, Thanaporn Sichanugrist
- **Posted on arXiv:** 05 November 2025
- **Published in Phys.Rev.Lett.:** 08 July 2026

3.10.2 Lattice Field Theories - Scalar/Fermion Theories

3.10.2.1 Hierarchical qubit maps and hierarchically implemented quantum error correction [242]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 04 September 2021
- **Published in Phys.Rev.A:** 15 December 2021

3.10.3 Lattice Field Theories - Gauge Theories

3.10.3.1 Quantum error correction with gauge symmetries [288]

- **Authors:** Abhishek Rajput, Alessandro Roggero, Nathan Wiebe
- **Posted on arXiv:** 09 December 2021
- **Published in npj Quantum Inf.:** 25 April 2023

3.10.3.2 Mitigating depolarizing noise on quantum computers with noise-estimation circuits [289]

- **Authors:** Miroslav Urbanek, Benjamin Nachman, Vincent R. Pascuzzi, Andre He, Christian W. Bauer, Wibe A. de Jong
- **Posted on arXiv:** 15 March 2021
- **Published in Phys.Rev.Lett.:** 2021

3.10.3.3 Robustness of Gauge Digitization to Quantum Noise [290]

- **Authors:** Erik J. Gustafson, Henry Lamm
- **Posted on arXiv:** 24 January 2023

3.10.3.4 Quantum error thresholds for gauge-redundant digitizations of lattice field theories [291]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 26 February 2024
- **Published in Phys.Rev.D:** 01 September 2024

3.10.3.5 Symmetry verification for noisy quantum simulations of non-Abelian lattice gauge theories [292]

- **Authors:** Edoardo Ballini, Julius Mildenberger, Matteo M. Wauters, Philipp Hauke
- **Posted on arXiv:** 10 December 2024
- **Published in Quantum 9, 1802 (2025):** 22 July 2025

3.10.4 Lattice Field Theories - Schwinger Model

3.10.4.1 Measurement error mitigation in quantum computers through classical bit-flip correction [304]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn, Paolo Stornati, Xiaoyang Wang
- **Posted on arXiv:** 07 July 2020
- **Published in Phys. Rev. A 105, 062404 (2022):** 03 June 2022

3.10.5 Crosslists

3.10.5.1 Quantum error mitigation for Fourier moment computation [359]

- **Authors:** Oriel Kiss, Michele Grossi, Alessandro Roggero
- **Posted on arXiv:** 23 January 2024
- **Published in Phys. Rev. D 111, 034504 (2025):** 01 February 2025

3.11 Quantum Machine Learning - Supervised Methods

3.11.1 Anomaly Detection

3.11.1.1 Unravelling physics beyond the standard model with classical and quantum anomaly detection [33]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

3.11.2 Event Classification

3.11.2.1 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [189]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevior
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

3.11.2.2 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [190]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

3.11.2.3 Higgs analysis with quantum classifiers [191]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in nan:** 2021

3.11.2.4 Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets [192]

- **Authors:** Zhelun Li, Lento Nagano, Koji Terashi
- **Posted on arXiv:** 17 May 2024
- **Published in Phys.Rev.Res.:** 10 October 2024

3.11.2.5 From Reachability to Learnability: Geometric Design Principles for Quantum Neural Networks [193]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky
- **Posted on arXiv:** 03 March 2026

3.11.3 Lattice Field Theories - Gauge Theories

3.11.3.1 Quantum data learning for quantum simulations in high-energy physics [214]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

3.11.4 Track Reconstruction

3.11.4.1 Reconstructing charged particle track segments with a quantum-enhanced support vector machine [454]

- **Authors:** Philippa Duckett, Gabriel Facini, Marcin Jastrzebski, Sarah Malik, Tim Scanlon, Sebastien Rettie
- **Posted on arXiv:** 14 December 2022
- **Published in Phys.Rev.D:** 01 March 2024

3.11.5 Crosslists

3.11.5.1 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.11.5.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.11.5.3 Quantum-centric Supercomputing for Physics Research [170]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.12 Quantum Machine Learning - Unsupervised Methods

3.12.1 Anomaly Detection

3.12.1.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [34]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

3.12.2 Detector Technologies and Simulations

3.12.2.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [161]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.12.2.2 Running the Dual-PQC GAN on noisy simulators and real quantum hardware [162]

- **Authors:** Su Yeon Chang, Edwin Agnew, Elías F. Combarro, Michele Grossi, Steven Herbert, Sofia Vallecorsa
- **Posted on arXiv:** 30 May 2022
- **Published in nan:** 2023

3.12.2.3 Precise image generation on current noisy quantum computing devices [163]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borras, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

3.12.3 Event Classification

3.12.3.1 Guided graph compression for quantum graph neural networks [194]

- **Authors:** Mikel Casals, Vasilis Belis, Elias F. Combarro, Eduard Alarcón, Sofia Vallecorsa, Michele Grossi
- **Posted on arXiv:** 11 June 2025
- **Published in Mach.Learn.Sci.Tech.:** 12 September 2025

3.12.4 Event Generation

3.12.4.1 Style-based quantum generative adversarial networks for Monte Carlo events [208]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

3.12.4.2 Quantum integration of elementary particle processes [209]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

3.12.4.3 Generative invertible quantum neural networks [210]

- **Authors:** Armand Rousselot, Michael Spannowsky
- **Posted on arXiv:** 24 February 2023
- **Published in SciPost Phys. 16, 146 (2024):** 04 June 2024

3.12.4.4 Quantum-centric Supercomputing for Physics Research [170]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.12.5 Jet Reconstruction

3.12.5.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [229]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

3.12.5.2 Quantum clustering and jet reconstruction at the LHC [230]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

3.12.6 Crosslists

3.12.6.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [159]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.12.6.2 Unsupervised event classification with graphs on classical and photonic quantum computers [31]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

3.12.6.3 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.12.6.4 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico

Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.12.6.5 Machine learning for anomaly detection in particle physics [9]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klaeboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

3.13 Quantum Machine Learning - Variational Quantum Algorithms

3.13.1 Anomaly Detection

3.13.1.1 Anomaly detection in high-energy physics using a quantum autoencoder [35]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

3.13.1.2 Quantum anomaly detection for collider physics [36]

- **Authors:** Sulaiman Alvi, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 16 June 2022
- **Published in JHEP:** 22 February 2023

3.13.2 Beyond the Standard Model

3.13.2.1 Hybrid Quantum-Classical Graph Convolutional Network [41]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

3.13.3 Event Classification

3.13.3.1 Event Classification with Quantum Machine Learning in High-Energy Physics [43]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

3.13.3.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [44]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

3.13.3.3 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [195]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

3.13.3.4 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [196]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

3.13.3.5 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [197]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

3.13.3.6 Quantum Metric Learning for New Physics Searches at the LHC [47]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

3.13.3.7 Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics [198]

- **Authors:** Eyup B. Unlu, Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva
- **Posted on arXiv:** 01 February 2024
- **Published in Axioms v. 13, no 3, (2024) 187:** 13 March 2024

3.13.3.8 Quantum Vision Transformers for Quark–Gluon Classification [199]

- **Authors:** Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva, Eyup B. Unlu
- **Posted on arXiv:** 16 May 2024
- **Published in Axioms 2024, 13(5), 323:** 13 May 2024

3.13.4 Event Generation

3.13.4.1 Partonic collinear structure by quantum computing [211]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

3.13.4.2 Unsupervised quantum circuit learning in high energy physics [212]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.13.4.3 Quantum Wishlist: Lessons from Parton Showers [213]

- **Authors:** Masahito Yamazaki
- **Posted on arXiv:** 25 February 2025
- **Published in nan:** 24 October 2025

3.13.5 Jet Reconstruction

3.13.5.1 Quantum Machine Learning for b-jet charge identification [231]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

3.13.6 Lattice Field Theories - Gauge Theories

3.13.6.1 SU(2) hadrons on a quantum computer via a variational approach [293]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

3.13.7 Lattice Field Theories - Schwinger Model

3.13.7.1 Quantum-classical computation of Schwinger model dynamics using quantum computers [305]

- **Authors:** N. Klco, E.F. Dumitrescu, A.J. McCaskey, T.D. Morris, R.C. Pooser, M. Sanz, E. Solano, P. Lougovski, M.J. Savage
- **Posted on arXiv:** 08 March 2018
- **Published in Phys. Rev. A 98, 032331 (2018):** 29 September 2018

3.13.7.2 Quantum-Classical Simulation of Quantum Field Theory by Quantum Circuit Learning [306]

- **Authors:** Kazuki Ikeda
- **Posted on arXiv:** 27 November 2023
- **Published in Annalen Phys.:** 01 June 2025

3.13.8 Neutrino Physics

3.13.8.1 Collective neutrino oscillations on a quantum computer with hybrid quantum-classical algorithm [332]

- **Authors:** Pooja Siwach, Kaytlin Harrison, A. Baha Balantekin
- **Posted on arXiv:** 17 August 2023
- **Published in Phys.Rev.D:** 15 October 2023

3.13.8.2 Three-flavor supernova neutrino simulation using a hybrid quantum-classical algorithm with qutrits [333]

- **Authors:** Daniel J. Heimsoth, Pooja Siwach, A. Baha Balantekin
- **Posted on arXiv:** 01 May 2026
- **Published in Phys. Rev. D 114, 043055 (2026):** 15 August 2026

3.13.9 Nuclear Many-Body Physics - Nuclear Structure

3.13.9.1 Cloud Quantum Computing of an Atomic Nucleus [178]

- **Authors:** E.F. Dumitrescu, A.J. McCaskey, G. Hagen, G.R. Jansen, T.D. Morris, T. Papenbrock, R.C. Pooser, D.J. Dean, P. Lougovski
- **Posted on arXiv:** 11 January 2018
- **Published in Phys. Rev. Lett. 120, 210501 (2018):** 24 May 2018

3.13.9.2 Improving Hamiltonian encodings with the Gray code [180]

- **Authors:** Olivia Di Matteo, Anna McCoy, Peter Gysbers, Takayuki Miyagi, R.M. Woloshyn, Petr Navrátil
- **Posted on arXiv:** 11 August 2020
- **Published in Phys. Rev. A 103, 042405 (2021):** 03 April 2021

3.13.9.3 Lipkin model on a quantum computer [339]

- **Authors:** Michael J. Cervia, A.B. Balantekin, S.N. Coppersmith, Calvin W. Johnson, Peter J. Love, C. Poole, K. Robbins, M. Saffman
- **Posted on arXiv:** 08 November 2020
- **Published in Phys. Rev. C 104, 024305 (2021):** 03 August 2021

3.13.9.4 Variational approaches to constructing the many-body nuclear ground state for quantum computing [340]

- **Authors:** I. Stetcu, A. Baroni, J. Carlson
- **Posted on arXiv:** 12 October 2021
- **Published in Phys. Rev. C 105, 064308 (2022):** 21 June 2022

3.13.9.5 Simulating excited states of the Lipkin model on a quantum computer [341]

- **Authors:** Manqoba Q. Hlatshwayo, Yinu Zhang, Herlik Wibowo, Ryan LaRose, Denis Lacroix, Elena Litvinova
- **Posted on arXiv:** 02 March 2022
- **Published in Phys.Rev.C:** 18 August 2022

3.13.9.6 Solving nuclear structure problems with the adaptive variational quantum algorithm [342]

- **Authors:** A.M. Romero, J. Engel, Ho Lun Tang, Sophia E. Economou
- **Posted on arXiv:** 03 March 2022
- **Published in Phys. Rev. C 105, 064317 (2022):** 27 June 2022

3.13.9.7 Quantum computing of the ^{6}Li nucleus via ordered unitary coupled cluster [343]

- **Authors:** Oriel Kiss, Michele Grossi, Pavel Lougovski, Federico Sanchez, Sofia Vallecorsa, Thomas Papenbrock
- **Posted on arXiv:** 02 May 2022
- **Published in Phys. Rev. C 106, 034325 (2022):** 29 September 2022

3.13.9.8 QCSH: A full quantum computer nuclear shell-model package [344]

- **Authors:** Peng Lv, Shi-Jie Wei, Hao-Nan Xie, Gui-Lu Long
- **Posted on arXiv:** 24 May 2022
- **Published in Sci.China Phys.Mech.Astron.:** 06 March 2023

3.13.9.9 Subatomic Many-Body Physics Simulations on a Quantum Frequency Processor [182]

- **Authors:** Hsuan-Hao Lu, Natalie Klco, Joseph M. Lukens, Titus D. Morris, Aaina Bansal, Andreas Ekström, Gaute Hagen, Thomas Papenbrock, Andrew M. Weiner, Martin J. Savage, Pavel Lougovski
- **Published in nan:** 2019

3.13.9.10 Nuclear shell-model simulation in digital quantum computers [345]

- **Authors:** A. Pérez-Obiol, A.M. Romero, J. Menéndez, A. Rios, A. García-Sáez, B. Juliá-Díaz
- **Posted on arXiv:** 07 February 2023
- **Published in Sci.Rep.:** 29 July 2023

3.13.9.11 Deep quantum circuit simulations of low-energy nuclear states [346]

- **Authors:** Ang Li, Alessandro Baroni, Ionel Stetcu, Travis S. Humble
- **Posted on arXiv:** 26 October 2023
- **Published in Eur.Phys.J.A:** 14 May 2024

3.13.9.12 Exploiting symmetries in nuclear Hamiltonians for ground state preparation [347]

- **Authors:** Joe Gibbs, Zoë Holmes, Paul Stevenson
- **Posted on arXiv:** 15 February 2024
- **Published in Quantum Machine Intelligence:** 30 January 2025

3.13.9.13 Comparison of variational quantum eigensolvers in light nuclei [348]

- **Authors:** Miquel Carrasco-Codina, Emanuele Costa, Antonio Márquez Romero, Javier Menéndez, Arnau Rios
- **Posted on arXiv:** 18 July 2025
- **Published in Phys.Rev.C:** 23 February 2026

3.13.9.14 Toward scalable quantum computations of atomic nuclei [349]

- **Authors:** Chenyi Gu, Matthias Heinz, Oriel Kiss, Thomas Papenbrock
- **Posted on arXiv:** 19 July 2025
- **Published in Phys. Rev. C 113, 034321 (2026):** 24 March 2026

3.13.9.15 Bridging quantum computing and nuclear structure: Atomic nuclei on a trapped-ion quantum computer [350]

- **Authors:** Sota Yoshida, Takeshi Sato, Takumi Ogata, Masaaki Kimura
- **Posted on arXiv:** 25 September 2025
- **Published in Phys. Rev. Research 8, 013134 (2026):** 06 February 2026

3.13.10 Nuclear Many-Body Physics - Nuclear Reactions and Response

3.13.10.1 Quantum simulations of nuclear resonances with variational methods [363]

- **Authors:** Ashutosh Singh, Pooja Siwach, P. Arumugam
- **Posted on arXiv:** 15 April 2025
- **Published in Phys.Rev.C:** 25 August 2025

3.13.11 Track Reconstruction

3.13.11.1 Quantum convolutional neural networks for high energy physics data analysis [45]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

3.13.11.2 Hybrid quantum classical graph neural networks for particle track reconstruction [455]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

3.13.11.3 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [456]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in nan:** 2023

3.13.11.4 Particle track reconstruction with noisy intermediate-scale quantum computers [457]

- **Authors:** Tim Schwägerl, Cigdem Issever, Karl Jansen, Teng Jian Khoo, Stefan Kühn, Cenk Tüysüz, Hannsjörg Weber
- **Posted on arXiv:** 23 March 2023
- **Published in J.Phys.Conf.Ser.:** 2026

3.13.11.5 Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm [458]

- **Authors:** Hideki Okawa
- **Posted on arXiv:** 16 October 2023

3.13.12 Crosslists

3.13.12.1 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

3.13.12.2 Simulating Lattice Gauge Theories within Quantum Technologies [253]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.13.12.3 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.13.12.4 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [161]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.13.12.5 Higgs analysis with quantum classifiers [191]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in nan:** 2021

3.13.12.6 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [20]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.13.12.7 Quantum Computing for Data Analysis in High-Energy Physics [22]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.13.12.8 Quantum Computing Applications in Future Colliders [5]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.13.12.9 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham

Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao

- **Posted on arXiv:** 28 February 2023

3.13.12.10 Quantum Computing for High-Energy Physics: State of the Art and Challenges [29]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023

- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.13.12.11 Precise image generation on current noisy quantum computing devices [163]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borras, Dirk Krücker, Michele Grossi, Valle Varo

- **Posted on arXiv:** 11 July 2023

- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

3.13.12.12 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [295]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage

- **Posted on arXiv:** 08 August 2023

- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

3.13.12.13 Quantum Algorithms in Particle Physics [215]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.13.12.14 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [275]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

3.13.12.15 Steps toward quantum simulations of hadronization and energy loss in dense matter [296]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

3.14 Quantum Measurement and Tomography

3.14.1 Beyond the Standard Model

3.14.1.1 Emergent symmetry in a two-Higgs-doublet model from quantum information and nonstabilizerness [42]

- **Authors:** Giorgio Busoni, John Gargalionis, Ewan N.V. Wallace, Martin J. White
- **Posted on arXiv:** 02 June 2025
- **Published in Phys. Rev. D. 112 (2025) 035022:** 01 August 2025

3.14.2 Nuclear Many-Body Physics - Nuclear Structure

3.14.2.1 Shadow-based quantum subspace algorithm for the nuclear shell model [351]

- **Authors:** Ruyu Yang, Tianren Wang, Bing-Nan Lu, Ying Li, Xiaosi Xu
- **Posted on arXiv:** 15 June 2023
- **Published in Phys.Rev.A:** 21 January 2025

3.14.2.2 Quantum complexity fluctuations from nuclear and hypernuclear forces [352]

- **Authors:** Caroline E. P. Robin, Martin J. Savage
- **Posted on arXiv:** 16 May 2024
- **Published in Phys.Rev.C 112, 044004 (2025):** 17 October 2025

3.14.3 Nuclear Many-Body Physics - Nuclear Reactions and Response

3.14.3.1 Antiflatness and nonlocal nonstabilizerness in two-particle scattering processes [364]

- **Authors:** Caroline E.P. Robin, Martin J. Savage
- **Posted on arXiv:** 27 October 2025
- **Published in Phys. Rev. D 114, 014007 (2026):** 01 July 2026

3.14.4 Quantum Correlations at Colliders

3.14.4.1 The magic of top quarks [431]

- **Authors:** Chris D. White, Martin J. White
- **Posted on arXiv:** 10 December 2024
- **Published in nan:** 29 January 2026

3.14.4.2 Probing new physics in the top sector using quantum information [432]

- **Authors:** Rafael Aoude, Hannah Banks, Chris D. White, Martin J. White
- **Posted on arXiv:** 18 May 2025
- **Published in Phys.Rev.D 113 (2026) 11, 115066:** 01 June 2026

3.14.4.3 Spin versus nonstabilizerness in gluon and graviton scattering [433]

- **Authors:** John Gargalionis, Nathan Moynihan, Sokratis Trifinopoulos, Ewan N.V. Wallace, Chris D. White, Martin J. White
- **Posted on arXiv:** 20 August 2025
- **Published in Phys.Rev.D:** 01 January 2026

3.14.4.4 Z-boson quantum tomography at next-to-leading order [434]

- **Authors:** Morgan Del Gratta, Federica Fabbri, Michele Grossi, Fabio Maltoni, Davide Pagani, Giovanni Pelliccioli, Alessandro Vicini
- **Posted on arXiv:** 24 September 2025
- **Published in JHEP:** 04 February 2026

3.14.4.5 Automated computation of spin-density matrices and quantum observables for collider physics [435]

- **Authors:** Valentin Durupt, Fabio Maltoni, Olivier Mattelaer
- **Posted on arXiv:** 20 October 2025
- **Published in JHEP:** 14 April 2026

3.14.5 Quantum Correlations in Scattering Amplitudes

3.14.5.1 Quantum magic in quantum electrodynamics [443]

- **Authors:** Qiaofeng Liu, Ian Low, Zhewei Yin
- **Posted on arXiv:** 04 March 2025
- **Published in Phys.Rev.D:** 01 March 2026

3.14.6 Crosslists

3.14.6.1 Entanglement and quantum tomography with top quarks at the LHC [386]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

3.14.6.2 Quantum SMEFT tomography: Top quark pair production at the LHC [181]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

3.14.6.3 Quantum state tomography, entanglement detection and Bell violation prospects in weak decays of massive particles [398]

- **Authors:** Rachel Ashby-Pickering, Alan J. Barr, Agnieszka Wierzchucka
- **Posted on arXiv:** 28 September 2022
- **Published in JHEP:** 02 May 2023

3.14.6.4 Full quantum tomography of top quark decays [413]

- **Authors:** J.A. Aguilar-Saavedra
- **Posted on arXiv:** 22 February 2024
- **Published in Phys.Lett.B:** 04 July 2024

3.14.6.5 Quantum Complexity and New Directions in Nuclear Physics and High-Energy Physics Phenomenology [11]

- **Authors:** Caroline E.P. Robin, Martin J. Savage
- **Posted on arXiv:** 29 April 2026

3.15 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

3.15.1 Dark Matter - Particle-like Dark Matter

3.15.1.1 Trapped Electrons and Ions as Particle Detectors [49]

- **Authors:** Daniel Carney, Hartmut Häffner, David C. Moore, Jacob M. Taylor
- **Posted on arXiv:** 12 April 2021
- **Published in Phys Rev Lett, Vol. 127, No. 6, 2021. "Editor's suggestion":** 05 August 2021

3.15.1.2 Millicharged Dark Matter Detection with Ion Traps [50]

- **Authors:** Dmitry Budker, Peter W. Graham, Harikrishnan Ramani, Ferdinand Schmidt-Kaler, Christian Smorra, Stefan Ulmer
- **Posted on arXiv:** 11 August 2021
- **Published in PRX Quantum:** 01 February 2022

3.15.2 Dark Matter - Wave-like Dark Matter

3.15.2.1 The Global Network of Optical Magnetometers for Exotic physics (GNOME): A novel scheme to search for physics beyond the Standard Model [66]

- **Authors:** Szymon Pustelny, Derek F. Jackson Kimball, Chris Pankow, Micah P. Ledbetter, Przemyslaw Wlodarczyk, Piotr Wcislo, Maxim Pospelov, Joshua R. Smith, Jocelyn Read, Wojciech Gawlik, Dmitry Budker
- **Posted on arXiv:** 22 March 2013
- **Published in Annalen der Physik 525, 659 (2013):** 2013

3.15.2.2 The cosmic axion spin precession experiment (CASPEr): a dark-matter search with nuclear magnetic resonance [67]

- **Authors:** Antoine Garcon, Deniz Aybas, John W. Blanchard, Gary Centers, Nataniel L. Figueroa, Peter W. Graham, Derek F. Jackson Kimball, Surjeet Rajendran, Marina Gil Sendra, Alexander O. Sushkov, Lutz Trahms, Tao Wang, Arne Wickenbrock, Teng Wu, Dmitry Budker
- **Posted on arXiv:** 16 July 2017
- **Published in Quantum Science and Technology 2017:** 15 December 2017

3.15.2.3 Overview of the Cosmic Axion Spin Precession Experiment (CASPEr) [68]

- **Authors:** Derek F. Jackson Kimball, S. Afach, D. Aybas, J.W. Blanchard, D. Budker, G. Centers, M. Engler, N.L. Figueroa, A. Garcon, P.W. Graham, H. Luo, S. Rajendran, M.G. Sendra, A.O. Sushkov, T. Wang, A. Wickenbrock, A. Wilzewski, T. Wu
- **Posted on arXiv:** 09 November 2017
- **Published in Springer Proc.Phys.:** 2020

3.15.2.4 Precision Metrology Meets Cosmology: Improved Constraints on Ultralight Dark Matter from Atom-Cavity Frequency Comparisons [69]

- **Authors:** Colin J. Kennedy, Eric Oelker, John M. Robinson, Tobias Bothwell, Dhruv Kedar, William R. Milner, G. Edward Marti, Andrei Derevianko, Jun Ye
- **Posted on arXiv:** 20 August 2020
- **Published in Phys. Rev. Lett. 125, 201302 (2020):** 12 November 2020

3.15.2.5 Search for Axionlike Dark Matter Using Solid-State Nuclear Magnetic Resonance [70]

- **Authors:** Deniz Aybas, Janos Adam, Emmy Blumenthal, Alexander V. Gramolin, Dorian Johnson, Annalies Kleyheeg, Samer Afach, John W. Blanchard, Gary P. Centers, Antoine Garcon, Martin Engler, Nataniel L. Figueroa, Marina Gil Sendra, Arne Wickenbrock, Matthew Lawson, Tao Wang, Teng Wu, Haosu Luo, Hamdi Mani, Philip Mauskopf, Peter W. Graham, Surjeet Rajendran, Derek F. Jackson Kimball, Dmitry Budker, Alexander O. Sushkov
- **Posted on arXiv:** 04 January 2021
- **Published in Phys. Rev. Lett. 126, 141802 (2021):** 10 April 2021

3.15.2.6 Search for Dark-Matter-Induced Oscillations of Fundamental Constants Using Molecular Spectroscopy [71]

- **Authors:** R. Oswald, A. Nevsky, V. Vogt, S. Schiller, N.L. Figueroa, K. Zhang, O. Tretiak, D. Antypas, D. Budker, A. Banerjee, G. Perez
- **Posted on arXiv:** 12 November 2021
- **Published in Phys.Rev.Lett.:** 15 July 2022

3.15.2.7 Measuring the stability of fundamental constants with a network of clocks [72]

- **Authors:** G. Barontini, L. Blackburn, V. Boyer, F. Butuc-Mayer, X. Calmet, J.R. Crespo Lopez-Urrutia, E.A. Curtis, B. Darquie, J. Dunningham, N.J. Fitch, E.M. Forgan, K. Georgiou, P. Gill, R.M. Godun, J. Goldwin, V. Guarrera, A.C. Harwood, I.R. Hill, R.J. Hendricks, M. Jeong, M.Y.H. Johnson, M. Keller, L.P. Kozhiparambil Sajith, F. Kuipers, H.S. Margolis, C. Mayo, P. Newman, A.O. Parsons, L. Prokhorov, B.I. Robertson, J. Rodewald, M.S. Safronova, B.E. Sauer, M. Schioppo, N. Sherrill, Y.V. Stadnik, K. Szymaniec, M.R. Tarbutt, R.C. Thompson, A. Tofful, J. Tunesi, A. Vecchio, Y. Wang, S. Worm
- **Posted on arXiv:** 20 December 2021
- **Published in EPJ Quantum Technology volume 9, Article number: 12 (2022):** 11 May 2022

3.15.2.8 Intensity interferometry for ultralight bosonic dark matter detection [73]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball
- **Posted on arXiv:** 05 February 2022
- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

3.15.2.9 One-Electron Quantum Cyclotron as a Milli-eV Dark-Photon Detector [74]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Roni Harnik, Thomas G. Myers, Harikrishnan Ramani, Benedict A.D. Sukra, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 12 August 2022
- **Published in Phys. Rev. Lett. 129, 261801 (2022):** 23 December 2022

3.15.2.10 Search for Ultralight Dark Matter from Long-Term Frequency Comparisons of Optical and Microwave Atomic Clocks [75]

- **Authors:** Takumi Kobayashi, Akifumi Takamizawa, Daisuke Akamatsu, Akio Kawasaki, Akiko Nishiyama, Kazumoto Hosaka, Yusuke Hisai, Masato Wada, Hajime Inaba, Takehiko Tanabe, Masami Yasuda
- **Posted on arXiv:** 12 December 2022
- **Published in Phys. Rev. Lett. 129, 241301 (2022):** 07 December 2022

3.15.2.11 Improved Limits on the Coupling of Ultralight Bosonic Dark Matter to Photons from Optical Atomic Clock Comparisons [76]

- **Authors:** M. Filzinger, S. Dörscher, R. Lange, J. Klose, M. Steinel, E. Benkler, E. Peik, C. Lisdat, N. Huntemann
- **Posted on arXiv:** 09 January 2023
- **Published in Phys.Rev.Lett.:** 22 June 2023

3.15.2.12 Oscillating Nuclear Charge Radii as Sensors for Ultralight Dark Matter [77]

- **Authors:** Abhishek Banerjee, Dmitry Budker, Melina Filzinger, Nils Huntemann, Gil Paz, Gilad Perez, Sergey Porsev, Marianna Safronova
- **Posted on arXiv:** 25 January 2023
- **Published in Phys. Rev. Lett. 135 (2025) 22, 223001:** 24 November 2025

3.15.2.13 Analysis of atomic-clock data to constrain variations of fundamental constants [78]

- **Authors:** Nathaniel Sherrill, Adam O. Parsons, Charles F.A. Baynham, William Bowden, E. Anne Curtis, Richard Hendricks, Ian R. Hill, Richard Hobson, Helen S. Margolis, Billy I. Robertson, Marco Schioppo, Krzysztof Szymaniec, Alexandra Tofful, Jacob Tunesi, Rachel M. Godun, Xavier Calmet
- **Posted on arXiv:** 09 February 2023
- **Published in New J.Phys.:** 06 September 2023

3.15.2.14 Frequency ratio measurements at 18-digit accuracy using an optical clock network [79]

- **Authors:** Kyle Beloy, Martha I. Bodine, Tobias Bothwell, Samuel M. Brewer, Sarah L. Bromley, Jwo-Sy Chen, Jean-Daniel Deschênes, Scott A. Diddams, Robert J. Fasano, Tara M. Fortier, Youssef S. Hassan, David B. Hume, Dhruv Kedar, Colin J. Kennedy, Isaac Khader, Amanda Koepke, David R. Leibrandt, Holly Leopardi, Andrew D. Ludlow, William F. McGrew, William R. Milner, Nathan R. Newbury, Daniele Nicolodi, Eric Oelker, Thomas E. Parker, John M. Robinson, Stefania Romisch, Stefan A. Schäffer, Jeffrey A. Sherman, Laura C. Sinclair, Lindsay Sonderhouse, William C. Swann, Jian Yao, Jun Ye, Xiaogang Zhang
- **Posted on arXiv:** 29 May 2020
- **Published in Nature:** 24 March 2021

3.15.2.15 Detection of bosenovae with quantum sensors on Earth and in space [80]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

3.15.2.16 Rydberg-atom-based single-photon detection for haloscope axion searches [81]

- **Authors:** Eleanor Graham, Sumita Ghosh, Yuqi Zhu, Xiran Bai, Sidney B. Cahn, Elsa Durcan, Michael J. Jewell, Danielle H. Speller, Sabrina M. Zacarias, Laura T. Zhou, Reina H. Maruyama
- **Posted on arXiv:** 23 October 2023
- **Published in Phys.Rev.D:** 01 February 2024

3.15.2.17 Quantum entanglement of ions for light dark matter detection [82]

- **Authors:** Asuka Ito, Ryuichiro Kitano, Wakutaka Nakano, Ryoto Takai
- **Posted on arXiv:** 20 November 2023
- **Published in JHEP:** 16 February 2024

3.15.2.18 Nuclear interferometer for ultralight dark matter detection [83]

- **Authors:** Hannah Banks, Elina Fuchs, Matthew McCullough
- **Posted on arXiv:** 15 July 2024
- **Published in Phys. Rev. D, 2026:** 01 July 2026

3.15.2.19 Highly excited electron cyclotron for QCD axion and dark-photon detection [84]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Harikrishnan Ramani, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 07 October 2024
- **Published in Phys.Rev.D 111 (2025) 7, 075022:** 01 April 2025

3.15.2.20 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [85]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

3.15.2.21 Sensitively searching for microwave dark photons with atomic ensembles [86]

- **Authors:** Suirong He, De He, Yufen Li, Li Gao, Xianing Feng, Hao Zheng, L.F. Wei
- **Posted on arXiv:** 01 December 2024
- **Published in Phys.Rev.D:** 15 September 2025

3.15.2.22 Searches for exotic spin-dependent interactions with spin sensors [87]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

3.15.2.23 Spin squeezing of macroscopic nuclear spin ensembles [88]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

3.15.2.24 Detecting Dark Matter Using Optically Trapped Rydberg Atom Tweezer Arrays [89]

- **Authors:** So Chigusa, Taiyo Kasamaki, Toshi Kusano, Takeo Moroi, Kazunori Nakayama, Naoya Ozawa, Yoshiro Takahashi, Atsuhiro Umemoto, Amar Vutha
- **Posted on arXiv:** 17 July 2025
- **Published in Phys. Rev. Lett. 136, 151801 (2026):** 14 April 2026

3.15.2.25 On the Speed-up of Wave-like Dark Matter Searches with Entangled Qubits [90]

- **Authors:** Arushi Bodas, Sohitri Ghosh, Roni Harnik
- **Posted on arXiv:** 13 October 2025

3.15.2.26 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [91]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

3.15.2.27 A thorium-229 optical nuclear clock with feedback loop [92]

- **Authors:** L. Toscani De Col, T. Riebner, I. Morawetz, F. Schneider, N. Sempelmann, J. Schlachet-Lépinay, F. Schaden, M. Bartokos, G.A. Kazakov, K. Beeks, B. Gerstenecker, M. Pimon, S. Lahs, A. Hellerschmied, T. Lercher, J. Premper, A. Niessner, M. Matus, H. Denker, M. Cizek, O. Cip, V. Lal, G. Zitzer, V. Petrov, J. Tiedau, M.V. Okhapkin, E. Peik, T. Schumm
- **Posted on arXiv:** 03 June 2026

3.15.3 Detector Technologies and Simulations

3.15.3.1 Quantum Systems for Enhanced High Energy Particle Physics Detectors [164]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

3.15.3.2 Quantum sensing for particle physics [153]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.15.4 Gravitational Waves

3.15.4.1 Exploring gravity with the MIGA large scale atom interferometer [216]

- **Authors:** B. Canuel, A. Bertoldi, L. Amand, E. Pozzo di Borgo, B. Fang, R. Geiger, J. Gillot, S. Henry, J. Hinderer, D. Holleville, G. Lefèvre, M. Merzougui, N. Mielec, T. Monfret, S. Pelisson, M. Prevedelli, S. Reynaud, I. Riou, Y. Rogister, S. Rosat, E. Cormier, A. Landragin, W. Chaibi, S. Gaffet, P. Bouyer
- **Posted on arXiv:** 07 March 2017
- **Published in Scientific Reports (2018) 8:14064:** 14 September 2018

3.15.4.2 Mid-band gravitational wave detection with precision atomic sensors [217]

- **Authors:** Peter W. Graham, Jason M. Hogan, Mark A. Kasevich, Surjeet Rajendran, Roger W. Romani
- **Posted on arXiv:** 06 November 2017

3.15.4.3 Matter-wave Atomic Gradiometer InterferometricSensor (MAGIS-100) at Fermilab [146]

- **Authors:** Jon Coleman
- **Posted on arXiv:** 02 December 2018
- **Published in PoS:** 28 November 2018

3.15.4.4 ZAIGA: Zhaoshan Long-baseline Atom Interferometer Gravitation Antenna [147]

- **Authors:** Ming-Sheng Zhan, Jin Wang, Wei-Tou Ni, Dong-Feng Gao, Gang Wang, Ling-Xiang He, Run-Bing Li, Lin Zhou, Xi Chen, Jia-Qi Zhong, Biao Tang, Zhan-Wei Yao, Lei Zhu, Zong-Yuan Xiong, Si-Bin Lu, Geng-Hua Yu, Qun-Feng Cheng, Min Liu, Yu-Rong Liang, Peng Xu, Xiao-Dong He, Min Ke, Zheng Tan, Jun Luo
- **Posted on arXiv:** 21 March 2019
- **Published in Int.J.Mod.Phys.D:** 02 July 2019

3.15.4.5 SAGE: A Proposal for a Space Atomic Gravity Explorer [148]

- **Authors:** Guglielmo M. Tino, Angelo Bassi, Giuseppe Bianco, Kai Bongs, Philippe Bouyer, Luigi Cacciapuoti, Salvatore Capozziello, Xuzong Chen, Maria L. Chiofalo, Andrei Derevianko, Wolfgang Ertmer, Naceur Gaaloul, Patrick Gill, Peter W. Graham, Jason M. Hogan, Luciano Iess, Mark A. Kasevich, Hidetoshi Katori, Carsten Klempt, Xuanhui Lu, Long-Sheng Ma, Holger Müller, Nathan R. Newbury, Chris W. Oates, Achim Peters, Nicola Poli, Ernst M. Rasel, Gabriele Rosi, Albert Roura, Christophe Salomon, Stephan Schiller, Wolfgang Schleich, Dennis Schlippert, Florian Schreck, Christian Schubert, Fiodor Sorrentino, Uwe Sterr, Jan W. Thomsen, Giuseppe Vallone, Flavio Vetrano, Paolo Villoresi, Wolf von Klitzing, David Wilkowski, Peter Wolf, Jun Ye, Nan Yu, Mingsheng Zhan
- **Posted on arXiv:** 08 July 2019
- **Published in Eur. Phys. J. D 73, 228 (2019):** 15 November 2019

3.15.4.6 ELGAR—a European Laboratory for Gravitation and Atom-interferometric Research [218]

- **Authors:** B. Canuel, S. Abend, P. Amaro-Seoane, F Badaracco, Q. Beaufils, A. Bertoldi, K. Bongs, P. Bouyer, C. Braxmaier, W. Chaibi, N. Christensen, F. Fitzek, G. Flouris, N. Gaaloul, S. Gaffet, C.L. Garrido Alzar, R. Geiger, S. Guellati-Khelifa, K. Hammerer, J. Harms, J. Hinderer, J. Junca, S. Katsanevas, C. Klempt, C. Kozanitis, M. Krutzik, A. Landragin, I. Làzaro Roche, B. Leykauf, Y-H. Lien, S. Loriani, S. Merlet,

M. Merzougui, M. Nofrarias, P. Papadakos, F. Pereira, A. Peters, D. Plexousakis, M. Prevedelli, E. Rasel, Y. Rogister, S. Rosat, A. Roura, D.O. Sabulsky, V. Schkolnik, D. Schlippert, C. Schubert, L. Sidorenkov, J.-N. Siemß, C.F. Sopuerta, F. Sorrentino, C. Struckmann, G.M. Tino, G. Tsagkatakis, A. Viceré, W. von Klitzing, L. Woerner, X. Zou

- **Posted on arXiv:** 09 November 2019
- **Published in Class.Quant.Grav.:** 31 October 2020

3.15.4.7 AION: An Atom Interferometer Observatory and Network [149]

- **Authors:** L. Badurina, E. Bentine, D. Blas, K. Bongs, D. Bortoletto, T. Bowcock, K. Bridges, W. Bowden, O. Buchmueller, C. Burrage, J. Coleman, G. Elertas, J. Ellis, C. Foot, V. Gibson, M.G. Haehnelt, T. Harte, S. Hedges, R. Hobson, M. Holynski, T. Jones, M. Langlois, S. Lellouch, M. Lewicki, R. Maiolino, P. Majewski, S. Malik, J. March-Russell, C. McCabe, D. Newbold, B. Sauer, U. Schneider, I. Shipsey, Y. Singh, M.A. Uchida, T. Valenzuela, M. van der Grinten, V. Vaskonen, J. Vossebeld, D. Weatherill, I. Wilmut
- **Posted on arXiv:** 26 November 2019
- **Published in JCAP:** 06 May 2020

3.15.4.8 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [150]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

3.15.4.9 Matter-wave Atomic Gradiometer Interferometric Sensor (MAGIS-100) [151]

- **Authors:** Mahiro Abe, Philip Adamson, Marcel Borcean, Daniela Bortoletto, Kieran Bridges, Samuel P. Carman, Swapan Chattopadhyay, Jonathon Coleman, Noah M. Curfman, Kenneth DeRose, Tejas Deshpande, Savas Dimopoulos, Christopher J. Foot, Josef C. Frisch, Benjamin E. Garber, Steve Geer, Valerie Gibson, Jonah Glick, Peter W. Graham, Steve R. Hahn, Roni Harnik, Leonie Hawkins, Sam Hindley, Jason M. Hogan, Yijun Jiang (姜一君), Mark A. Kasevich, Ronald J. Kellett, Mandy Kiburg, Tim Kovachy, Joseph D. Lykken, John March-Russell, Jeremiah Mitchell, Martin Murphy, Megan Nantel, Lucy E. Nobrega, Robert K. Plunkett, Surjeet Rajendran, Jan Rudolph, Natasha Sachdeva, Murtaza Safdari, James K. Santucci, Ariel G. Schwartzman, Ian Shipsey, Hunter Swan, Linda R. Valerio, Arvydas Vasonis, Yiping Wang, Thomas Wilkason

- **Posted on arXiv:** 06 April 2021
- **Published in Quantum Sci. Technol. 6, 044003 (2021):** 16 July 2021

3.15.4.10 Prospective sensitivities of atom interferometers to gravitational waves and ultralight dark matter [219]

- **Authors:** Leonardo Badurina, Oliver Buchmueller, John Ellis, Marek Lewicki, Christopher McCabe, Ville Vaskonen
- **Posted on arXiv:** 05 August 2021
- **Published in Phil. Trans. R. Soc. A.380 20210060, 2022:** 07 February 2022

3.15.4.11 Letter of Intent: AICE – 100m Atom Interferometer Experiment at CERN [220]

- **Authors:** Charles Baynham, Andrea Bertoldi, Diego Blas, Oliver Buchmueller, Sergio Calatroni, Vassilis Charmandaris, Maria Luisa Marilu Chiofalo, Pierre Cladé, Jonathon Coleman, Fabio Di Pumpo, John Ellis, Naceur Gaaloul, Saïda Guellati-Khelifa, Tiffany Harte, Richard Hobson, Michael Holynski, Samuel Lellouch, Lucas Lombriser, Elias Lopez Asamar, Michele Maggiore, Christopher McCabe, Jeremiah Mitchell, Ernst M. Rasel, Federico Sanchez Nieto, Wolfgang Schleich, Dennis Schlippert, Ulrich Schneider, Steven Schramm, Marcelle Soares-Santos, Guglielmo M. Tino, Jonathan N. Tinsley, Tristan Valenzuela, Maurits van der Grinten, Wolf von Klitzing
- **Posted on arXiv:** 15 September 2025

3.15.4.12 A prototype differential atom interferometer for fundamental physics [157]

- **Authors:** C.F.A. Baynham, R. Hobson, O. Buchmüller, D. Evans, L. Hawkins, L. Iannizzotto Venezze, A. Josset, D. Lee, E. Pasatembou, B.E. Sauer, M.R. Tarbutt, T. Walker, O. Ennis, U. Chauhan, A. Brzakalik, S. Dey, S. Hedges, B. Stray, M. Langlois, K. Bongs, T. Hird, S. Lellouch, M. Holynski, B. Bostwick f, J. Chen, Z. Eyler, V. Gibson, T.L. Harte, C.C. Hsu, M. Karzazi, C. Lu, B. Millward, J. Mitchell, N. Mouelle, B. Panchumarthi g, J. Scheper, U. Schneider, X. Su, X. Suh, Y. Tang, K. Tkalčec, M. Zeuneri, S. Zhang, Y. Zhij, Y. Zhi, L. Badurina, A. Beniwal l, D. Blas, J. Carlton, J. Ellis, C. McCabe, G. Parish, D. Pathak Govardhan, V. Vaskonenn, T. Bowcock, K. Bridges, A. Carroll, J. Coleman, G. Elertas, S. Hindley, C. Metelko, H. Throssell, J.N. Tinsley, E. Bentine, M. Booth, D. Bortoletto, N. Callaghan, C. Foot, C. Gómez-Monedero, K. Hughes, A. James, T. Leese, A. Lowe, J. March-Russell, J. Sander, J. Schelfhout, I. Shipsey, D. Weatherill, D. Wood, M.G. Bason, K. Hussain, H. Labiad, A.L. Marchant, T.C. Thornton, T. Valenzuela, S.N. Balashov, P. Majewski, D. Newbold, M.G.D. van der Grinten, Z. Pan, Z. Tam, I. Wilmut, K. Clarke, A. Vick
- **Posted on arXiv:** 12 April 2025
- **Published in Nature:** 17 June 2026

3.15.5 Neutrino Physics

3.15.5.1 Superradiant interactions for relic detection with entangled nuclear spins [156]

- **Authors:** Marios Galanis, Onur Hosten, Asimina Arvanitaki, Savas Dimopoulos
- **Posted on arXiv:** 28 August 2025
- **Published in Phys.Rev.A:** 10 June 2026

3.15.6 Crosslists

3.15.6.1 Dark Matter Direct Detection with Accelerometers [121]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

3.15.6.2 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

3.15.6.3 Quantum Sensing for High Energy Physics [12]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell,

Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.15.6.4 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar

- **Posted on arXiv:** 13 March 2019

3.15.6.5 AEDGE: Atomic Experiment for Dark Matter and Gravity Exploration in Space [14]

- **Authors:** Yousef Abou El-Neaj, Cristiano Alpigiani, Sana Amairi-Pyka, Henrique Araújo, Antun Balaž, Angelo Bassi, Lars Bathe-Peters, Baptiste Battelier, Aleksandar Belić, Elliot Bentine, José Bernabeu, Andrea Bertoldi, Robert Bingham, Diego Blas, Vasiliki Bolpasi, Kai Bongs, Sougato Bose, Philippe Bouyer, Themis Bowcock, William Bowden, Oliver Buchmueller, Clare Burrage, Xavier Calmet, Benjamin Canuel, Laurentiu-Ioan Caramete, Andrew Carroll, Giancarlo Cella, Vassilis Charmandaris, Swapan Chattopadhyay, Xuzong Chen, Maria Luisa Chiofalo, Jonathon Coleman, Joseph Cotter, Yanou Cui, Andrei Derevianko, Albert De Roeck, Goran S. Djordjevic, Peter Dornan, Michael Doser, Ioannis Drougkakis, Jacob Dunningham, Ioana Dutan, Sajan Easo, Gedminas Elertas, John Ellis, Mai El Sawy, Farida Fassi, Daniel Felea, Chen-Hao Feng, Robert Flack, Chris Foot, Ivette Fuentes, Naceur Gaaloul, Alexandre Gauguet, Remi Geiger, Valerie Gibson, Gian Giudice, Jon Goldwin, Oleg Grachov, Peter W. Graham, Dario Grasso, Maurits van der Grinten, Mustafa Gündogan, Martin G. Haehnelt, Tiffany Harte, Aurelien Hees, Richard Hobson, Jason Hogan, Bodil Holst, Michael Holynski, Mark Kasevich, Bradley J. Kavanagh, Wolf von Klitzing, Tim Kovachy, Benjamin Krikler, Markus Krutzik, Marek Lewicki, Yu-Hung Lien, Miaoyuan Liu, Giuseppe

Gaetano Luciano, Alain Magnon, Mohammed Attia Mahmoud, Sarah Malik, Christopher McCabe, Jeremiah Mitchell, Julia Pahl, Debapriya Pal, Saurabh Pandey, Dimitris Papazoglou, Mauro Paternostro, Bjoern Penning, Achim Peters, Marco Prevedelli, Vishnupriya Puthiya-Veettil, John Quenby, Ernst Rasel, Sean Ravenhall, Jack Ringwood, Albert Roura, Dylan Sabulsky, Muhammed Sameed, Ben Sauer, Stefan Alaric Schäffer, Stephan Schiller, Vladimir Schkolnik, Dennis Schlippert, Christian Schubert, Haifa Rejeb Sfar, Armin Shayeghi, Ian Shipsey, Carla Signorini, Yeshpal Singh, Marcelle Soares-Santos, Fiodor Sorrentino, Timothy Sumner, Konstantinos Tassis, Silvia Tentindo, Guglielmo Maria Tino, Jonathan N. Tinsley, James Unwin, Tristan Valenzuela, Georgios Vasilakis, Ville Vaskonen, Christian Vogt, Alex Webber-Date, Andŕ e} Wenzlawski, Patrick Windpassinger, Marian Woltmann, Efe Yazgan, Ming-Sheng Zhan, Xinhao Zou, Jure Zupan

- **Posted on arXiv:** 02 August 2019
- **Published in EPJ Quant.Technol.:** 04 March 2020

3.15.6.6 Simulating Lattice Gauge Theories within Quantum Technologies [253]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.15.6.7 Quantum Sensors with Matter Waves for GW Observation [4]

- **Authors:** Andrea Bertoldi, Philippe Bouyer, Benjamin Canuel
- **Posted on arXiv:** 27 October 2020
- **Published in nan:** 2022

3.15.6.8 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [19]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.15.6.9 Snowmass2021 Cosmic Frontier White Paper: Future Gravitational-Wave Detector Facilities [21]

- **Authors:** Stefan W. Ballmer, Rana Adhikari, Leonardo Badurina, Duncan A. Brown, Swapan Chattopadhyay, Matthew Evans, Peter Fritschel, Evan Hall, Jason M. Hogan, Karan Jani, Tim Kovachy, Kevin Kuns, Ariel Schwartzman, Daniel Sigg, Bram Slagmolen, Salvatore Vitale, Christopher Wipf

- **Posted on arXiv:** 15 March 2022

3.15.6.10 New Horizons: Scalar and Vector Ultralight Dark Matter [23]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.15.6.11 Quantum Networks for High Energy Physics [24]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.15.6.12 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [27]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.15.6.13 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao
- **Posted on arXiv:** 28 February 2023

3.15.6.14 Quantum Sensors for High Energy Physics [30]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.15.6.15 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [154]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis
- **Posted on arXiv:** 07 August 2024
- **Published in Phys.Rev.D:** 01 March 2025

3.15.6.16 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [155]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

3.15.6.17 Quantum-Error-Correction-Inspired Noise Mitigation for Wavelike Dark Matter Searches with Quantum Sensors [65]

- **Authors:** Hajime Fukuda, Takeo Moroi, Thanaporn Sichanugrist
- **Posted on arXiv:** 05 November 2025
- **Published in Phys.Rev.Lett.:** 08 July 2026

3.16 Quantum Sensors - Optical/Photonic Sensors

3.16.1 Dark Matter - Wave-like Dark Matter

3.16.1.1 Analysis of single-photon and linear amplifier detectors for microwave cavity dark matter axion searches [93]

- **Authors:** S.K. Lamoreaux, K.A. van Bibber, K.W. Lehnert, G. Carosi
- **Posted on arXiv:** 15 June 2013
- **Published in Phys.Rev.D:** 23 August 2013

3.16.1.2 Laser Interferometers as Dark Matter Detectors [94]

- **Authors:** Evan D. Hall, Rana X. Adhikari, Valery V. Frolov, Holger Müller, Maxim Pospelov, Rana X Adhikari
- **Posted on arXiv:** 03 May 2016
- **Published in Phys.Rev.D:** 23 October 2018

3.16.1.3 Accelerating dark-matter axion searches with quantum measurement technology [95]

- **Authors:** Huaixiu Zheng, Matti Silveri, R.T. Brierley, S.M. Girvin, K.W. Lehnert
- **Posted on arXiv:** 08 July 2016

3.16.1.4 A Search for Invisible Axion Dark Matter with the Axion Dark Matter Experiment [96]

- **Authors:** N. Du, N. Force, R. Khatiwada, E. Lentz, R. Ottens, L.J. Rosenberg, G. Rybka, G. Carosi, N. Woollett, D. Bowring, A.S. Chou, A. Sonnenschein, W. Wester, C. Boutan, N.S. Oblath, R. Bradley, E.J. Daw, A.V. Dixit, J. Clarke, S.R. O'Kelley, N. Crisosto, J.R. Gleason, S. Jois, P. Sikivie, I. Stern, N.S. Sullivan, D.B. Tanner, G.C. Hilton
- **Posted on arXiv:** 16 April 2018
- **Published in Phys.Rev.Lett.:** 09 April 2018

3.16.1.5 Searching for Ultralight Dark Matter with Optical Cavities [97]

- **Authors:** Andrew A. Geraci, Colin Bradley, Dongfeng Gao, Jonathan Weinstein, Andrei Derevianko
- **Posted on arXiv:** 01 August 2018
- **Published in Phys. Rev. Lett. 123, 031304 (2019):** 18 July 2019

3.16.1.6 Squeezed vacuum used to accelerate the search for a weak classical signal [98]

- **Authors:** M. Malnou, D.A. Palken, B.M. Brubaker, Leila R. Vale, Gene C. Hilton, K.W. Lehnert
- **Posted on arXiv:** 17 September 2018
- **Published in Phys.Rev.X:** 04 May 2019

3.16.1.7 Extended Search for the Invisible Axion with the Axion Dark Matter Experiment [99]

- **Authors:** T. Braine, R. Cervantes, N. Crisosto, N. Du, S. Kimes, L.J. Rosenberg, G. Rybka, J. Yang, D. Bowring, A.S. Chou, R. Khatiwada, A. Sonnenschein, W. Wester, G. Carosi, N. Woollett, L.D. Duffy, R. Bradley, C. Boutan, M. Jones, B.H. LaRoque, N.S. Oblath, M.S. Taubman, J. Clarke, A. Dove, A. Eddins, S.R. O'Kelley, S. Nawaz, I. Siddiqi, N. Stevenson, A. Agrawal, A.V. Dixit, J.R. Gleason, S. Jois, P. Sikivie, N.S. Sullivan, D.B. Tanner, E. Lentz, E.J. Daw, J.H. Buckley, P.M. Harrington, E.A. Henriksen, K.W. Murch

- **Posted on arXiv:** 18 October 2019
- **Published in Phys. Rev. Lett. 124, 101303 (2020):** 11 March 2020

3.16.1.8 A quantum-enhanced search for dark matter axions [100]

- **Authors:** K.M. Backes, D.A. Palken, S. Al Kenany, B.M. Brubaker, S.B. Cahn, A. Droster, Gene C. Hilton, Sumita Ghosh, H. Jackson, S.K. Lamoreaux, A.F. Leder, K.W. Lehnert, S.M. Lewis, M. Malnou, R.H. Maruyama, N.M. Rapidis, M. Simanovskaia, Sukhman Singh, D.H. Speller, I. Urdinaran, Leila R. Vale, E.C. van Assendelft, K. van Bibber, H. Wang
- **Posted on arXiv:** 04 August 2020
- **Published in Nature:** 10 February 2021

3.16.1.9 Searching for Dark Matter with a Superconducting Qubit [101]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

3.16.1.10 Cavity Entanglement and State Swapping to Accelerate the Search for Axion Dark Matter [102]

- **Authors:** K. Wurtz, B.M. Brubaker, Y. Jiang, E.P. Ruddy, D.A. Palken, K.W. Lehnert
- **Posted on arXiv:** 08 July 2021
- **Published in PRX Quantum:** 10 December 2021

3.16.1.11 Search for Invisible Axion Dark Matter in the 3.3–4.2 μeV Mass Range [103]

- **Authors:** C. Bartram, T. Braine, E. Burns, R. Cervantes, N. Crisosto, N. Du, H. Korandla, G. Leum, P. Mohapatra, T. Nitta, L.J. Rosenberg, G. Rybka, J. Yang, John Clarke, I. Siddiqi, A. Agrawal, A.V. Dixit, M.H. Awida, A.S. Chou, M. Hollister, S. Knirck, A. Sonnenschein, W. Wester, J.R. Gleason, A.T. Hipp, S. Jois, P. Sikivie, N.S. Sullivan, D.B. Tanner, E. Lentz, R. Khatiwada, G. Carosi, N. Robertson, N. Woollett, L.D. Duffy, C. Boutan, M. Jones, B.H. LaRoque, N.S. Oblath, M.S. Taubman, E.J. Daw, M.G. Perry, J.H. Buckley, C. Gaikwad, J. Hoffman, K.W. Murch, M. Goryachev, B.T. McAllister, A. Quiskamp, C. Thomson, M.E. Tobar
- **Posted on arXiv:** 12 October 2021
- **Published in Phys. Rev. Lett. 127, 261803 (2021):** 23 December 2021

3.16.1.12 Entangled Sensor-Networks for Dark-Matter Searches [104]

- **Authors:** Anthony J. Brady, Christina Gao, Roni Harnik, Zhen Liu, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 10 March 2022
- **Published in PRX Quantum 3, 030333 (2022):** 01 September 2022

3.16.1.13 Detecting Hidden Photon Dark Matter Using the Direct Excitation of Transmon Qubits [105]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 07 December 2022
- **Published in Phys.Rev.Lett.:** 22 November 2023

3.16.1.14 New results from HAYSTAC's phase II operation with a squeezed state receiver [106]

- **Authors:** M.J. Jewell, A.F. Leder, K.M. Backes, Xiran Bai, K. van Bibber, B.M. Brubaker, S.B. Cahn, A. Droster, Maryam H. Esmat, Sumita Ghosh, Eleanor Graham, Gene C. Hilton, H. Jackson, Claire Laffan, S.K. Lamoreaux, K.W. Lehnert, S.M. Lewis, M. Malnou, R.H. Maruyama, D.A. Palken, N.M. Rapidis, E.P. Ruddy, M. Simanovskaia, Sukhman Singh, D.H. Speller, Leila R. Vale, H. Wang, Yuqi Zhu
- **Posted on arXiv:** 23 January 2023
- **Published in Phys.Rev.D:** 01 April 2023

3.16.1.15 Stimulated Emission of Signal Photons from Dark Matter Waves [107]

- **Authors:** Ankur Agrawal, Akash V. Dixit, Tanay Roy, Srivatsan Chakram, Kevin He, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 05 May 2023
- **Published in Phys. Rev. Lett. 132, 140801 (2024):** 03 April 2024

3.16.1.16 Quantum measurements in fundamental physics: a user's manual [108]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023
- **Published in SciPost Phys. Rev. 2 (2026):** 2026

3.16.1.17 Quantum Enhancement in Dark Matter Detection with Quantum Computation [109]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 17 November 2023
- **Published in Phys.Rev.Lett.:** 08 July 2024

3.16.1.18 Quantum-Enhanced Sensing of Axion Dark Matter with a Transmon-Based Single Microwave Photon Counter [110]

- **Authors:** C. Braggio, L. Balembois, R. Di Vora, Z. Wang, J. Travesedo, L. Pallegoix, G. Carugno, A. Ortolan, G. Ruoso, U. Gambardella, D. D'Agostino, P. Bertet, E. Flurin
- **Posted on arXiv:** 04 March 2024
- **Published in Phys.Rev.X:** 01 April 2025

3.16.1.19 Search for QCD axion dark matter with transmon qubits and quantum circuit [111]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 29 July 2024
- **Published in Phys.Rev.D:** 01 December 2024

3.16.1.20 Axion bounds from quantum technology [112]

- **Authors:** Martin Bauer, Sreemanti Chakraborti, Guillaume Rostagni
- **Posted on arXiv:** 12 August 2024
- **Published in JHEP:** 05 May 2025

3.16.1.21 ADMX Axion Dark Matter Bounds around $3.3\,\mu$eV with Dine-Fischler-Srednicki-Zhitnitsky Discovery Ability [113]

- **Authors:** C. Goodman, M. Guzzetti, C. Hanretty, L.J. Rosenberg, G. Rybka, J. Sinnis, D. Zhang, John Clarke, I. Siddiqi, A.S. Chou, M. Hollister, S. Knirck, A. Sonnenschein, T.J. Caligiure, J.R. Gleason, A.T. Hipp, P. Sikivie, M.E. Solano, N.S. Sullivan, D.B. Tanner, R. Khatiwada, G. Carosi, C. Cisneros, N. Du, N. Robertson, N. Woollett, L.D. Duffy, C. Boutan, T. Braine, E. Lentz, N.S. Oblath, M.S. Taubman, E.J. Daw, C. Mostyn, M.G. Perry, C. Bartram, T.A. Dyson, S. Ruppert, M.O. Withers, C.L. Kuo, B.T. McAllister, J.H. Buckley, C. Gaikwad, J. Hoffman, K. Murch, M. Goryachev, E. Hartman, A. Quiskamp, M.E. Tobar

- **Posted on arXiv:** 27 August 2024
- **Published in Phys. Rev. Lett. 134, 111002 (2025):** 21 March 2025

3.16.1.22 Quantum-enhanced dark matter detection with in-cavity control: mitigating the Rayleigh curse [114]

- **Authors:** Haowei Shi, Anthony J. Brady, Wojciech Górecki, Lorenzo Maccone, Roberto Di Candia, Quntao Zhuang
- **Posted on arXiv:** 06 September 2024
- **Published in npj Quantum Information 11, 48 (2025):** 17 March 2025

3.16.1.23 Transmon qubit modeling and characterization for Dark Matter search [115]

- **Authors:** Roberto Moretti, Danilo Labranca, Pietro Campana, Rodolfo Carobene, Marco Gobbo, Manuel A. Castellanos-Beltran, David Olaya, Peter F. Hopkins, Leonardo Banchi, Matteo Borghesi, Alessandro Candido, Stefano Carrazza, Hervè Atsè Corti, Alessandro D'Elia, Marco Faverzani, Elena Ferri, Angelo Nucciotti, Luca Origo, Andrea Pasquale, Alex Stephane Piedjou Komnang, Alessio Rettaroli, Simone Tocci, Claudio Gatti, Andrea Giachero
- **Posted on arXiv:** 09 September 2024
- **Published in IEEE Trans.Quantum Eng.:** 2026

3.16.1.24 Dark Matter Axion Search with HAYSTAC Phase II [116]

- **Authors:** Xiran Bai, M.J. Jewell, J. Echevers, K. van Bibber, A. Droster, Maryam H. Esmat, Sumita Ghosh, Eleanor Graham, H. Jackson, Claire Laffan, S.K. Lamoreaux, A.F. Leder, K.W. Lehnert, S.M. Lewis, R.H. Maruyama, R.D. Nath, N.M. Rapidis, E.P. Ruddy, M. Silva-Feaver, M. Simanovskaia, Sukhman Singh, D.H. Speller, Sabrina Zacarias, Yuqi Zhu
- **Posted on arXiv:** 13 September 2024
- **Published in Phys. Rev. Lett. 134, 151006 (2025):** 18 April 2025

3.16.1.25 Flux-Tunable Cavity for Dark Matter Detection [117]

- **Authors:** Fang Zhao, Ziqian Li, Akash V. Dixit, Tanay Roy, Andrei Vrajitoarea, Riju Banerjee, Alexander Anferov, Kan-Heng Lee, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 12 January 2025
- **Published in Phys.Rev.Lett.:** 13 November 2025

3.16.1.26 Search for Axion Dark Matter from 1.1 to 1.3 GHz with ADMX [118]

- **Authors:** G. Carosi, C. Cisneros, N. Du, S. Durham, N. Robertson, C. Goodman, M. Guzzetti, C. Hanretty, K. Enzian, L.J. Rosenberg, G. Rybka, J. Sinnis, D. Zhang, John Clarke, I. Siddiqi, A.S. Chou, M. Hollister, A. Sonnenschein, S. Knirck, T.J. Caligiure, J.R. Gleason, A.T. Hipp, P. Sikivie, M.E. Solano, N.S. Sullivan, D.B. Tanner, R. Khatiwada, L.D. Duffy, C. Boutan, T. Braine, E. Lentz, N.S. Oblath, M.S. Taubman, E.J. Daw, C. Mostyn, M.G. Perry, C. Bartram, J. Laurel, A. Yi, T.A. Dyson, S. Ruppert, M.O. Withers, C.L. Kuo, B.T. McAllister, J.H. Buckley, C. Gaikwad, J. Hoffman, K. Murch, M. Goryachev, E. Hartman, A. Quiskamp, M.E. Tobar
- **Posted on arXiv:** 09 April 2025
- **Published in Phys. Rev. Lett. 135, 191001 (2025):** 03 November 2025

3.16.1.27 Quantum Enhanced Dark-Matter Search with Entangled Fock States in High-Quality Cavities [119]

- **Authors:** Benjamin Freiman, Xinyuan You, Andy C.Y. Li, Raphael Cervantes, Taeyoon Kim, Anna Grasselino, Roni Harnik, Yao Lu
- **Posted on arXiv:** 30 October 2025

3.16.2 Gravitational Waves

3.16.2.1 Electromagnetic cavities as mechanical bars for gravitational waves [152]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D'Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

3.16.2.2 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [155]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

3.16.3 Neutrino Physics

3.16.3.1 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [154]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis
- **Posted on arXiv:** 07 August 2024
- **Published in Phys.Rev.D:** 01 March 2025

3.16.4 Crosslists

3.16.4.1 Quantum Sensing for High Energy Physics [12]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

3.16.4.2 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan

Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar

- **Posted on arXiv:** 13 March 2019

3.16.4.3 Simulating Lattice Gauge Theories within Quantum Technologies [253]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller

- **Posted on arXiv:** 31 October 2019

- **Published in Eur.Phys.J.D:** 04 August 2020

3.16.4.4 Intensity interferometry for ultralight bosonic dark matter detection [73]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball

- **Posted on arXiv:** 05 February 2022

- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

3.16.4.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [19]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova

- **Posted on arXiv:** 14 March 2022

3.16.4.6 New Horizons: Scalar and Vector Ultralight Dark Matter [23]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M.

Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.16.4.7 Quantum Networks for High Energy Physics [24]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.16.4.8 Quantum computing hardware for HEP algorithms and sensing [26]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti

- **Posted on arXiv:** 18 April 2022

3.16.4.9 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [27]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage

- **Posted on arXiv:** 14 September 2022

3.16.4.10 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao
- **Posted on arXiv:** 28 February 2023

3.16.4.11 Quantum sensing for particle physics [153]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.16.4.12 Detection of bosenovae with quantum sensors on Earth and in space [80]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

3.16.4.13 Rydberg-atom-based single-photon detection for haloscope axion searches [81]

- **Authors:** Eleanor Graham, Sumita Ghosh, Yuqi Zhu, Xiran Bai, Sidney B. Cahn, Elsa Durcan, Michael J. Jewell, Danielle H. Speller, Sabrina M. Zacarias, Laura T. Zhou, Reina H. Maruyama
- **Posted on arXiv:** 23 October 2023
- **Published in Phys.Rev.D:** 01 February 2024

3.16.4.14 Quantum Sensors for High Energy Physics [30]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.16.4.15 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [85]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

3.16.4.16 Quantum-Error-Correction-Inspired Noise Mitigation for Wavelike Dark Matter Searches with Quantum Sensors [65]

- **Authors:** Hajime Fukuda, Takeo Moroi, Thanaporn Sichanugrist
- **Posted on arXiv:** 05 November 2025
- **Published in Phys.Rev.Lett.:** 08 July 2026

3.16.4.17 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [91]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

3.17 Quantum Sensors - Optomechanical Sensors

3.17.1 Dark Matter - Particle-like Dark Matter

3.17.1.1 Search for Kilogram-scale Dark Matter with Precision Displacement Sensors [51]

- **Authors:** Akio Kawasaki
- **Posted on arXiv:** 26 August 2018
- **Published in Phys. Rev. D 99, 023005 (2019):** 07 January 2019

3.17.1.2 Proposal for gravitational direct detection of dark matter [52]

- **Authors:** Daniel Carney, Sohitri Ghosh, Gordan Krnjaic, Jacob M. Taylor
- **Posted on arXiv:** 01 March 2019
- **Published in Phys.Rev.D:** 14 October 2020

3.17.1.3 Search for composite dark matter with optically levitated sensors [53]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

3.17.1.4 Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors [54]

- **Authors:** Gadi Afek, Daniel Carney, David C. Moore
- **Posted on arXiv:** 05 November 2021
- **Published in Phys.Rev.Lett.:** 09 March 2022

3.17.1.5 Models of ultraheavy dark matter visible to macroscopic mechanical sensing arrays [55]

- **Authors:** Carlos Blanco, Bahaa Elshimy, Rafael F. Lang, Robert Orlando
- **Posted on arXiv:** 29 December 2021
- **Published in Phys.Rev.D:** 01 June 2022

3.17.1.6 Optomechanical dark matter instrument for direct detection [56]

- **Authors:** Christopher G. Baker, Warwick P. Bowen, Peter Cox, Matthew J. Dolan, Maxim Goryachev, Glen Harris
- **Posted on arXiv:** 16 June 2023
- **Published in Phys. Rev. D 110, 043005 (2024):** 02 August 2024

3.17.1.7 Probing levitodynamics with multi-stochastic forces and the simple applications on the dark matter detection in optical levitation experiment [57]

- **Authors:** Xi Cheng, Ji-Heng Guo, Wenyu Wang, Bin Zhu
- **Posted on arXiv:** 07 December 2023
- **Published in Nucl. Phys. B 1010 (2025) 116780:** 19 December 2024

3.17.1.8 Dark Matter Searches with Levitated Sensors [58]

- **Authors:** Eva Kilian, Markus Rademacher, Jonathan M.H. Gosling, Julian H. Iacoponi, Fiona Alder, Marko Toroš, Antonio Pontin, Chamkaur Ghag, Sougato Bose, Tania S. Monteiro, P.F. Barker
- **Posted on arXiv:** 31 January 2024
- **Published in AVS Quantum Sci. 6, 030503 (2024):** 2024

3.17.1.9 Probing light particles with optically trapped sensors through nucleon scattering [59]

- **Authors:** Bhaskar Dutta, Dilip Kumar Ghosh, Sk Jeesun
- **Posted on arXiv:** 31 January 2025
- **Published in Phys.Rev.D:** 01 January 2026

3.17.1.10 Mechanical sensors for ultraheavy dark matter searches via long-range forces [60]

- **Authors:** Juehang Qin, Dorian W.P. Amaral, Sunil A. Bhave, Erqian Cai, Daniel Carney, Rafael F. Lang, Shengchao Li, Alberto M. Marino, Giacomo Marocco, Claire Marvinney, Jared R. Newton, Jacob M. Taylor, Christopher Tunnell
- **Posted on arXiv:** 14 March 2025
- **Published in Phys.Rev.D 112 (2025) 7, 072003:** 01 October 2025

3.17.2 Dark Matter - Wave-like Dark Matter

3.17.2.1 Sound of Dark Matter: Searching for Light Scalars with Resonant-Mass Detectors [120]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Ken Van Tilburg
- **Posted on arXiv:** 07 August 2015
- **Published in Phys.Rev.Lett.:** 22 January 2016

3.17.2.2 Dark Matter Direct Detection with Accelerometers [121]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

3.17.2.3 Ultralight dark matter detection with mechanical quantum sensors [122]

- **Authors:** Daniel Carney, Anson Hook, Zhen Liu, Jacob M. Taylor, Yue Zhao
- **Posted on arXiv:** 13 August 2019
- **Published in New Journal of Physics, Volume 23, February 2021:** 02 March 2021

3.17.2.4 Searching for scalar dark matter with compact mechanical resonators [123]

- **Authors:** Jack Manley, Russell Stump, Dalziel Wilson, Daniel Grin, Swati Singh
- **Posted on arXiv:** 16 October 2019
- **Published in Phys. Rev. Lett. 124, 151301 (2020):** 17 April 2020

3.17.2.5 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [124]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

3.17.2.6 Searching for vector dark matter with an optomechanical accelerometer [125]

- **Authors:** Jack Manley, Mitul Dey Chowdhury, Daniel Grin, Swati Singh, Dalziel J. Wilson
- **Posted on arXiv:** 09 July 2020
- **Published in Phys. Rev. Lett. 126, 061301 (2021):** 11 February 2021

3.17.2.7 Entanglement-enhanced optomechanical sensor array with application to dark matter searches [126]

- **Authors:** Anthony J. Brady, Xin Chen, Kewen Xiao, Yi Xia, Jack Manley, Mitul Dey Chowdhury, Zhen Liu, Roni Harnik, Dalziel J. Wilson, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 13 October 2022
- **Published in Commun.Phys.:** 01 September 2023

3.17.2.8 Axion detection with optomechanical cavities [127]

- **Authors:** Clara Murgui, Yikun Wang, Kathryn M. Zurek
- **Posted on arXiv:** 15 November 2022
- **Published in Phys.Lett.B:** 27 June 2024

3.17.2.9 Superfluid helium ultralight dark matter detector [128]

- **Authors:** M. Hirschel, V. Vadakkumbatt, N.P. Baker, F.M. Schweizer, J.C. Sankey, S. Singh, J.P. Davis
- **Posted on arXiv:** 14 September 2023
- **Published in Physical Review D 109, 095011 (2024):** 01 May 2024

3.17.2.10 Vector wave dark matter and terrestrial quantum sensors [129]

- **Authors:** Dorian W.P. Amaral, Mudit Jain, Mustafa A. Amin, Christopher Tunnell
- **Posted on arXiv:** 04 March 2024
- **Published in JCAP:** 21 June 2024

3.17.3 Gravitational Waves

3.17.3.1 Quantum interactions between a laser interferometer and gravitational waves [221]

- **Authors:** Belinda Pang, Yanbei Chen
- **Posted on arXiv:** 28 August 2018
- **Published in Phys. Rev. D 98, 124006 (2018):** 11 December 2018

3.17.3.2 Superconducting Levitated Detector of Gravitational Waves [222]

- **Authors:** Daniel Carney, Gerard Higgins, Giacomo Marocco, Michael Wentzel
- **Posted on arXiv:** 02 August 2024
- **Published in Carney (2025) Phys. Rev. Lett. 134, 181402:** 06 May 2025

3.17.4 Neutrino Physics

3.17.4.1 Requirements on quantum superpositions of macro-objects for sensing neutrinos [334]

- **Authors:** Eva Kilian, Marko Toroš, Frank F. Deppisch, Ruben Saakyan, Sougato Bose
- **Posted on arXiv:** 27 April 2022
- **Published in Phys. Rev. Research 5, 023012 (2023):** 07 April 2023

3.17.4.2 Searches for Massive Neutrinos with Mechanical Quantum Sensors [335]

- **Authors:** Daniel Carney, Kyle G. Leach, David C. Moore
- **Posted on arXiv:** 12 July 2022
- **Published in Phys.Rev.X:** 01 February 2023

3.17.5 Crosslists

3.17.5.1 Quantum Sensing for High Energy Physics [12]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

3.17.5.2 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

3.17.5.3 Mechanical quantum sensing in the search for dark matter [15]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

3.17.5.4 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [19]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.17.5.5 New Horizons: Scalar and Vector Ultralight Dark Matter [23]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz,

S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.17.5.6 Quantum Networks for High Energy Physics [24]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.17.5.7 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [27]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage

- **Posted on arXiv:** 14 September 2022

3.17.5.8 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev,

Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao

- **Posted on arXiv:** 28 February 2023

3.17.5.9 Quantum sensing for particle physics [153]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.17.5.10 Quantum Sensors for High Energy Physics [30]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.17.5.11 Quantum measurements in fundamental physics: a user's manual [108]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023
- **Published in SciPost Phys. Rev. 2 (2026):** 2026

3.17.5.12 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [91]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

3.18 Quantum Sensors - Solid State Sensors

3.18.1 Dark Matter - Particle-like Dark Matter

3.18.1.1 Dark matter direct detection with quantum dots [61]

- **Authors:** Carlos Blanco, Rouven Essig, Marivi Fernandez-Serra, Harikrishnan Ramani, Oren Slone
- **Posted on arXiv:** 10 August 2022
- **Published in Phys.Rev.D:** 01 May 2023

3.18.1.2 Dark Matter Induced Power in Quantum Devices [62]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 17 October 2022
- **Published in Phys. Rev. Lett. 132, 121801 (2024):** 22 March 2024

3.18.1.3 Transmon Qubit constraints on dark matter-nucleon scattering [63]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 30 April 2024
- **Published in JHEP:** 25 July 2024

3.18.1.4 Quantum parity detectors: A qubit-based particle-detection scheme with meV thresholds for rare-event searches [64]

- **Authors:** Karthik Ramanathan, Brandon J. Sandoval, John E. Parker, Lalit M. Joshi, Andrew D. Beyer, Pierre M. Echternach, Serge Rosenblum, Sunil R. Golwala
- **Posted on arXiv:** 27 May 2024
- **Published in APS Open Science:** 01 May 2026

3.18.2 Dark Matter - Wave-like Dark Matter

3.18.2.1 Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor [130]

- **Authors:** Xing Rong, Mengqi Wang, Jianpei Geng, Xi Qin, Maosen Guo, Man Jiao, Yijin Xie, Pengfei Wang, Pu Huang, Fazhan Shi, Yi-Fu Cai, Chongwen Zou, Jiangfeng Du
- **Posted on arXiv:** 12 June 2017
- **Published in Nature Commun.:** 21 February 2018

3.18.2.2 First Results from ABRACADABRA-10 cm: A Search for Sub-μeV Axion Dark Matter [131]

- **Authors:** Jonathan L. Ouellet, Chiara P. Salemi, Joshua W. Foster, Reyco Henning, Zachary Bogorad, Janet M. Conrad, Joseph A. Formaggio, Yonatan Kahn, Joe Minervini, Alexey Radovinsky, Nicholas L. Rodd, Benjamin R. Safdi, Jesse Thaler, Daniel Winklehner, Lindley Winslow
- **Posted on arXiv:** 29 October 2018
- **Published in Phys. Rev. Lett. 122, 121802 (2019):** 30 March 2019

3.18.2.3 Search for axion-like dark matter with ferromagnets [132]

- **Authors:** Alexander V. Gramolin, Deniz Aybas, Dorian Johnson, Janos Adam, Alexander O. Sushkov
- **Posted on arXiv:** 06 March 2020
- **Published in Nature Physics 17, 79-84 (2021):** 17 August 2020

3.18.2.4 Search for Low-Mass Axion Dark Matter with ABRACADABRA-10 cm [133]

- **Authors:** Chiara P. Salemi, Joshua W. Foster, Jonathan L. Ouellet, Andrew Gavin, Kaliroe M.W. Pappas, Sabrina Cheng, Kate A. Richardson, Reyco Henning, Yonatan Kahn, Rachel Nguyen, Nicholas L. Rodd, Benjamin R. Safdi, Lindley Winslow
- **Posted on arXiv:** 12 February 2021
- **Published in Phys. Rev. Lett. 127, 081801 (2021):** 17 August 2021

3.18.2.5 Axion search with quantum nondemolition detection of magnons [134]

- **Authors:** Tomonori Ikeda, Asuka Ito, Kentaro Miuchi, Jiro Soda, Hisaya Kurashige, Yutaka Shikano
- **Posted on arXiv:** 17 February 2021
- **Published in Phys. Rev. D 105, 102004 (2022):** 15 May 2022

3.18.2.6 Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors [135]

- **Authors:** P.-H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov, V.M. Acosta
- **Posted on arXiv:** 29 December 2021
- **Published in Physical Review Research 4, 023162 (2022):** 31 May 2022

3.18.2.7 Ultimate precision limit of noise sensing and dark matter search [136]

- **Authors:** Haowei Shi, Quntao Zhuang
- **Posted on arXiv:** 29 August 2022
- **Published in npj Quantum Inf.:** 20 March 2023

3.18.2.8 Light dark matter search with nitrogen-vacancy centers in diamonds [137]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 24 February 2023
- **Published in JHEP:** 12 March 2025

3.18.2.9 Axion detection via superfluid ^{3}He ferromagnetic phase and quantum measurement techniques [138]

- **Authors:** So Chigusa, Dan Kondo, Hitoshi Murayama, Risshin Okabe, Hiroyuki Sudo
- **Posted on arXiv:** 17 September 2023
- **Published in JHEP:** 26 September 2024

3.18.2.10 Nuclear spin metrology with nitrogen vacancy center in diamond for axion dark matter detection [139]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 09 July 2024
- **Published in Phys. Rev. D 111 (2025) 075028:** 01 April 2025

3.18.2.11 Listening for new physics with quantum acoustics [140]

- **Authors:** Ryan Linehan, Tanner Trickle, Christopher R. Conner, Sohitri Ghosh, Tongyan Lin, Mukul Sholapurkar, Andrew N. Cleland
- **Posted on arXiv:** 22 October 2024
- **Published in Phys.Rev.D:** 01 December 2025

3.18.2.12 Entanglement-enhanced ac magnetometry in the presence of Markovian noise [141]

- **Authors:** Thanaporn Sichanugrist, Hajime Fukuda, Takeo Moroi, Kazunori Nakayama, So Chigusa, Norikazu Mizuochi, Masashi Hazumi, Yuichiro Matsuzaki
- **Posted on arXiv:** 28 October 2024
- **Published in Phys.Rev.A:** 07 April 2025

3.18.2.13 Probing Sub-eV Dark Photon, Scalar and Axion-like Particle Dark Matters with Transmon Qubits [142]

- **Authors:** Wei Chao, Yu Gao, Ming-Jie Jin, Xiao-Sheng Liu, Xi-Lei Sun
- **Posted on arXiv:** 30 December 2024
- **Published in Phys.Dark Univ. 50 (2025) 102171:** 15 November 2025

3.18.2.14 Piezoelectric bulk acoustic resonators for dark photon detection [143]

- **Authors:** Tanner Trickle
- **Posted on arXiv:** 09 January 2025
- **Published in Phys.Rev.D:** 01 September 2025

3.18.2.15 Background suppression in quantum sensing of dark matter via collective entangled-state projection [144]

- **Authors:** Shion Chen, Hajime Fukuda, Yutaro Iiyama, Yuya Mino, Takeo Moroi, Mikio Nakahara, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 02 October 2025
- **Published in Phys.Rev.D:** 01 March 2026

3.18.2.16 Hybrid-spin decoupling for noise-resilient DC quantum sensing [145]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 20 November 2025

3.18.3 Detector Technologies and Simulations

3.18.3.1 Impact of ionizing radiation on superconducting qubit coherence [165]

- **Authors:** Antti P. Vepsäläinen, Amir H. Karamlou, John L. Orrell, Akshunna S. Dogra, Ben Loer, Francisca Vasconcelos, David K. Kim, Alexander J. Melville, Bethany M. Niedzielski, Jonilyn L. Yoder, Simon Gustavsson, Joseph A. Formaggio, Brent A. VanDevender, William D. Oliver
- **Posted on arXiv:** 24 January 2020
- **Published in Nature:** 26 August 2020

3.18.3.2 Correlated charge noise and relaxation errors in superconducting qubits [166]

- **Authors:** C.D. Wilen, S. Abdullah, N.A. Kurinsky, C. Stanford, L. Cardani, G. D'Imperio, C. Tomei, L. Faoro, L.B. Ioffe, C.H. Liu, A. Opremcak, B.G. Christensen, J.L. DuBois, R. McDermott
- **Posted on arXiv:** 10 December 2020
- **Published in Nature 594, 369–373 (2021):** 16 June 2021

3.18.3.3 Reducing the impact of radioactivity on quantum circuits in a deep-underground facility [167]

- **Authors:** Laura Cardani, Francesco Valenti, Nicola Casali, Gianluigi Catelani, Thibault Charpentier, Massimiliano Clemenza, Ivan Colantoni, Angelo Cruciani, G. D'Imperio, Luca Gironi, Lukas Grünhaupt, Daria Gusenkova, Fabio Henriques, Marc Lagoin, Maria Martinez, Giorgio Pettinari, Claudia Rusconi, Oliver Sander, C. Tomei, Alexey V. Ustinov, Marc Weber, Wolfgang Wernsdorfer, Marco Vignati, Stefano Pirro, Ioan M. Pop
- **Posted on arXiv:** 05 May 2020
- **Published in Nature Commun.:** 12 May 2021

3.18.3.4 Synchronous detection of cosmic rays and correlated errors in superconducting qubit arrays [168]

- **Authors:** Patrick M. Harrington, Mingyu Li, Max Hays, Wouter Van De Pontseele, Daniel Mayer, H. Douglas Pinckney, Felipe Contipelli, Michael Gingras, Bethany M. Niedzielski, Hannah Stickler, Jonilyn L. Yoder, Mollie E. Schwartz, Jeffrey A. Grover, Kyle Serniak, William D. Oliver, Joseph A. Formaggio
- **Posted on arXiv:** 05 February 2024
- **Published in Nature Commun.:** 11 July 2025

3.18.3.5 Cosmic-ray-induced correlated errors in superconducting qubit array [169]

- **Authors:** Xue-Gang Li, Jun-Hua Wang, Yao-Yao Jiang, Guang-Ming Xue, Xiao-Xia Cai, Jun Zhou, Ming Gong, Zhao-Feng Liu, Shuang-Yu Zheng, Deng-Ke Ma, Mo Chen, Wei-Jie Sun, Shuang Yang, Fei Yan, Yi-Rong Jin, Xue-Feng Ding, Hai-Feng Yu
- **Posted on arXiv:** 06 February 2024
- **Published in Nature Commun.:** 20 May 2025

3.18.4 Crosslists

3.18.4.1 Quantum Sensing for High Energy Physics [12]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

3.18.4.2 Opportunities for Nuclear Physics & Quantum Information Science [13]

- **Authors:** Ian C. Cloët, Matthew R. Dietrich, John Arrington, Alexei Bazavov, Michael Bishof, Adam Freese, Alexey V. Gorshkov, Anna Grassellino, Kawtar Hafidi, Zubin Jacob, Michael McGuigan, Yannick Meurice, Zein-Eddine Meziani, Peter Mueller, Christine Muschik, James Osborn, Matthew Otten, Peter Petreczky, Tomas Polakovic, Alan Poon, Raphael Pooser, Alessandro Roggero, Mark Saffman, Brent VanDevender, Jiehang Zhang, Erez Zohar
- **Posted on arXiv:** 13 March 2019

3.18.4.3 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [124]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu

- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

3.18.4.4 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [19]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.18.4.5 New Horizons: Scalar and Vector Ultralight Dark Matter [23]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou
- **Posted on arXiv:** 28 March 2022

3.18.4.6 Quantum Networks for High Energy Physics [24]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

3.18.4.7 Quantum Systems for Enhanced High Energy Particle Physics Detectors [164]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

3.18.4.8 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [27]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.18.4.9 Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 [28]

- **Authors:** Douglas Beck, Joseph Carlson, Zohreh Davoudi, Joseph Formaggio, Sofia Quaglioni, Martin Savage, Joao Barata, Tanmoy Bhattacharya, Michael Bishof, Ian Cloet, Andrea Delgado, Michael DeMarco, Caleb Fink, Adrien Florio, Marianne Francois, Dorota Grabowska, Shannon Hoogerheide, Mengyao Huang, Kazuki Ikeda, Marc Illa, Kyungseon Joo, Dmitri Kharzeev, Karol Kowalski, Wai Kin Lai, Kyle Leach, Ben Loer, Ian Low, Joshua Martin, David Moore, Thomas Mehen, Niklas Mueller, James Mulligan, Pieter Mumm, Francesco Pederiva, Rob Pisarski, Mateusz Ploskon, Sanjay Reddy, Gautam Rupak, Hersh Singh, Maninder Singh, Ionel Stetcu, Jesse Stryker, Paul Szypryt, Semeon Valgushev, Brent VanDevender, Samuel Watkins, Christopher Wilson, Xiaojun Yao, Andrei Afanasev, Akif Baha Balantekin, Alessandro Baroni, Raymond Bunker, Bipasha Chakraborty, Ivan Chernyshev, Vincenzo Cirigliano, Benjamin Clark, Shashi Kumar Dhiman, Weijie Du, Dipangkar Dutta, Robert Edwards, Abraham Flores, Alfredo Galindo-Uribarri, Ronald Fernando Garcia Ruiz, Vesselin Gueorguiev, Fanqing Guo, Erin Hansen, Hector Hernandez, Koichi Hattori, Philipp Hauke, Morten Hjorth-Jensen, Keith Jankowski, Calvin Johnson, Denis Lacroix, Dean Lee, Huey-Wen Lin, Xiaohui Liu, Felipe J. Llanes-Estrada, John Looney, Misha Lukin, Alexis Mercenne, Jeff Miller, Emil Mottola, Berndt Mueller, Benjamin Nachman, John Negele, John Orrell, Amol Patwardhan, Daniel Phillips, Stephen Poole, Irene Qualters, Mike Rumore, Thomas Schaefer, Jeremy Scott, Rajeev Singh, James Vary, Juan-Jose Galvez-Viruet, Kyle Wendt, Hongxi Xing, Liang Yang, Glenn Young, Fanyi Zhao
- **Posted on arXiv:** 28 February 2023

3.18.4.10 Quantum sensing for particle physics [153]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.18.4.11 Quantum Sensors for High Energy Physics [30]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.18.4.12 Quantum measurements in fundamental physics: a user's manual [108]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023
- **Published in SciPost Phys. Rev. 2 (2026):** 2026

3.18.4.13 Searches for exotic spin-dependent interactions with spin sensors [87]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

3.18.4.14 Spin squeezing of macroscopic nuclear spin ensembles [88]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

3.18.4.15 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [91]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

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